OECD Economic Surveys

Denmark

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OECD ECONOMIC SURVEYS

2001-2002

Denmark

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BASIC STATISTICS OF DENMARK

THE LAND

Area (sq. km.)	43 094
Agricultural area (sq. km, 1997)	26 470

Population of major cities, 2000, thousands:	
Copenhagen	499
Århus	275
Odense	176
Ålborg	154

THE PEOPLE

Population, 2000, thousands	5 337
Number of inhabitants per sq. km	124
Population, annual net natural increase (average 1996-2000, thousands)	7.7
Natural increase rate per 1 000 inhabitants, 2000	1.7

Total employment, 2000, thousands	2 686
By sector:	
Agriculture	102
Industry	436
Construction	166
Market services	1 021
Community, social and personal services	948

THE PRODUCTION

Gross domestic product, 2000	
Kr billion	1 296.1
Per capita (US$)	30 027

Gross fixed capital formation, 2000	
Kr billion	283.0
Per cent of GDP	21.8
Per capita (US$)	6 555

THE GOVERNMENT

Public consumption, 2000	
Per cent of GDP	25.1
General government current revenue	
Per cent of GDP	53.1
Public gross fixed capital investment	
Per cent of GDP	1.7

Composition of Parliament:	Number of seats
Social Democrats	52
Liberals	56
Danish People's Party	22
Conservatives	16
Socialist People's Party	12
Social Liberals	9
Left Alliance	4
Christian Democrats	4
North Atlantic	4
Total	179

Last general elections: 20.11.2001

Next general elections: 20.11.2005 (at the latest)

THE FOREIGN TRADE

Exports, 2000	
Exports of goods and services	
Per cent of GDP	43.8
Main merchandise exports (per cent of total)	
Agricultural products	10.3
of which: Products of animal origin	8.4
Manufactured products	73.8
of which: Machinery and instruments	27.1
Other manufactured products	46.7
Other products	15.9

Imports, 2000	
Imports of goods and services	
Per cent of GDP	37.9
Main merchandise imports (per cent of total)	
Intermediate goods for agriculture	2.4
Intermediate goods for other sectors	41.1
Fuels and lubricants	3.6
Capital goods	14.0
Transport equipment	6.2
Consumer goods	27.4

THE CURRENCY

Monetary unit: Krone

Currency units per US$	
Year 2001	8.321
December 2001	8.332

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is published on the responsibility of the Economic and Development Review Committee of the OECD, which is charged with the examination of the economic situation of Member countries.

•

The economic situation and policies of Denmark were reviewed by the Committee on 7 January 2002. The draft report was then revised in the light of the discussions and given final approval as the agreed report of the whole Committee on 23 January 2002.

•

The Secretariat's draft report was prepared for the Committee by Deborah Roseveare and Steen Daugaard under the supervision of Peter Jarrett.

•

The previous Survey of Denmark was issued in July 2000.

Assessment and recommendations

International developments have slowed Denmark's economy…

The Danish economy was operating at close to its potential when the international downturn occurred. The underlying pace of activity slowed somewhat last year, although the estimated GDP growth of 1¼ per cent in 2001 overstates the apparent deceleration, since output in 2000 had been boosted by rebuilding work following the hurricane in December 1999. Export volume growth weakened sharply to around 4 per cent, and business investment shrank as sales prospects deteriorated. Nevertheless, increases in public and private consumption underpinned aggregate demand, total employment continued to rise, and unemployment fell further. With the labour market remaining tight, hourly earnings growth reached 4¼ per cent in the third quarter.

… but activity is expected to rebound around mid-year

The Danish economy is likely to be quite weak in the first half of 2002, as for most countries. But activity is expected to accelerate around the middle of the year as the international situation improves, according to the OECD's central scenario published in *Economic Outlook* 70. This should gradually translate into a more favourable climate for investment. Private consumption will be supported by steady growth in real disposable incomes; both wage increases and inflation are expected to moderate as the output gap widens. Nevertheless, the household saving ratio is projected to rise a little further, given uncertainties about job prospects as unemployment increases slightly. By 2003 the economy should again be growing at around its potential rate of about 2¼ per cent, though this would not be enough to eliminate the slack. Uncertainties stemming from the current external environment are unusually large and skewed downwards. On the other hand, domestic risks

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appear relatively small but may be biased towards stronger growth and labour market pressures.

Monetary policy, effectively set at the euro area level, is supportive…

Monetary policy remains dedicated to maintaining the value of the Danish Krone against the euro in the ERMII. Since the referendum in September 2000, when Danes voted against adopting the euro, exchange rate movements *vis-à-vis* the euro have remained extremely modest around the central parity. The Danish National Bank has followed the European Central Bank (ECB) and has lowered the discount rate by 1½ percentage points since May 2001. These reductions have helped to support activity and attenuate the slowdown. As the Danish economy is currently more closely aligned with the cycle in the rest of Europe than in the past, interest rates are now able to play a more useful and appropriate stabilising role; the further projected easing of monetary policy by the ECB should help address the current unfavourable international environment.

… while fiscal policy remains appropriately neutral

With relatively little excess capacity and monetary settings underpinning activity, the present neutral fiscal stance is appropriate. The overall general government surplus reached an estimated 2 per cent of GDP in 2001, and the new government is expected to follow through on its stated intention to pursue a similarly prudent course, while allowing automatic stabilisers to work. Growth in real public consumption is less benign, having risen by around 1¾ per cent in 2001, and the local government budgets agreed for 2002 imply considerable expansion of education, health and elderly care. Structurally, Denmark already stands out with a ratio of public consumption to GDP of 25 per cent, which is 5 percentage points higher than the EU average and more than 10 percentage points higher than in most other OECD countries. Adjusted for differences in tax treatment, net social expenditure is largely on a par with levels in most other EU countries, but the degree of redistribution involved is higher, above all in the area of public pensions. Channelling a relatively high proportion of resources through the public sector seems to enjoy strong public support in Denmark. However, as in other countries this comes at some cost, as the economic distortions of tax financing rise more than proportionally.

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A binding expenditure ceiling would reinforce the central budget framework…

Public expenditure policy is analysed in depth in this *Survey* to assess whether current Danish policy processes and institutional frameworks encourage high efficiency and ensure overall spending control. The current pragmatic approach to macroeconomic management has been effective in achieving sound public finances. Nevertheless, although the overall expenditure level has been reduced, largely reflecting reduced expenditures on unemployment benefits and lower interest payments, there has been no firm mechanism to prevent public consumption and tax rates from creeping upwards. The new government's "tax freeze" deliberately shifts attention to the prioritisation of expenditures, which should enhance spending control, if implemented effectively. A binding ceiling on aggregate central government spending could encourage more explicit prioritisation of competing spending proposals and reduce the scope for last-minute amendments. Such a ceiling should be set in nominal terms, encompass grants to local government and, preferably, be approved by parliament before the annual negotiations with local governments get underway. The framework could also be strengthened if performance and outcome evaluations were more closely linked to the budget process. In addition to the assessments of the government, the presently fragmented approaches to external evaluation could be strengthened by centralising them in an independent agency.

… and control over expenditure by lower levels of government should be reinforced

A key feature of the Danish public sector is the more extensive devolution of responsibilities to local governments than elsewhere, which puts more than half of public expenditure beyond the immediate control of the central government. At the same time, scope for local governments to effectively reflect local preferences has been hampered by a reluctance at the national level to allow significant local differences to emerge. The system of budget agreements between the central government and the associations of local governments has proved inadequate, particularly in the second half of the 1990s. More often than not, actual spending levels and locally-determined income tax rates have exceeded those agreed upon. This has partly counteracted national economic measures to improve the incentives to work and develop skills. To avoid this clash of priorities and make the agreements binding on all parties, it could be necessary to restrict

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local governments' rights to raise income tax rates. This could be achieved via a formal tax "freeze" or by imposing financial penalties on local authorities that increase income tax rates. To offset this, the current limits on the scope for counties and municipalities to fund themselves through property taxes and user charges could be eased. In addition, individual local governments could be allowed to vary the basic personal allowance in the tax system; unlike changes in income tax rates, this would modify the tax base without distorting incentives to work and acquire human capital.

Competition and market signals should play a greater role in the provision of public services

Public services are mostly sheltered from competition, and market signals and consumer choice are relatively unimportant in the allocation of resources. The use of outsourcing, user charges and client choice should be increased to generate higher efficiency, rein in demand and facilitate expenditure control at the margin. In particular, pricing public services more consistently would reveal consumer preferences more clearly, while greater recourse to private providers may make it easier for a market for supplementary services to emerge, relieving pressures to upgrade the range of services financed from taxes. Several economic, political and institutional barriers need to be overcome. The large number of rather small municipalities probably impedes the spread of outsourcing, due to both diseconomies of scale and the inherently fragmented decision process. The current mix of user charges could be better aligned with the likely price sensitivities of demand for various services, and their use broadened. Greater reliance on user charges could also help governments to respond to expected future increases in demand for services without compromising long-run fiscal sustainability or resorting to steadily rising tax rates.

Spending on education and childcare needs better management; health and elderly care involve difficult choices about private/public financing

Several spending areas merit attention, some of which exemplify the broader expenditure issues and future expenditure challenges:

– *Educational spending* is high by international standards but is not matched by correspondingly better performance. More stringent expenditure control is needed in primary and lower secondary education where unit costs have trended upwards over a long period. Introducing modest tuition fees for tertiary education could encourage a faster pass-through and make

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more resources available for enhancing quality. In addition, subsidies for training of employees should be reduced.

– *Expenditure on pre-school childcare and after-school services* has surged over the past decade. Since the average effective fees were reduced in the early 1990s, coverage rates have risen rapidly, despite declining labour force participation rates and more generous parental leave schemes. Linking charges more closely to children's hours of attendance could reduce excessive demand, although potential labour supply effects should be taken into account, along with possible losses of revenue.

– *Health care and elderly care (including home help) costs* have not risen faster than GDP, in large part because of efficient supply-side management and rationing. However, strains are becoming more visible. Private insurance arrangements offering better-quality services are becoming more popular, even though they compromise the principle of "free and equal access", which is a high political priority. Yet expanding public resources to provide everyone with the level of service sought by those groups most willing to pay could prove expensive. Policymakers need to address more explicitly where the boundary should lie between public and private financing.

Future growth prospects depend on lifting participation rates and adopting reforms that offer long-run benefits

Dealing effectively with the pressures facing the public sector becomes even more important because demographic developments will soon start acting as a brake on potential growth. This presents two other broad challenges:

– Boosting the labour supply will require higher age-specific participation rates. Since virtually all non-employed adults receive some kind of public income transfer, this would be easier to achieve if further reforms of the transfer system were made, for example involving stricter eligibility criteria, less generous replacement rates and greater incentives to complete tertiary education more expeditiously.

– The particular sensitivity of policymakers to first-round winners and losers of policy changes needs to

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be overcome, because it inhibits reforms that could stimulate a more efficient allocation of resources even for changes that would not affect low-income households or widen income distribution. The long-standing system of rent controls in the housing market and subsidies to social and co-operative housing provide examples where concerns for the immediate losers from reforms have been placed ahead of the longer-term benefits, even though current policies are not particularly well targeted on lower-income groups.

Tax reforms should focus on reducing marginal tax rates on labour income and addressing anomalies

Excessive concerns about immediate distribution consequences have also stood in the way of reforms that would reduce the average marginal tax rate on labour, thereby boosting potential output by improving the incentives to work. While reducing the bottom personal income tax rate is politically attractive because it leaves income distribution largely undisturbed, it is an inefficient instrument in this context. In contrast, raising the threshold for the top state (*i.e.* central) tax rate from barely above the average production worker (APW) wage to around 30 per cent above the APW wage would improve work incentives at a relatively modest cost since the induced labour supply response could partly compensate the immediate revenue loss, while reducing the average marginal tax rate by some 2 percentage points. A wider tax reform designed to reduce marginal tax rates on labour across the board is also warranted, but rate cuts should be matched with offsetting revenue or spending proposals, given that robust fiscal surpluses need to be maintained. While the tax reforms over the past 15 years have generally made the system more neutral, a few notable exceptions remain:

– The taxation of car ownership is extremely heavy, making purchase prices up to three times higher than the corresponding pre-tax prices. As a result, the car stock is relatively old, while many low-income households find it difficult to afford a car. Pollution would be reduced by modernisation of the car fleet. Since other externalities associated with car use would be better dealt with *via* road pricing, efforts should be made to develop an effective system for doing so.

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– Taxation of capital has become overly complicated, in part because distribution policy has taken precedence over efficiency and simplicity. Tax rates on positive net capital income, which currently match the progressive rates on personal income, should be better aligned with the lower flat rate on negative capital income (*i.e.* interest payments) although care should be taken to avoid encouraging reclassification of earned income. Taxation of share income should also be simplified.

A more "inclusive" labour market is laudable, but difficult to achieve

Creating a more "inclusive" labour market with better employment opportunities for people such as immigrants, long-term clients of the social welfare system and those with reduced working capacity, has moved up the policy agenda. This approach is designed to lift labour supply in the medium term and extends the successful "workfare" philosophy applied to other groups in the labour market. The reform of the early retirement (disability) pension system is rather audacious in tightening eligibility criteria significantly on the assumption that those left out will be able to obtain jobs on special terms, where their reduced ability to work is compensated by wage subsidies (so-called *flexi-jobs*). Although the number of such positions has recently risen, it seems optimistic to expect continued expansion on the scale required. At the least, it will require favourable economic conditions overall and greater acceptance by other workers. Moreover, the pension reform and the recently introduced generous out-of-work benefits for the would-be employed in *flexi-jobs* tend to weaken the economic incentives to opt for a *flexi-job* and may need to be reconsidered.

Successful integration of immigrants requires better-quality training programmes and stronger incentives

Integration of immigrants in the labour market remains another challenge. It is made more difficult because inflows are dominated by refugees and those arriving under family reunion rules, who are often from culturally distant countries. Nevertheless, the mediocre record to date reflects several factors, including inadequate educational and linguistic skills, unequal access to employment opportunities, and the poverty trap that results from high replacement rates and access to universal social benefits irrespective of past labour-market experience. Considering the complexity

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of the issue, a coherent strategy for improving integration would need to involve several measures:

- Improving the quality of the now compulsory three-year introduction courses. Non-attendance seems to remain a problem, and activation is not taking place on the scale envisaged in the Integration Act.
- Making greater efforts to make immigrants' home-country qualifications usable in Denmark, perhaps through some kind of state certification of competence.
- Strengthening the economic incentives to work. One possibility might be an extension of targeted wage subsidies to employers, or in-work benefits to employees available to immigrants for a limited number of years. This would also encourage individuals to acquire the skills necessary to gain a foothold in the labour market.

The whole electricity market should be made subject to market forces

Another area where non-economic objectives have complicated the way markets operate is the promotion of renewable sources of energy, particularly wind. With the proposed "green certificates" scheme still on hold, the electricity market remains segmented into two parts: *i*) the Public Service Obligation (PSO) covering energy from renewable sources, which users are required to purchase at a high fixed tariff; and *ii*) the balance of the market where prices are determined by supply and demand. As a result, the price for wind energy is currently around three times higher than Nordpool spot prices. Furthermore, this arrangement also provides large economic rents to owners of modern vintages of wind turbines, as the cost of generation has fallen sharply in recent years to a level where producers could compete without support. Addressing these distortions in a way that ensures greater competition, a more efficient allocation of resources and an integration of environmental objectives would best be handled by a three-pronged strategy:

- Allowing electricity users complete freedom to buy the cheapest power available on the market, which would in turn encourage the search for lower generation costs.

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– Addressing the financial commitments made to existing turbine-owners by shifting them back onto the budget as subsidies and phasing them out over a fixed period of time.
– Using the recently introduced CO_2 emissions cap-and-trade system to influence relative prices in favour of renewables energy, and reducing the allowable emissions ceiling more sharply, if stronger incentives are required.

It would also make sense to extend the tradable CO_2 emissions quota system to cover the whole economy instead of just electricity generation; this would help Denmark meet its objectives for reducing greenhouse gas emissions in the least cost way. In addition, the natural gas market should be liberalised more rapidly.

Summing up

Denmark has experienced an economic slowdown, reflecting the weak international climate, although the impact of the downturn has been cushioned by the composition of exports. The pace of activity is expected to accelerate in the second half of 2002, leading to growth at around potential rates by 2003, with the balance of domestic risks possibly on the upside. Against this backdrop, fiscal policy should remain neutral, allowing automatic fiscal stabilisers and interest rates, effectively determined at the euro area level, to support growth. In a medium-term perspective, boosting labour supply should remain a top policy priority, given the relative tightness of the labour market and future pressures from demographic developments. The "inclusive labour market" approach is laudable, but insufficient, and further structural measures to increase labour market participation and reduce unemployment are needed. Raising the threshold for the top personal income tax rate would be effective in encouraging work for reasonably modest fiscal cost, while lowering taxes on positive capital income would provide stronger incentives to save and invest and make the tax system more neutral. The public sector is large, by political choice; but ensuring that it does not involve excessive economic costs would be facilitated by rethinking some of its characteristics. Attaining the optimal level and mix of public services is complicated by the extensive devolution of responsibilities to local authorities. Better institutional

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arrangements are needed to deliver firmer expenditure control, which could include making the agreements between central and local governments a more effective and enforceable instrument, changing local governments' taxing powers and implementing a binding nominal spending target at the central government level. More extensive recourse to user charges and competition among providers would also help to ensure better value for money. Overall, the Danish economy is in good shape but would benefit from greater efforts to improve the supply side through further structural reforms.

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I. Steering a steady macroeconomic course

Recent developments

Close to capacity and well balanced

The Danish economy has been performing close to its potential in recent years. The so-called "Whitsun package" of tax measures, implemented over the period 1999 to 2002, has achieved both a "soft landing" and a subsequent re-balancing of activity towards export markets and business investment (Figure 1). Private consumption was weak until early 2001 and price inflation remained within reasonable bounds, especially once the impact of energy prices is excluded. A robust fiscal surplus of 2 to 3 per cent of GDP has been maintained throughout. Moreover, the current account surplus has grown steadily since 1999, in sharp contrast to 1997 and 1998 when vigorous domestic demand and the resulting capacity constraints led to significant losses of export market shares and a sharp deterioration of the external balance. However, the labour market has remained tight; unemployment fell below the OECD's estimates of the structural rate in 1997 and has continued to edge down until late 2001 (Figure 2). The combination of a turnaround in the external balance and the further inroads that have been made on unemployment illustrate the extent to which the economy has been successfully re-oriented.

Recent slowdown has been mild

Against this background of macroeconomic stability, growth has slowed in Denmark. However, the apparent sharp slowdown in GDP growth from 3 per cent in 2000 to only 1¼ per cent in 2001 includes the effects of the nation-wide hurricane in December 1999, which may in itself account for as much as 1 to 1½ percentage point of the measured deceleration in growth.[1] Hurricane-adjusted GDP growth in 2001 would be around 1¾ per cent, which brings Denmark's rate of expansion close to the European Union average.

Exports held up well in international markets

Nevertheless, the international slowdown served to dampen the economy. Following double-digit growth rates in 2000, world trade contracted by around

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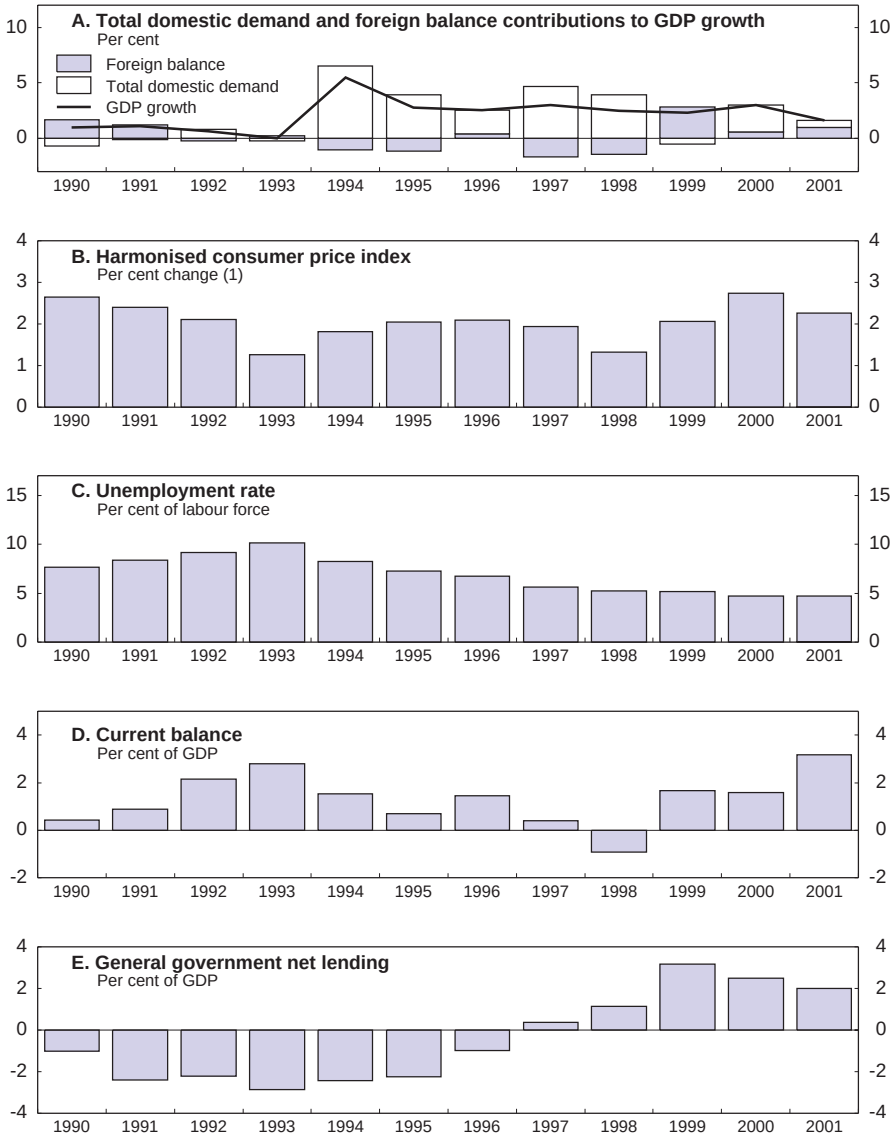
Figure 1. **Macroeconomic performance**



Note: 2001 figures are estimates.
1. Before 1994, national CPI.
Source: OECD.

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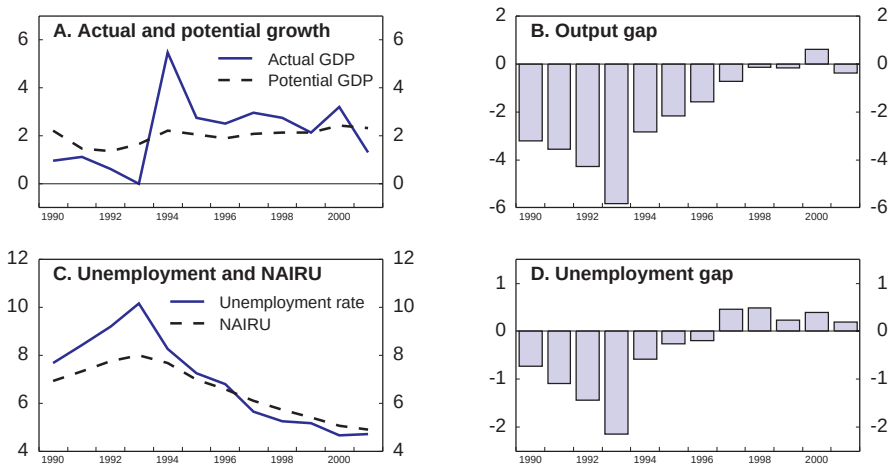
Figure 2. **Output and unemployment gaps**

Per cent



Source: OECD.

2 per cent at an annualised rate in the first six months of 2001 and probably continued to shrink in the second half. Against this background, growth in Danish exports slowed to an estimated rate of 4 per cent for 2001, down from the 11½ per cent expansion in 2000 (Figure 3). This implies strong export market performance with a gain of market shares of some 5 per cent.[2] The explanation lies in the industrial structure with many small and medium-sized companies and in the composition of Danish exports, which in the current slowdown has left the country less exposed to variations in international activity than some others:

- Pork exports, as well as being relatively insensitive to economic fluctuations, have benefited from both a trend shift in consumer preferences away from beef and the temporary effects of the foot and mouth outbreak in the United Kingdom and the Netherlands.
- Exports of wind turbines have soared, and although sales still comprise a small proportion of total exports, industry estimates indicate that they may have risen by around 50 per cent in 2001.
- Demand for pharmaceuticals and other biotechnology products are relatively immune from cyclical developments, while Denmark has continued to broaden and deepen its specialisation in them.[3]

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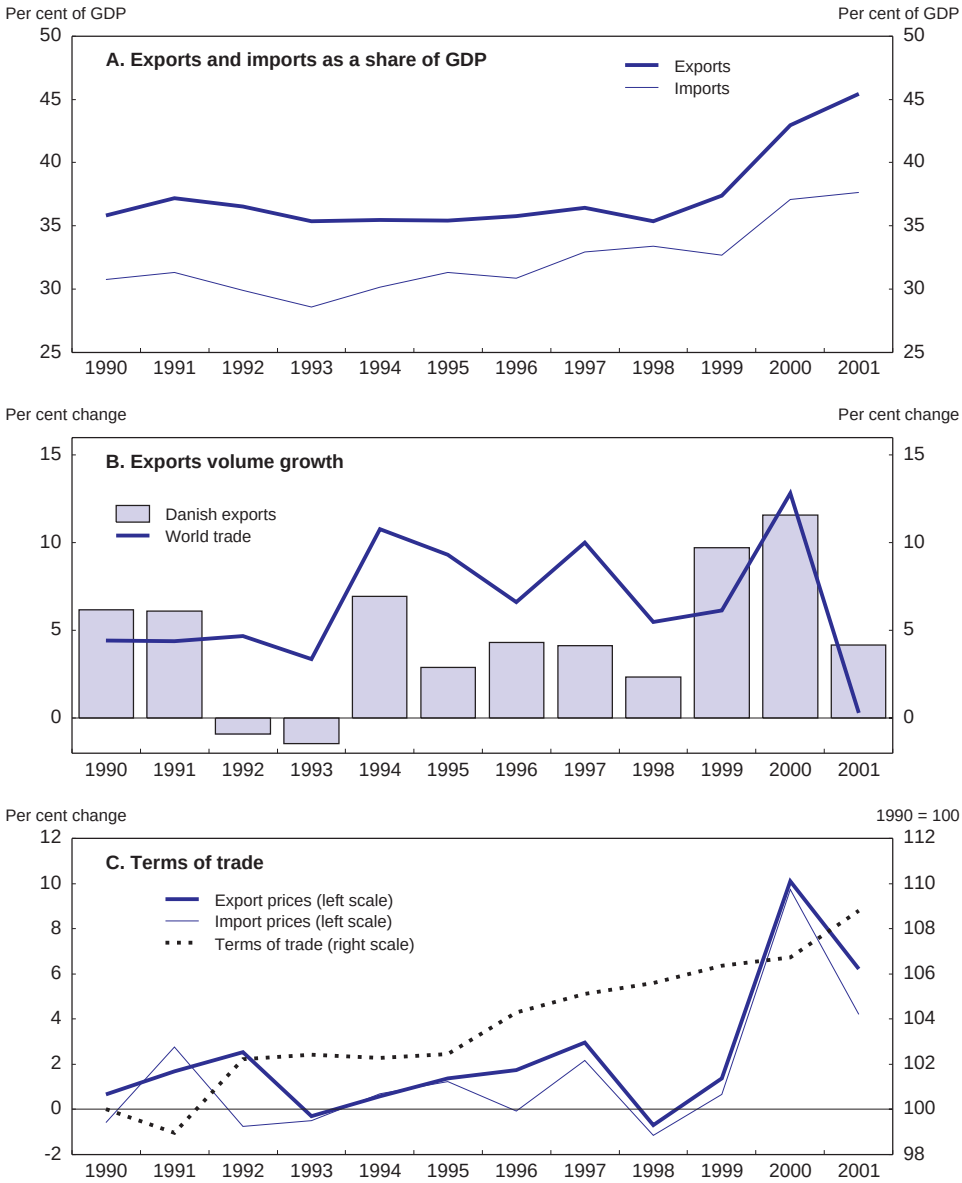
Figure 3. **Exports and imports**



Note: Data for 2001 are OECD estimates.
Source: OECD.

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Towards the end of 2001, some sectors were particularly hard hit by the aftermath of the 11 September terrorist attacks in the United States, with the slump in freight, travel and tourism services directly affecting Denmark along with most other countries. Other knock-on effects from those events on demand for Danish exports remain difficult to assess.

Business investment was cut back

As in other countries, weaker business investments have played a big part in the slowdown. Faced with a rapidly deteriorating global outlook, and sinking business confidence (Figure 4), private non-residential capital outlays shrank in 2001. This came on the back of a major expansion in 2000, when business investment grew by 11¼ per cent and the capital stock increased by 3½ per cent. Some slackening of pace was not surprising, but the turnaround was sharp: these investments are estimated to have declined by 1½ per cent for 2001 as a whole, with machinery, equipment and transport falling more than 3 per cent and non-residential construction around 6 per cent. The rise in interest rates over the course of 2000 may have also acted as a brake on investment decisions, but real rates of return for Danish enterprises that year were at an all-time high (Pedersen, 2001). Financing expansion through equity also became more difficult in 2001, as the Danish stock market declined.

Housing starts fell though house prices rose

Housing investment plummeted in 2001 by 14 per cent, mainly reflecting the completion of the repair work required after the hurricane of December 1999, which pushed overall housing investment up by 11 per cent in 2000. But housing starts, measured in square metres, also fell by around 5 per cent in 2001, while single-family housing starts flattened out from a trend decline (Figure 5).[4] This is somewhat surprising, given the continued rise in real house prices, at least until the middle of 2001 and might indicate that present house price levels are not far from the point at which supply and demand are balanced. Normally, once the price of existing housing rises above the cost of construction (known as Tobin's q) additional housing investment would be expected to occur (see previous S*urvey*). On the basis of prices and construction activity, the Ministry of Finance currently estimates the level of house prices to be around 5 to 10 per cent above the equilibrium. House prices have been boosted by a strong shift towards floating rate mortgages, for which interest rates have been lower than for long-term fixed rate mortgages (see Box 1). It is possible that some new construction activity was postponed while resources were diverted to repair work and the sector was capacity constrained, although employment patterns within the sector in the early part of 2000 lend only limited support to this argument as the additional workers on repairs and maintenance were generally not offset by lower employment

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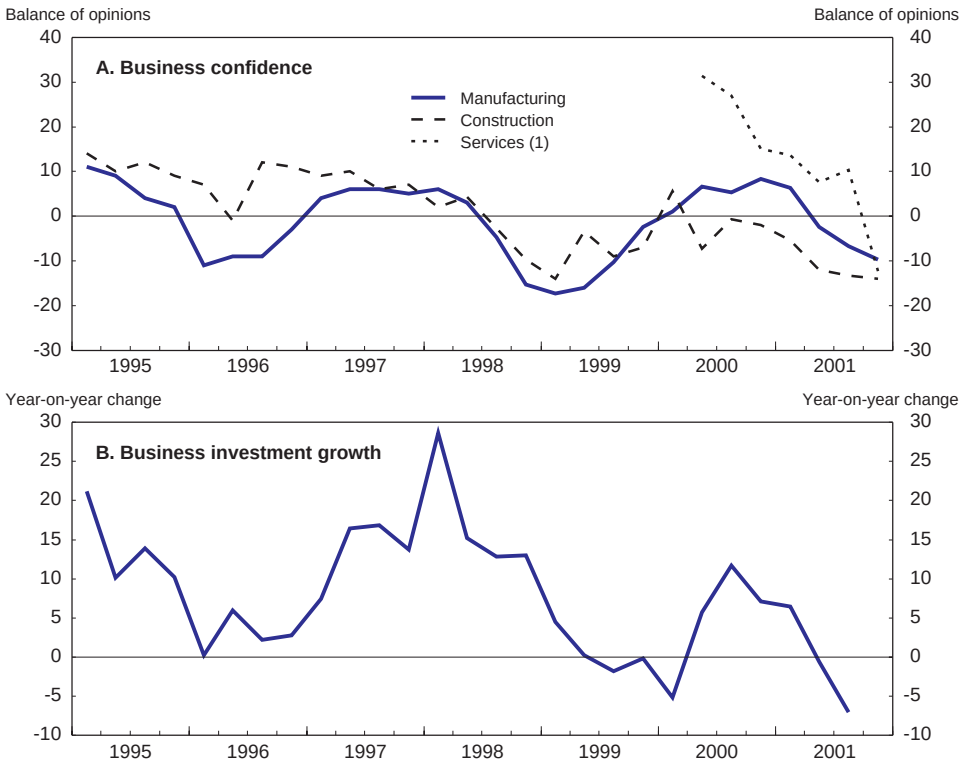
Figure 4. **Business confidence and investment**

Per cent



1. Not seasonally adjusted.
Source: Statistics Denmark.

elsewhere in the sector. Another factor explaining low housing starts may be the scarcity of suitable building sites in the Copenhagen area, where real estate pressures have been the most severe because of the buoyant job market, and where stringent environmental and planning rules may also be inhibiting development.

Private consumption finally picked up

Private consumption resumed modest growth of almost 1 per cent in 2001 after two very subdued years following the Whitsun package, which was expressly

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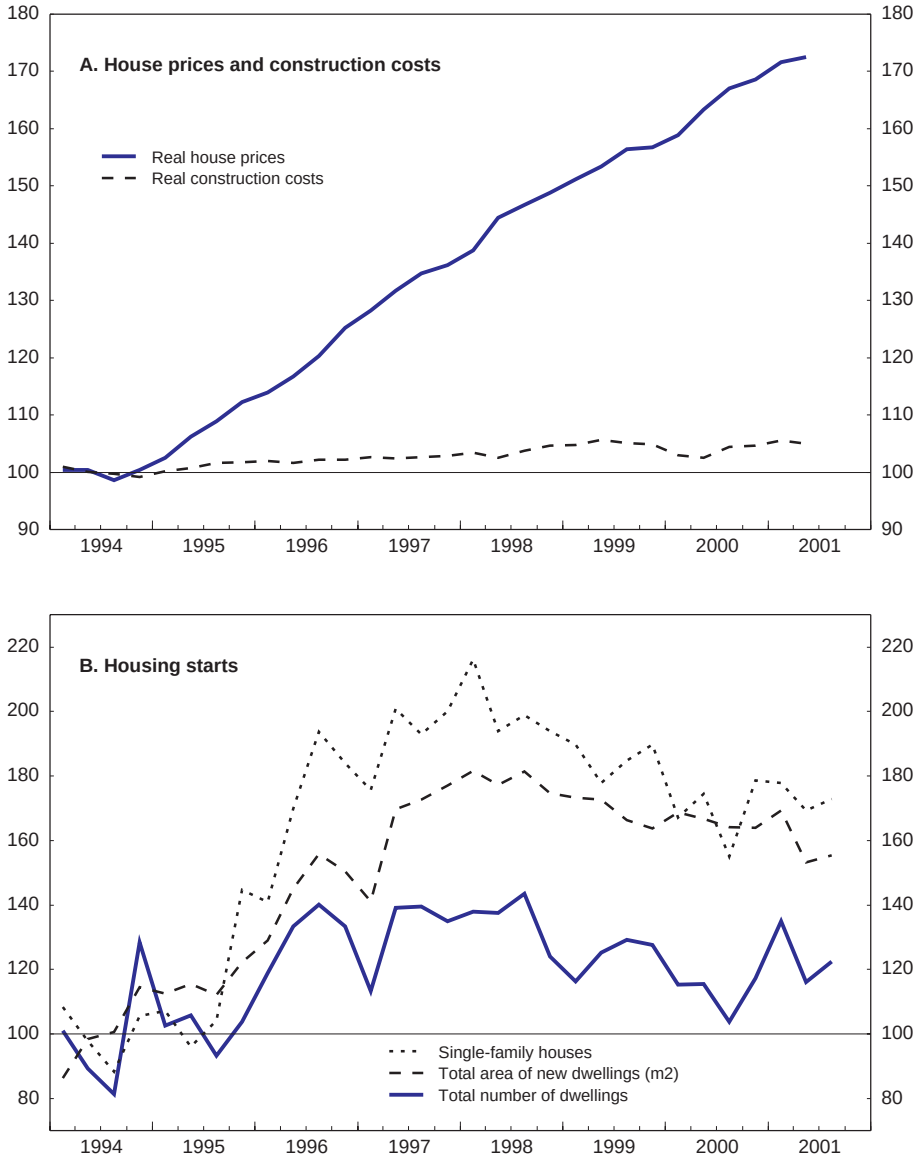
Figure 5. **Housing indicators**
1994 = 100



Source: Statistics Denmark.

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Box 1. Flexible interest rate mortgages and house prices

Early redemptions of fixed interest rate mortgages soared in 1998 and 1999 as mortgage bond rates fell; for example, the 30-year bond rate in 1998 was more than 2 percentage points lower than the rate applying to a 30-year bond taken out three years previously. In addition, many of these early redemptions were replaced by new variable rate contracts, which have only been offered by the mortgage credit institutes since 1999. These instruments typically involve refinancing of shorter-term (usually one to two year) mortgage bonds within an overall long-term financing arrangement with a mortgage credit institution. The gains from shifting from a fixed 30-year bond to a flexible rate instrument are estimated to have averaged 3 percentage points in 1999, 2 percentage points in 2000 and 2¼ percentage points in 2001. As a result, around 30 to 35 per cent of new loans taken out in each of the past 3 years are floating rate mortgages and one quarter of the stock of nominal loans by the mortgage credit institutes is of this type.

Shifts in interest rates can now be incorporated into these mortgage instruments and therefore will affect households' disposable income directly whereas under fixed interest rates, households had an incentive to hold their fixed mortgages when interest rates were rising, but to refinance when rates fell. The popularity of the variable rate mortgage can thus in some ways be interpreted as a vote of confidence in the future conduct of monetary policy.

The effects of the development of flexible mortgages on house prices have been estimated using the DREAM national expectations model of Statistics Denmark. The immediate impact is calculated to be around 8½ per cent. This contrasts with overall price increases of 14 per cent for single-family dwellings and 30 per cent for apartments since the beginning of 1999. However, the estimated long-run effect is only around 2½ per cent, as the immediate increase in house prices is assumed to eventually bring forth a supply response.

Source: Ministry of Economic Affairs.

designed to dampen household spending (Figure 6). Private sector real disposable incomes grew by around 1¾ per cent in 2001 (Table 1) with households seeing much stronger inflows in 2001 than in the previous two years (although corporate positions were reversed).[5] Consumer confidence remained remarkably stable considering the economic slowdown in Denmark and the sharp deterioration in confidence elsewhere in Europe as international economic prospects declined. This may be because unemployment remained low, house prices kept increasing and Danish equity prices were less volatile – rising by much less than other European stocks in the upswing, but maintaining momentum for longer and easing more gently after their peak in late 2000 (Figure 7). Public consumption

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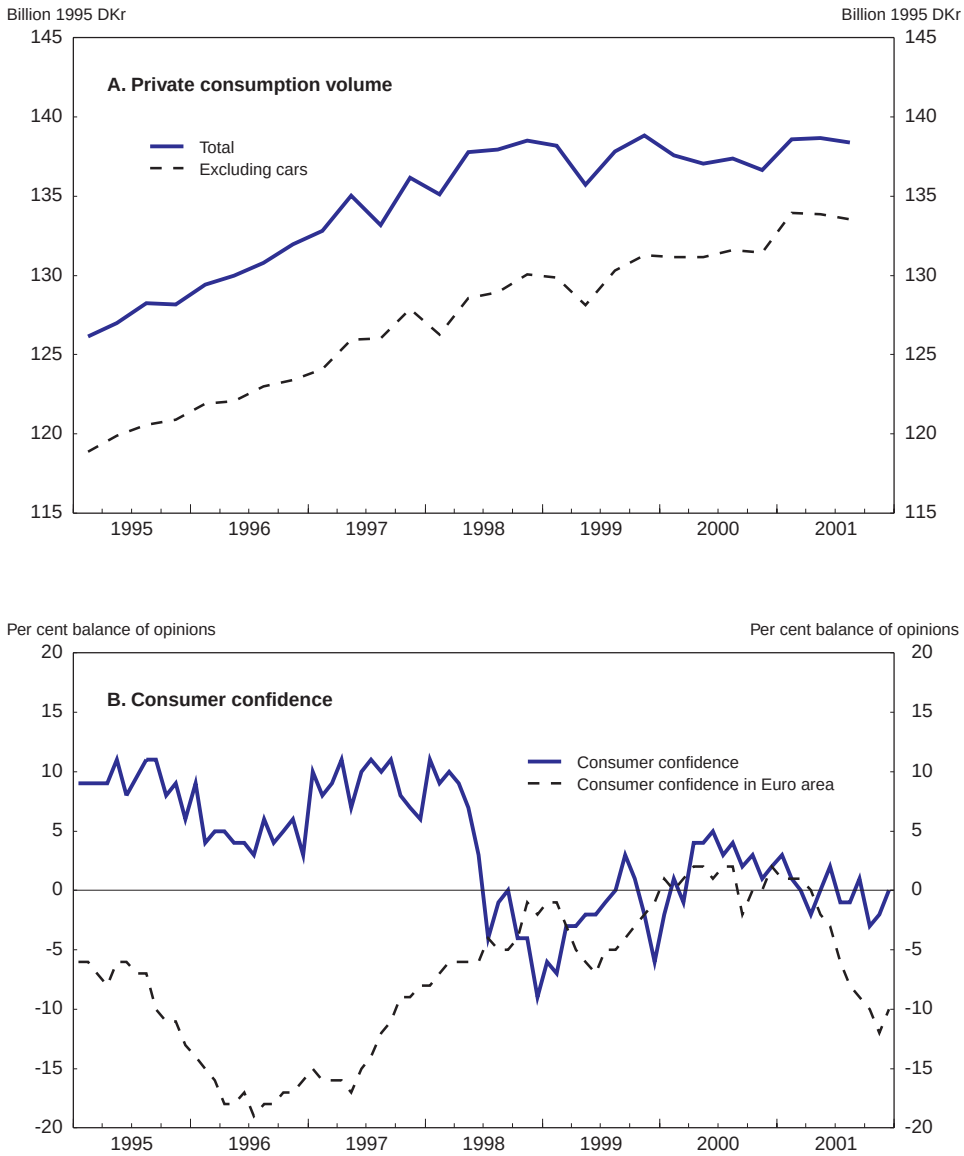
Figure 6. **Private consumption and confidence**



Source: Statistics Denmark; OECD.

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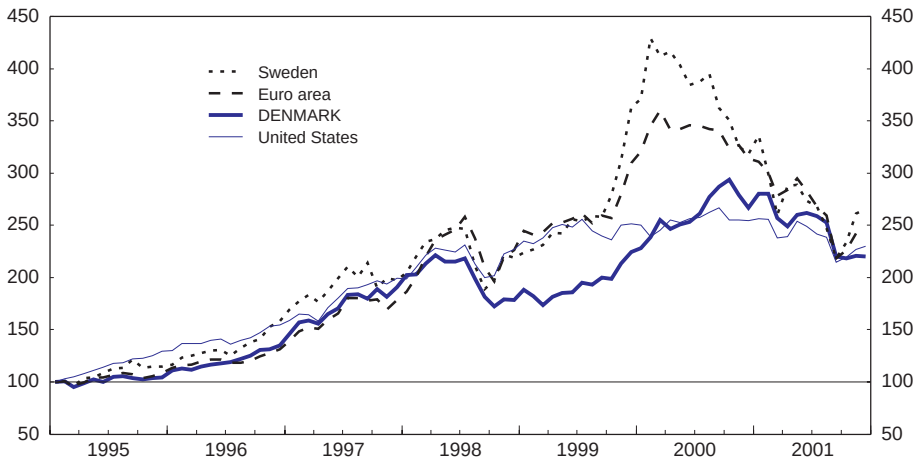
Table 1. **Nominal and real disposable incomes**

	1995	1996	1997	1998	1999	2000	2001[1]
	Percentage change						
Nominal income							
Private sector	4.0	3.6	4.2	2.7	5.1	5.6	3.6
Households	4.9	3.6	3.0	3.8	3.2	5.2	4.6
Corporations	0.7	3.6	8.4	–0.9	11.6	6.7	1.0
Nominal disposable income							
Private sector	5.3	2.7	4.4	2.1	2.2	7.0	4.4
Households	7.3	2.5	2.8	4.6	–0.3	6.0	7.4
Corporations	–0.2	3.5	9.2	–5.2	10.1	9.6	–3.0
Real disposable income[2]							
Private sector	3.3	0.6	2.1	0.7	–0.3	3.9	1.7
Households	5.3	0.4	0.5	3.2	–2.8	2.9	4.6
Corporations	–2.0	1.3	6.8	–6.4	7.3	6.5	–5.8

1. Estimates.
2. Deflated using private consumption deflator.
Source: Ministry of Finance.

Figure 7. **Equity prices**
January 1995 = 100



Source: OECD, *Main Economic Indicators*.

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grew in real terms by 1¾ per cent in 2001, again reflecting an expansion in education, health and care for the elderly (see Chapter III). Together, these two components were just sufficient to offset the fall in investment and keep final domestic demand at more or less the same level as in 2000.

Unemployment remains low

The labour market remains tight; the standardised unemployment rate was around 4¾ per cent in 2001, compared with a structural rate of 5 per cent, while the registered unemployment rate has continued falling slightly through the first three quarters of 2001, to reach its lowest level in 25 years. Employment growth was concentrated in the public sector, in line with the expansion of public services, while business employment remained more or less stable at 2000 levels. Aggregate business sector productivity growth (per person employed) dropped from 2¾ per cent in 2000 to less than 1½ per cent in 2001 (Figure 8). Since employment protection legislation is light in Denmark, a sharp decline might produce layoffs, and the fact that it has not done so suggests that some labour hoarding may be taking place. Although difficult to verify, this would be understandable firm behaviour against a background of general recruitment difficulties in 2000, especially for skilled labour, robust corporate earnings in 2000 and an expectation that the slowdown would be short-lived. In any case, the hourly productivity increases for 2000 and 2001, on average, remain above trend.

Wages have risen faster

Wage growth accelerated slightly in 2001, which also indicates that labour shortages have only slowly eased. Following convergence with euro-wide wage increases in 2000, the gap widened again over the course of 2001 to the disadvantage of Danish enterprises (Figure 8). With a somewhat bumpy pattern over time and across sectors, aggregate private sector hourly wage increases rose to around 4¼ per cent in 2001, up from the 3¾ per cent in the previous year. Technically, most of the wage acceleration was countered by lower working hours reflecting the agreement from the previous bargaining round to progressively establish an extra week of holiday (see previous S*urvey*). In consequence, wages paid per employee rose more moderately, at just under 4 per cent. Although not of great concern in the near term given recent export performance, higher wage increases than in the euro area underlines the fact that the labour market remains strained and implies a slow but steady erosion of wage competitiveness. Public sector wage increases edged down during the course of the year in counties and municipalities, but accelerated quite markedly at the central government level to reach a somewhat disquieting pace of 5½ per cent in the third quarter. However, these differences reflect in large part the particular profiles of the respective agreements.

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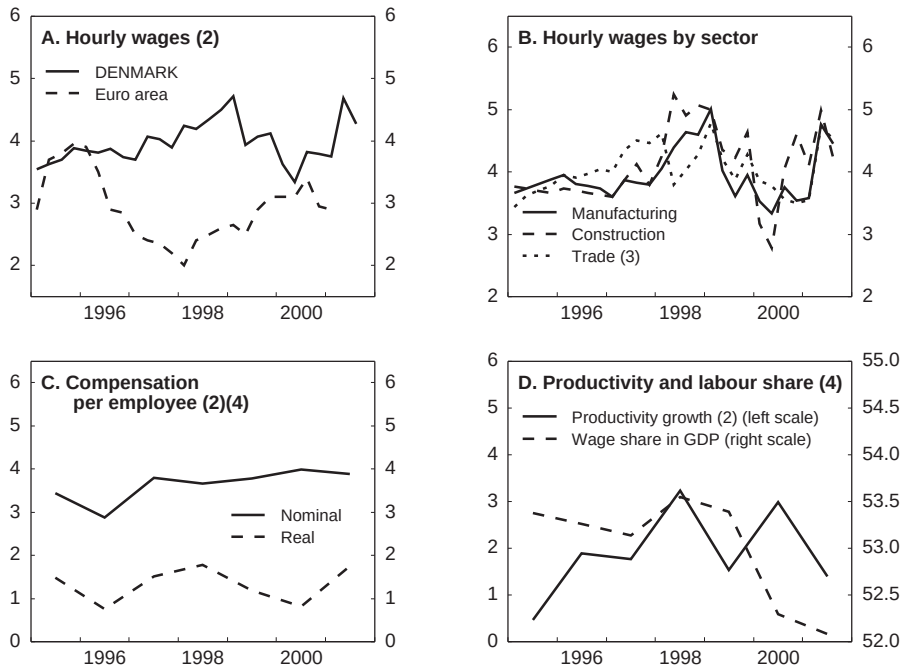
Figure 8. **Wages and productivity**
Year-on-year percentage change[1]



1. Except "Labour share" in Panel D.
2. Private sector.
3. Includes hotels and restaurants.
4. Data for 2001 are OECD estimates.

Source: Statistics Denmark; OECD.

Policy settings

Interest rates followed euro-area rates down

Denmark voted to retain the Danish krone rather than adopt the euro in a referendum in September 2000.[6] As a result, monetary policy remains dedicated to the objective of maintaining the krone fixed to the euro within narrow bands and the krone has in fact fluctuated by less than ½ per cent around its central parity. In the run-up to the referendum, the National Bank unilaterally raised its interest rates in order to send a clear signal that it would act to defend the krone's value, should that prove necessary. In fact, there was remarkably little financial turbulence in the wake of the referendum, and interest-rate differentials with the euro area quickly returned to previous low levels (Figure 9). The discount rate has

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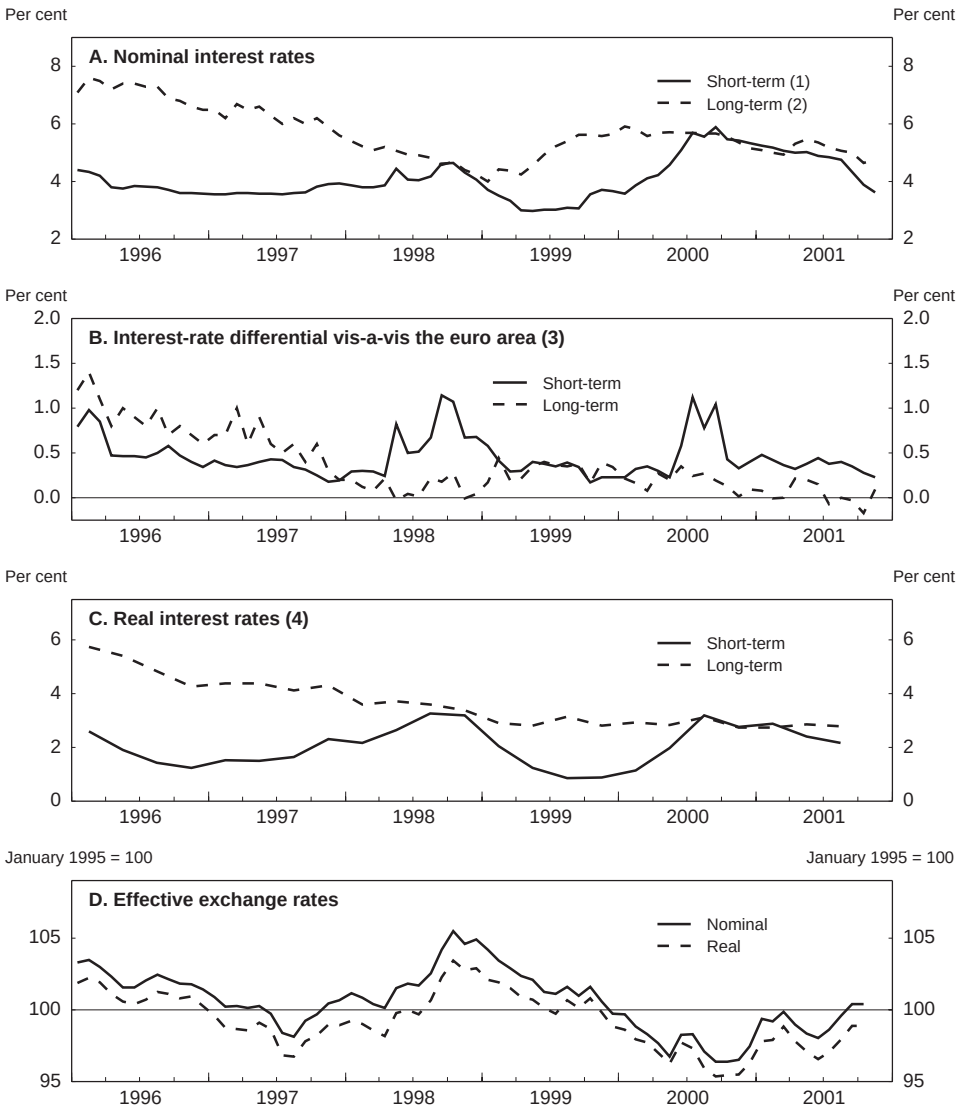
Figure 9. **Monetary conditions**



1. Interbank, 90-day.
2. Central government bonds, 10-year.
3. For the euro area before 1999, German government interest rates are used.
4. Deflated by the private consumption deflator change over 4 quarters.

Source: Statistics Denmark; OECD.

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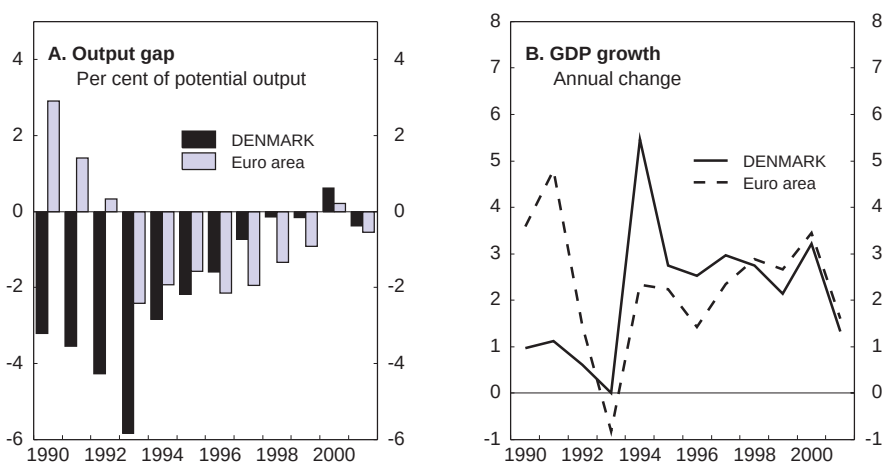
now been cut by 1½ percentage points since May 2001 as the Danish National Bank has mirrored the movements of the European Central Bank. Short-term interest rates peaked in September 2000 with the referendum, and have fallen steadily since then, although their effects on overall monetary conditions have been attenuated by an appreciation of the effective exchange rate of around 4 per cent over the same period. Nevertheless, lower interest rates have helped to support economic activity during the slowdown. Furthermore, the significant shift in mortgage housing financing arrangements in recent years has strengthened the transmission of shifts in official interest rates through to economic activity.

Denmark now closer to euro-area cycle

Monetary policy is effectively determined by the European Central Bank (ECB) and is now more suitable for Danish conditions. The soft landing achieved in Denmark in 1999, along with the elimination of negative output gaps in the euro area, has resulted in a much closer alignment between euro-area requirements and the domestic business cycle[7] (Figure 10). To the extent that Denmark remains more synchronised with the euro area in the future, interest rate decisions will continue to contribute a stabilising influence on activity.

Figure 10. **Converging output gaps and GDP growth**
Per cent



Source: OECD, *National Accounts* and Secretariat estimates.

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Fiscal policy was broadly neutral in 2001

The fiscal policy stance was broadly neutral in 2001. The 2001 budget passed in December 2000 provided for a net surplus amounting to 2.8 per cent of GDP – the same as the outcome achieved in 2000 (Table 2). It included a significant planned expansion of public consumption, involving additional public sector employment of some 10 thousand persons (1¼ per cent growth in the general government workforce). There were also some minor revenue initiatives at the central government level and increases in average tax rates for local governments. The Ministry of Finance now estimates the first year activity effect of budget initiatives to be around 0.2 per cent, but this will be offset by the accumulated activity effect of fiscal policy from earlier years, so that overall, fiscal policy has no effect on GDP growth in 2001. The same picture can be obtained by looking at the OECD's measure of the cyclically-adjusted net surplus, which shifted downwards slightly from 2.4 per cent to 2.2 per cent of potential GDP. Automatic stabilisers were allowed to work, as the actual budget surplus shrank to 2.0 per cent in 2001. Since registered unemployment continued falling slightly through most of 2001, this can largely be explained by the weakness in tax revenues and in particular the shortfall in taxes on pension fund yields due to the evolution of equity prices. It should be noted that the increase in the tax rate on pension fund earnings from shares from 5 to 15 per cent (see Chapter II) has significantly increased the sensitivity of public finances to fluctuations in equity prices. Public revenues generated may vary between 0 and 2 per cent of GDP, according to Ministry of Finance assessments. These fluctuations will be difficult to predict, although they do not, in themselves, warrant any change in fiscal policy strategy.

Table 2. **General government revenues and expenditures**

DKr billion

	2000 Final out-turn	2001			2002 As at August 2001
		As at December 2000	May 2001	August 2001	
Total expenditure	690.2	716.6	718.1	715.2	736.6
of which:					
Interest payments	55.2	52.6	52.5	51.2	49.3
Total revenues	726.9	754.4	747.2	745.6	772.313
Net surplus	36.8	37.7	29.1	30.4	35.7
Surplus as per cent of GDP	2.8	2.8	2.1	2.2	2.5
Projected GDP growth rate	3.0	1.8	1.5	1.4	2.0

Source: Ministry of Finance.

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A *similar stance seems likely for* 2002

The new government will present its draft budget for 2002 to Parliament at the end of January, and a law has been passed allowing the Danish administration to continue operating without a budget in place in the interim. The new administration has stated that it will continue to pursue the same medium-term objectives as the previous government had laid out in D*enmark* 2010 (Regeringen, 2001) reflecting a shared concern to prepare Denmark for the challenges of ageing population. It has also signalled that it would adopt a fiscal stance in line with that presented to parliament in the draft budget bill for 2002, which lapsed when the election was called. However, it is not yet clear exactly what this means, although a prudent approach would imply maintaining a neutral fiscal stance, allowing automatic stabilisers to work. Concrete expenditure plans and revenue projections are not yet available, although the new government has indicated some of its priorities in its "working programme" for the coming term of office (see Box 2).

L*ocal governments expand public consumption*

The June agreements with the local authorities on priorities for 2002 were rather generous, implying an increase in real operating expenditure of around 1¼ per cent in the municipalities and 2 per cent in the counties. Thus, overall real public consumption growth will be significantly above 1 per cent – the previous government's target – unless there are offsetting cutbacks at the central level. For the municipalities the agreed spending level was explicitly higher than what had earlier been prescribed in the four-year agreement covering the period between 1999 and 2002 (see Chapter III). In particular, significant increases were included for health and education services, and the municipalities' incentive to subsidise social housing was strengthened considerably. Following the June agreements, individual local governments adopted their budgets in mid-October. Unlike some recent years, local income tax rates were kept virtually unchanged, thereby complying with the agreements. Nonetheless, it would be premature to conclude that the budget co-operation system has become more robust. F*irst*, the 2002 budgets were determined shortly before the local government elections in November 2001, and there is strong empirical evidence of an election cycle in the taxing behaviour of municipalities and counties. S*econd*, given the generosity of the agreements and a subsequent windfall gain of some DKr 2 billion in extra revenues due to previously underestimated tax bases in 2000, it would have been reasonable to expect reductions in tax rates on average.

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Box 2. Measures announced by the new government

The new government's "working programme" for its first 100 days in office includes a range of specific measures with implications for fiscal policy. Offsetting measures to maintain the overall fiscal stance signalled will be set out in the budget at the end of January. Already announced measures are the following:

- A "tax-freeze", which means that all tax rates and some tax bases will be frozen in nominal levels. Of immediate impact, the DKr 1½ billion tax increases on businesses that were included in the draft budget bill presented by the previous government will not go ahead.
- An additional DKr 1½ billion would be spent on health care immediately and from July 2002, there would be a free choice of hospitals for treatment after two months waiting for treatment.
- DKr ½ billion would be added to home help in the elderly care sector.
- For children born from 1 January 2002, maternity leave would be extended from 28 weeks to 52 weeks. It could be taken any time up to the child's ninth birthday and could be used in discrete chunks or as a reduction in normal working hours. It would be paid at the full unemployment benefit level. (Parental leave would be abolished.)
- The 1 per cent special pension contribution will be redesigned. At present, all employees contribute with 1 per cent of their gross wages, while the benefits are paid out as a lump sum. The government wants to establish an actuarially fair system, where individuals' benefits match their contributions. If enacted, this will change the scheme from being a tax to being a private pension (and will therefore technically increase household disposable income). This may also induce a small stimulus to activity to the extent that higher income people are more likely to reduce other pension contributions and increase consumption than lower income people are to increase other savings to compensate for lower future benefits from the scheme.
- A "business package" of DKr ½ billion intended to strengthen business competitiveness.
- Tax credits for employer-paid health services, home computers for staff, and internet connections.
- Foreign aid would be cut back by DKr 1½ billion to 1.0 per cent of GDP.

Legislation enacting these measures has not yet been presented to parliament and it is too early to predict the overall "fiscal effect" of such measures. However, it should be noted that even where different elements might have completely offsetting contributions to the budget surplus, they could have quite distinct impacts on the domestic activity. As a case in point, the additional spending on health care would directly stimulate the economy whereas the same size cut in foreign aid will have limited impact in Denmark.

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The short-term outlook

International recovery will boost Danish prospects

The short-term outlook for Denmark to a large extent reflects the profile of the international recovery: OECD-wide growth is projected to remain weak through until mid-2002, but then pick up strongly (Table 3). The external improvement is expected to feed directly into Danish exports, which are expected to maintain market share, on the assumption of an unchanged nominal effective exchange rate since the beginning of November 2001, and moderation in wage growth. Danish private domestic demand components are predicted to be somewhat damped by the increased uncertainty about economic prospects, but continuing real household income growth of around 1½ per cent in both 2002 and 2003 should underpin private consumption growth of around 1 per cent. Both private consumption and investment will be boosted by lower interest rates in 2002, although monetary conditions then are expected to tighten somewhat as the European Central Bank is assumed to start reversing the recently provided stimulus towards the end of 2002. Overall, the Danish economy is projected to expand by 1¼ per cent in 2002 and 2¼ per cent in 2003 (Table 4).

Table 3. **International economic outlook**
Percentage change

	2000:2	2001:1	2001:2	2002:1	2002:2	2003:1	2003:2
	(Seasonally adjusted annual average)						
Real GDP							
United States	2.5	1.2	–0.6	–0.1	3.8	3.8	3.8
Japan	–0.6	0.1	–2.3	–0.6	0.2	0.9	1.1
European Union	2.4	1.9	0.7	1.4	2.5	3.0	3.2
Germany	1.4	0.9	–0.4	0.8	2.4	3.0	3.1
OECD	2.1	1.1	–0.3	0.7	2.9	3.2	3.3
Imports of goods and services							
United States	10.2	–4.8	–11.1	2.4	7.2	7.4	7.7
Japan	12.1	–1.9	–14.0	–5.8	–1.5	4.8	6.0
European Union	11.0	0.4	0.9	2.6	6.0	7.4	7.5
Germany	14.3	–1.9	2.2	3.1	5.8	6.8	7.5
OECD	9.8	–2.7	–6.2	0.5	5.9	7.2	7.6

Source: OECD (2001*a*), *Economic Outlook* No. 70.

Public employment growth will keep labour market tight

With the downturn lasting longer than initially expected, firms may start to lay off workers or at least stop new hires, and employment in the private sector is projected to be ¼ percentage point lower in 2002 before those jobs are regained

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Table 4. **Economic projections**

	2000 Current prices DKr billion	2001	2002	2003
		Percentage changes volume (1995 prices)		
Private consumption	613.3	0.9	0.9	1.2
Government consumption	325.8	1.7	1.1	1.3
Gross fixed capital formation	283.0	–3.4	1.4	3.6
Final domestic demand	1 222.2	0.1	1.1	1.8
Stockbuilding[1]	–1.8	0.1	0.0	0.1
Total domestic demand	1 220.2	0.2	1.1	1.8
Export of goods and services	567.4	4.2	3.3	6.7
Import of goods and services	491.4	1.8	2.9	6.1
Net exports[1]	76.0	1.1	0.3	0.6
GDP at market prices	1 296.1	1.3	1.3	2.3
GDP deflator	–	3.2	1.7	1.8
		Percentage changes		
Memorandum items:				
Private consumption deflator	3.0	2.1	1.7	1.8
Employment	0.7	0.2	0.0	0.2
Unemployment rate	4.7	4.7	5.0	5.0
Household savings ratio[2]	2.9	3.4	3.8	4.1
General government financial balance[3]	2.8	2.0	1.4	1.7
Current account balance[3]	1.6	3.2	3.1	3.3
Potential output growth	2.4	2.3	2.2	2.2
Output gap	0.6	–0.4	–1.2	–1.1
Interest rates short-term	5.0	4.5	3.3	4.1
Interest rates long-term	5.6	5.2	5.0	5.3
Effective exchange rate	–3.3	1.8	0.7	0.0
OECD import oil price (cif US$ per barrel)	28.0	24.5	21.5	25.0

1. Contributions to changes in real GDP (as percentage of real GDP in previous year), actual amount in first column.
2. As a percentage of disposable income.
3. As a percentage of GDP.

Source: OECD (2001*a*), *Economic Outlook* No. 70.

in 2003. But with employment in the general government sector projected to expand by more than ½ per cent in 2002 and almost as much in 2003, total employment would remain constant in 2002 despite the weak first half-year. This will keep the standardised unemployment rate from rising more than ¼ percentage point in 2002, which nevertheless brings it back to the NAIRU, easing labour market tightness to some extent, and moderating wage pressures. Hourly earnings are projected to slow to growth rates of around 3½ per cent over the outlook period.

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International risks skewed downwards

The current international situation is particularly uncertain and the risks are unusually large. The risks surrounding the central scenario presented in the *Economic Outlook* are on the downside and could involve such elements as: a deeper slump in OECD-wide consumption and investment; stalling non-OECD import demand; an upward shock in oil prices; or a realignment of exchange rates. Such shocks might be mitigated by further easing of monetary policy across the OECD area and some fiscal stimulus, but the net result for Denmark would be weaker export prospects.

Domestic risks are minor but biased upwards

Domestic risks for the Danish economy are currently small but may be biased upwards towards stronger growth and labour market pressures. With comfortable budget and current account surpluses, consistent policy settings and output close to its potential before the current downturn, there seems little reason (apart from external developments) why the economy should not recover easily from the present slow-down. The main downside risk would be more aggressive cut-backs in the private sector employment than embodied in these projections; as layoffs are easy in Denmark, jobs may turn out to be more sensitive to business sentiment or cost-cutting considerations than expected. This would spill over into lower private consumption and might conceivably exercise some downward pressure on house prices. However, the risk of a large fall in house prices seems rather remote and therefore only a small group of very recent house purchasers would be exposed to the risk of finding themselves with negative equity. The main upside risk is that as the recovery gathers pace, labour shortages will re-emerge causing wage pressures to start to mount. The continuing strong employment growth from the public sector built into the projections raises the demand for labour despite current weak business activity and adds urgency to efforts to expand the labour supply (see Chapter II). In the short term, plans to expand certain public services might need to be countered by cuts elsewhere in the public sector to avoid crowding out business expansion.

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II. Renewing the momentum for structural reforms

This chapter examines recent progress in implementing structural reforms, reviews obstacles and identifies areas where continuing efforts are needed to broaden and deepen the structural reform agenda. At the time of the previous S*urvey* in mid-2000, reforms of tax policy, the early retirement system, active labour-market policies (ALMPs) and competition and energy policies were being enacted. These have mostly been pursued as planned and have paid off in terms of further progress on a number of counts. In particular, unemployment has abated to its lowest rate in almost three decades, while the low-inflation environment remains well protected. Moreover, robust surpluses have been established on both the current account and the general government budget, the latter implying that Denmark – though already having high labour force participation rates – probably has a better starting point than many other countries for managing the inter-generational transfers implied by an ageing population.

The momentum for undertaking new reforms faded in recent years, in part because of the electoral timetable. This may be because most outstanding structural policy issues are not urgent; they do not risk prompting severe imbalances if not addressed in the near term. However, the current favourable economic situation should not be used as a pretext for postponing further structural reforms; rather, it should be seen as an opportunity for the new government to move ahead with reforms that could foster better resource allocation and strengthen productivity growth and potential output. Considerable scope for action remains in a number of areas, some of which have been analysed in depth in previous S*urveys*. The education system requires further quality improvements and greater economic efficiency; rent controls continue to prevent efficient allocation in the housing sector, and greater flexibility could strengthen resource allocation; and more vigorous competition should be encouraged in many industries, including the energy sector, construction, transport, retail trade and financial services.

This chapter focuses on a subset of structural policy issues. It starts out by discussing medium-term growth prospects. With unemployment unlikely to decline much further and demographic developments no longer leading to higher

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output, attention needs to be focused on productivity growth and increasing (already high) labour-force participation. The analysis moves on to review distributional policies; despite a remarkably equal distribution of income and consumption, equity concerns have been a major political barrier to efficiency-enhancing reforms across the spectrum of structural policy issues. Next, it revisits some of the main issues in Danish tax policies in the light of the completion in 2002 of phasing-in the current tax reform. The subsequent section provides an overview of policies to improve the employment opportunities for people at the margin of the labour market (including immigrants and people with reduced work ability) and a discussion of the recently adopted reform of the early retirement (disability) pension system. The chapter closes with a review of some competition and environment issues affecting the energy sector.

Medium-term growth prospects

Annual output growth averaged 2 per cent between 1980 and 2000, compared with 2¼ per cent in the euro area and 3 per cent in the United States. Having expanded vigorously at an average annual rate of 3¼ per cent from 1994 to 1998, output has slowed to an average growth rate of 2¼ per cent in the 1999 to 2001 period. Looking ahead, potential growth will depend on the evolution of trend productivity and the scope for increasing total hours worked. The latter is likely to be limited in the future, in contrast to the average annual rate of expansion of almost 1 per cent that has been experienced over the past six years. This is due to at least two factors:

- *First*, the large labour reserve in unemployment at the outset of the upswing in mid-1993 has now been absorbed. A substantial part of remaining unemployment is either frictional or recipients of non-core benefits (such as supplementary benefits for part-time workers and so-called vacation benefits[8]) and do not represent potential additions to the workforce. Some further small-scale reductions in unemployment might be possible, if such aspects are addressed or if generous benefits were reduced. However, this would require further structural reforms, and in any case the contributions to potential employment growth will be significantly lower than in the recent past.

- *Second*, demographic developments have supported higher labour supply considerably in the past but will cease doing so in the future. The contribution to the workforce from this source (*i.e.* the combined effect of changes in the size and the age composition of the working-age population) stagnated in the 1999 to 2001 period. This is in sharp contrast to the average annual contribution of 0.4 per cent between 1994 and 1998, and the demographic contribution is set to become slightly negative over the medium term and beyond. In the next five to ten years this is

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Table 5. **A mechanical decomposition of growth, 1980-2010**

	Average growth rates and contributions over the period				
	1980-2001	1994-1998	1999-2001	2002-2005[1]	2006-2010[1]
GDP at factor costs	1.9	3.1	2.6	1.9	1.9
of which:					
Hourly productivity	2.0	3.0	1.7	2.0	2.0
Employment (hours worked)	–0.1	0.1	0.8	–0.1	–0.1
of which:					
Unemployment[2]	0.0	1.2	0.5	–	–
Average annual working hours	–0.3	–0.7	0.1	–	–
Work force	0.2	–0.4	0.2	0.0	–0.2
Demographics	0.5	0.4	0.0	0.0	–0.2
Participation rates	–0.2	–0.5	0.2	–	–

1. The figures should not be interpreted as a projection of growth in potential output. They are compiled under the technical assumptions that hourly productivity growth remains unchanged at the average rate between 1980 and 2001; the population growth and composition evolves as suggested by Statistics Denmark's population projection; and unemployment, working hours and age-specific participation rates remain stable at the 2001 levels.
2. A positive sign indicates that unemployment has declined and contributed to higher growth.

Source: Statistics Denmark; OECD.

due to compositional shifts within the working-age population (the number of persons aged 15 to 66 years will continue to increase over this time horizon), with an increasing share composed of age groups with below-average participation rates.

Against this background, a mechanical decomposition that adjusts for changes in the demographic development but technically assumes that other determinants remain constant at present levels or past growth rates suggests that potential growth may decline to less than 2 per cent on average up to 2010 (Table 5). If so, this would be comparable to the average actual growth rate since 1980, but significantly slower than the rate of expansion since 1994. Although further reforms may facilitate a slight decrease in structural unemployment, this could easily be offset by a continuation of the trend decline in average working hours. The prospects of bringing about a higher rate of growth will depend mainly on the scope for increasing labour supply by raising age-specific labour force participation rates and lifting the trend productivity growth rate.

Increasing labour supply

The new government has stated that it will subscribe to the objectives of the previous government's medium-term strategy laid out in the *Denmark* 2010 programme (Regeringen, 2001*a*). A central target in this programme is to increase labour supply by 2¾ per cent by 2010. This implies that higher age-specific participation rates need to provide an increase in labour supply of some 4 per cent in order to

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offset the expected negative effects of demographics. If achieved, this would add up to ½ percentage point to the annual potential growth rate. Furthermore, it would imply an increase in overall labour-force participation to around 80 per cent of the working-age population, a level that has been achieved by very few OECD Member countries and then only temporarily. This underlines the ambitious character of the goal.

Since virtually all non-working adults receive some kind of public benefit, any increase in the participation rate must come from reducing the pool of transfer recipients by roughly the same amount (Table 6). The previous *Survey* questioned whether this was feasible without additional policy measures to reduce the generosity of the transfer system through stricter eligibility and/or lower benefit levels.[9] However, the authorities more recently concluded that the target has come within closer reach, an assessment underpinned by the recent reduced intake to early retirement schemes (Finansministeriet, 2001*a*). If sustained, the lower inflow rates are assumed to bring about an accumulated decline in the share of transfer recipients in the working-age population of 2½ percentage points by 2010, and an increase in the workforce of similar size. In that case the contribution that would need to materialise endogenously or through future policy measures to achieve the aforementioned increase of 4 percentage points would amount to 1½ per cent of the working-age population.

However, the risks still seem to be rather clearly on the side of a smaller total decline in transfer recipients than envisaged by the authorities. For example, it is optimistic to assume only a modest rebound in the inflow to the early retirement (disability) pension scheme from the current extraordinary low level (see below), when a decline is also assumed in recipients of benefits that are close short-term substitutes to early retirement pension.[10] Nonetheless, some decline in the number of transfer recipients is indeed likely, due to the ongoing phase-out of the early transitional withdrawal scheme and the reform of the early retirement benefit scheme from 1999, which up to now seems to have brought about the hoped-for moderately lower inflows. But a recent assessment by the Economic Council suggested a more modest increase in labour supply up to 2010 (less than 1 percentage point) than that of the government (Det økonomiske Råd, 2001*a*). To achieve the stated objectives in full, further policy decisions may need to be taken.[11] This could include discontinuing publicly paid leave schemes for the employed, further reducing early retirement subsidies and measures encouraging a faster pass-through in the education system, such as abolishing the voluntary tenth grade between compulsory schooling and upper-secondary education or charging tuition fees in tertiary education. Developments since the previous *Survey* have not been helpful on this account. On the contrary, broad parliamentary support has emerged for extending the maternity or parental leave scheme.[12] While in itself this would have only a relatively modest effect in reducing labour supply, it is nevertheless a step away from the stated objectives, and it does raise doubts about the commitment to the strategy.

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Table 6. **Transfer recipients aged 15-66 years**

Thousand persons

	1970	1980	1990	1995	2000	2001	2010[1]	Change 1990-2001	Change 1995-2001	Change 2000-2010
I. Transfers	307	635	798	972	873	874	747	76	–98	–126
Early retirement pension (disability)	205	236	251	271	264	261	224	10	–10	–40
Early retirement benefit[2]	0	54	98	138	181	184	134	86	46	–47
Unemployment	30	184	272	288	150	149	134	–123	–139	–16
Cash benefit[3]	34	79	52	65	69	72	67	20	7	–2
Educational leave[4]	0	0	0	44	41	36	35	36	–8	–6
Maternity and parental leave	9	16	32	78	54	54	55	22	–24	1
Sickness benefit	29	54	42	43	56	60	41	18	17	–15
Others	0	12	51	45	58	58	57	7	13	–1
II. Public-sector employment	405	693	771	770	813	819	863	48	49	50
III. Paid by the public, total	712	1 328	1 569	1 742	1 686	1 693	1 610	124	–49	–76
IV. Total employment	2 463	2 534	2 606	2 539	2 686	2 696	2 787	90	157	101
V. Work force	2 493	2 718	2 877	2 827	2 837	2 845	2 922	–32	18	85
VI. Working-age population[5]	3 247	3 417	3 561	3 617	3 652	3 656	3 747	–5	39	95
Dependency ratios										
A: I/VI	9.5	18.6	22.4	26.9	23.9	23.9	19.9	1.5	–3.0	–4.0
B: III/V	28.6	48.9	54.5	61.6	59.4	59.5	55.1	5.0	–2.1	–4.3
C: Non-employment ratio	24.1	25.8	26.8	29.8	26.5	26.3	26.2	–0.6	–3.5	–0.3
Tax-to-GDP ratio	43.2	44.1	47.1	49.3	48.2	47.8	46.1	0.7	–1.5	–2.1

1. Projected in the (previous) government's medium-term programme, *Denmark* 2010. For early retirement (disability) pension, the projection does not incorporate the recently adopted reform, which is discussed later in the chapter.
2. Includes recipients of early withdrawal benefit. Entrance into this scheme, which applied to persons aged 50 to 59, was abolished in 1996.
3. Cash benefit recipients (social assistance) registered as unemployed are included under "unemployment".
4. Includes all income-replacing transfers (except ordinary student grants) to adults engaged in educational programmes and not in work.
5. Persons aged 15 to 66 years. Despite the increase in the working-age population up to 2010, the contributions to labour supply will be negative, as a greater proportion will be in age groups with below-average participation rates.

Source: Statistics Denmark; Regeringen (2001); OECD.

Trends in productivity growth

Denmark has enjoyed higher growth rates in labour productivity (output per employed person) in the business sector during most of the 1990s than in both the United States and the euro area (Figure 11, Panel A). This has been a reversal of the pattern observed in the mid-1980s, when Danish productivity gains were particularly low. The higher average growth rate in the 1990s is strongly influenced by a remarkably strong increase in 1994 as Denmark came out of recession, when business-sector output per employed person rose by 8 per cent. In the most recent years, productivity gains have decelerated to around 2 per cent per year, which is close to the rates observed in the United States but still significantly higher than the EU-wide growth rate of around 1 per cent.

Hourly productivity growth (in manufacturing, construction and private services) has been somewhat more stable over the past two decades, averaging 2.1 per cent per year (Figure 11, Panel B). Labour productivity has not been correlated with capital deepening: consequently the contributions from multi-factor productivity and capital intensity have largely moved in opposite directions. A sectoral breakdown reveals considerable differences in productivity trends, particularly between manufacturing and construction. While productivity performance in manufacturing was dismal during most of the 1980s, outcomes have been much better in the 1990s (Figure 11, Panel C). The opposite pattern is observable in construction, where output per employed person has actually shrunk in most years of the past decade.

It is difficult to assess future prospects for aggregate trend productivity growth on the basis of recent trends. On the one hand, aggregate hourly productivity growth in the business sector has been fairly stable at around 2 per cent with no evidence of an acceleration in the second half of the 1990s. On the other hand, this masks significant and offsetting fluctuations within each of the main sectors. It is quite uncertain whether these fluctuations will continue, and, if they do, whether they will reinforce or offset each other. In the mechanical growth decomposition in Table 5 above, economy-wide hourly labour productivity growth has been assumed to remain at 2 per cent annually, which has been the average rate of increase over the past two decades. However, it could easily either accelerate (for example, due to greater diffusion of information and communication technology) or decelerate (for example, because of continued compositional shifts from manufacturing to services production).

The OECD's growth project has identified a number of factors that are likely to support output through productivity gains; a competitive business climate; well-functioning financial markets and intermediation; high research and development activity; good regulatory frameworks; and high (effective) human capital investment. In Denmark in the past, it has often proved difficult to co-ordinate policies in

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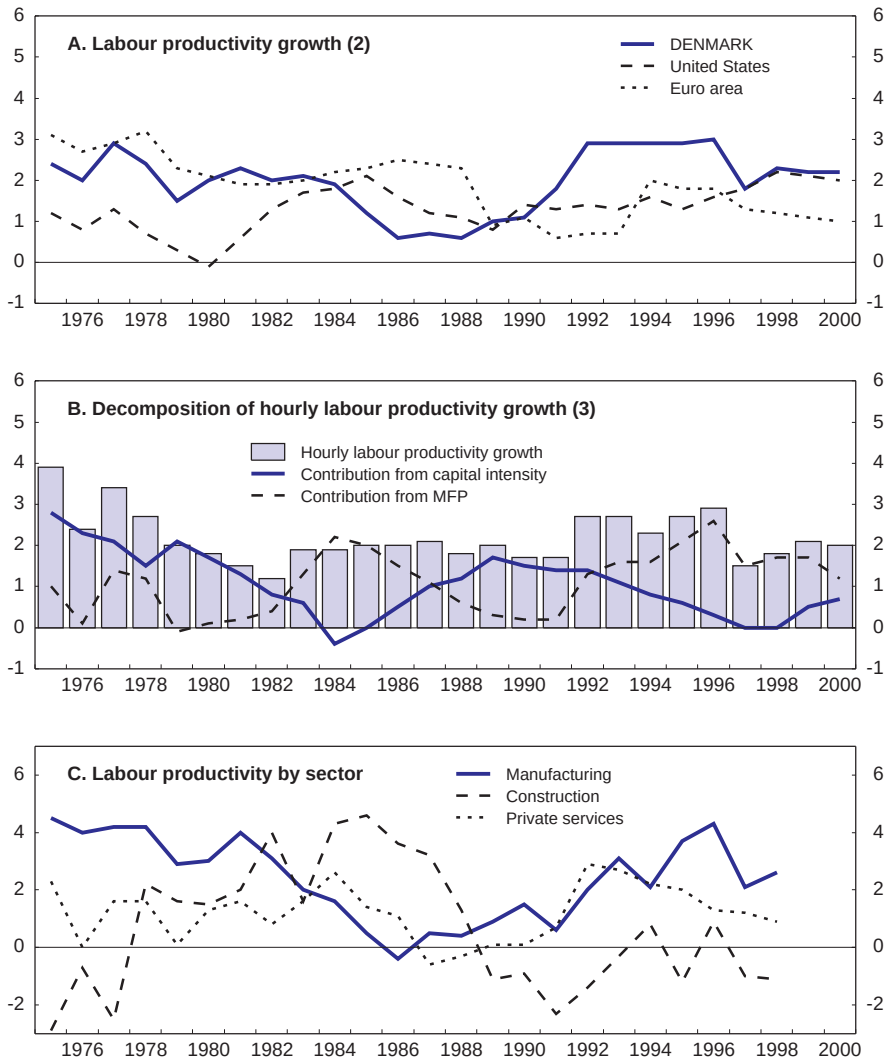
Figure 11. **Productivity trends**

Per cent[1]



1. All series are five-year centred moving averages. In Panels A and B data for 1999 and 2000 rely partly on OECD projections.
2. Output per employed person in the business sector.
3. Includes manufacturing, construction and private services.

Source: Statistics Denmark; OECD.

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different areas (taxation, education, energy, etc.) and to build support for giving precedence to growth over sectoral non-growth related objectives. For example, the previous government's business strategy laid out in the *.dk*21 programme emphasised several of the above factors and set ambitious targets for Denmark's performance, but it did not play a central role in shaping structural policies. The new government's reorganisation of Ministeries should facilitate the co-ordination of policies in the future.

In the Danish context, the scope for further action to promote higher productivity growth appears to be particularly big in three areas.

– *First*, the competitive climate could be enhanced. The legal framework should be strengthened by increasing financial penalties for violation of the competition rules. More vigorous competition is also called for in a number of specific sectors including some utilities sectors (energy and transports) and, for example, construction, building materials, food processing and retail trade. The Competition Authority's list of industries with signs of competition problems has not shortened since the previous S*urvey*, and the price level (excluding taxes) in Denmark remains some five per cent higher than in comparable European countries according to the authorities' assessment (Konkurrencestyrelsen, 2001).

– S*econd*, raising research and development (R&D) investment, currently close to the average EU level, could contribute to higher total factor productivity growth.[13] Bringing about such an increase is difficult to engineer and policy measures should focus primarily on ensuring good framework conditions. Promoting vigorous competition and high-quality education, important in their own right, are likely to have positive spillover effects on R&D. Ongoing efforts to improve the interplay between business activities and public-sector R&D should be further strengthened. Positive externalities may provide an argument for tax credits for business R&D activities[14] (Bassanini *et al.*, 2001; Det økonomiske Råd, 1997), but such measures would need to be weighed carefully against deadweight loss, reclassification of expenditures, administrative costs and small-country leakages. At present, the tax treatment of R&D in Denmark is considerably less favourable than in, for example, the United States, the Netherlands and France, although not much different from that in most other European countries (OECD, 2001*b*).

– T*hird*, considerable scope remains for improving outcomes of tertiary education by improving quality within existing resources. Barriers seem to exist that discourage student from embarking on natural science education; these should be identified and addressed.

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Distribution policies

Denmark has been highly successful in ensuring an equitable distribution of living standards, as indicated by traditional measures of inequality, such as absolute and relative poverty levels or the dispersion of disposable incomes. Moreover, starting from a position of already low inequality by international standards, Denmark was one of only a handful of countries in which the dispersion of disposable incomes did not increase from the mid-1980s to the mid-1990s (Burniaux *et al.*, 1998; Förster and Pellizzari, 2000). Consequently, disposable incomes were more evenly distributed in the mid-1990s than in any other country for which information is available (Figure 12).[15] Danish studies suggest that the income distribution has remained broadly stable even in the second half of the 1990s, with the Gini coefficient hovering around 21 per cent.

The high equality is partly due to the compressed wage structure and the comparatively high progressivity of the tax system (see below). In addition, the transfer system is more progressive, *i.e.* redistributive than in other countries. Heady *et al.* (2001) finds that the Danish transfer system is the most effective among the EU countries in reducing inequality. In part, this is due to the fact that the absolute ceilings on most benefits are fairly low relative to the average production worker's (APW) wage level. For example, the maximum after-tax unemployment benefit represents 66 per cent of the after-tax APW wage level in Denmark, which compares with 171 per cent in Germany, 100 per cent in the Netherlands, and 70 per cent in Sweden (Table 7). The relatively low cap implies a greater resemblance with a flat-rate system in contrast to the predominantly earnings-related transfer systems of many other European countries. In the Danish quasi-flat-rate system, "shadow" replacement rates are higher than in most other countries for low-income workers but lower for high-income workers. This is likely to increase perceived and measured redistribution. Part of the background for these differences is that social transfers are largely tax-financed in Denmark, whereas in other countries they are typically financed *via* social-security contributions, which generally link benefits closer to individuals' preceding labour-market status. This difference is most evident in the case of public old-age pensions, which in Denmark are unrelated to previous employment. However, private (actuarially fair) pension schemes play a more important role in supplementing public pensions than in most other countries.

The redistributive effects of public finances go beyond the impact of the tax and transfer systems; redistribution also takes place through public consumption (Table 8). Indeed, redistribution is caused by the different incidence of tax and the value of all public expenditure, respectively. The value of public services consumed by low-income households is double that of high-income households, a result that applies whether looking at all adults or only those aged 25 to 59 years.[16] While the average market income of the top 20 per cent was almost 10 times that of the bottom 20 per cent in 1997, the ratio of disposable income between the two

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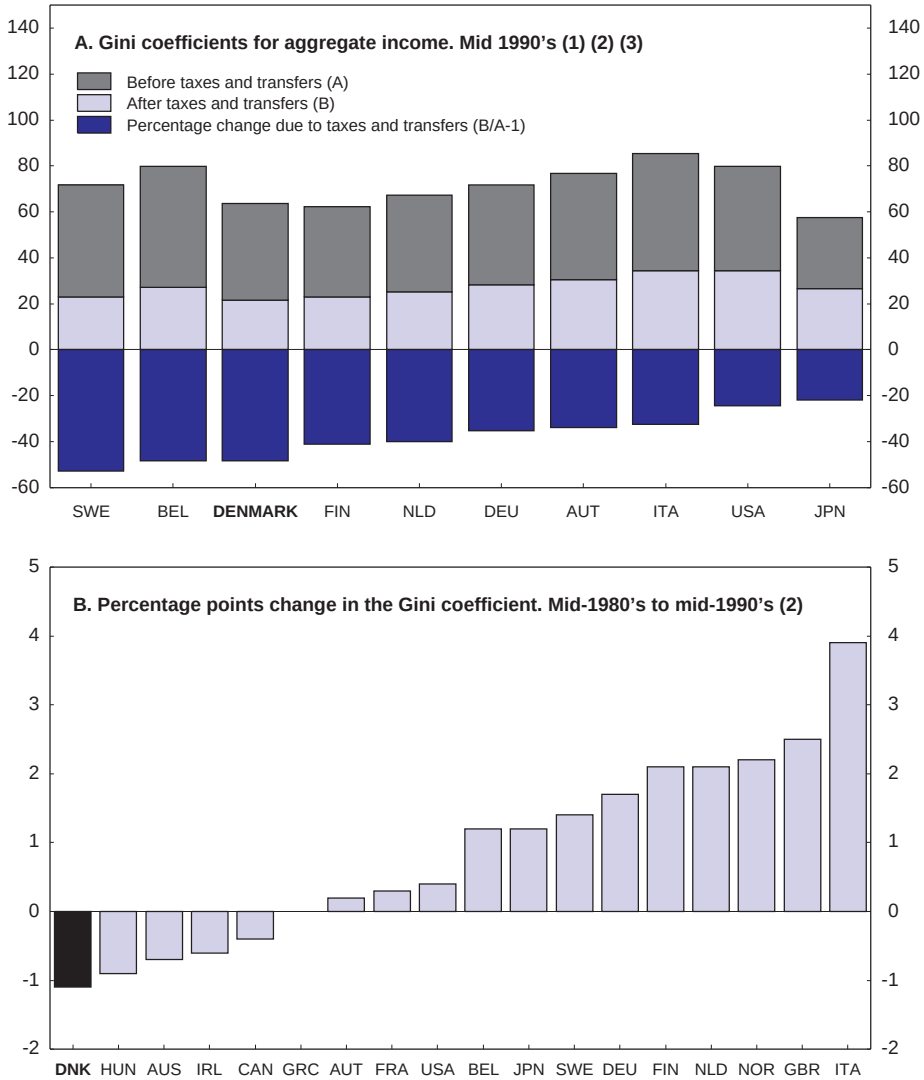
Figure 12. **Income distribution in perspective**



1. Data refer to 1993, 1994 or 1995 for all countries.
2. Household income, using an equivalence scale elasticity of 0.5.
3. Countries are ranked in descending order according to the scope of redistribution resulting from the tax and transfers systems.

Source: Burniaux *et al.* (1998); Förster and Pellizzari (2000).

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Table 7. **Unemployment benefit compensation in selected countries**
1997, per cent

	Statutory rate	Maximum benefit relative to the APW wage level[1]	Net replacement rates[2]			
			Singles		Couple, 2 children	
			67 per cent APW	APW	67 per cent APW	APW
Austria	57	75	57	57	79	73
Belgium	60	–	84	64	75	60
Denmark	**90**	**66**	**89**	**62**	**95**	**77**
Finland	90	62	72	60	94	84
France	75	–	83	71	86	74
Germany	60	171	69	60	74	74
Italy	80	56	36	36	52	54
Netherlands	70	100	92	75	90	85
Norway	62	70	65	66	74	73
Portugal	65	–	87	79	86	77
Spain	70	–	70	76	78	74
Sweden	80	70	77	72	90	84
United Kingdom	Flat	19	73	50	83	64
United States	50	68	59	60	51	61
Country mean[3]	70	76	72	63	79	72

1. Maximum unemployment benefit as a percentage of the wage of an average production worker (APW). Net of taxes.
2. Includes individual housing benefits, which are important particularly for the United Kingdom.
3. Simple average of the countries in the table.

Source: OECD.

groups was only 2.9 to 1 and total public and private consumption was only 60 per cent higher on average in the high-income group.[17] The average net contribution to public finances (*i.e.* total taxes paid less transfers and the value of public consumption received) of persons in the high-income group was DKr 120 000 in 1997, while it was DKr -80 000 for those in the bottom two income deciles.

The full distribution of disposable income across deciles reveals that total consumption is virtually constant between the second and the ninth deciles (Figure 13). Across these groups, the value of the extra public consumption received by persons in the lower income deciles is largely equivalent to their lower private consumption possibilities. The remaining differences in the value of total consumption between low- and high-income groups are therefore fully attributable to the first and tenth deciles. This might in part reflect high and involuntary medical care consumption by the elderly, who are typically found in the lower deciles. However, looking only at adults aged 25 to 59 years, total consumption even declines slightly between the second and the ninth deciles. Even though the character of public and private consumption is quite different, these results indicate that the scope of total short-term redistribution is very extensive indeed.

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Table 8. **Income and consumption for low- and high-income households**[1]
1997

	Average amounts		High/Low	Share of total	
	Low	High		Low	High
	Thousand DKr			Per cent	
All adults					
Market income	**31**	**297**	**9.6**	**4**	**41**
Transfers	53	14		25	7
Income taxes	–21	–129		6	39
Disposable income	**63**	**182**	**2.9**	**10**	**30**
Indirect taxes	–32	–53		15	24
Public consumption[2]	80	48		24	14
Total consumption	**111**	**177**	**1.6**	**15**	**24**
Net contribution	–80	120			
Adults aged 25 to 59 years					
Market income	65	320	**4.9**	8	44
Transfers	63	9		30	5
Income taxes	–37	–139		11	42
Disposable income	91	190		14	31
Indirect taxes	–41	–54	**2.1**	19	24
Public consumption[2]	95	46		29	13
Total consumption	145	182	**1.3**	20	25
Net contribution	–80	138			

1. The "low" and "high" income groups comprise the bottom and top two disposable income deciles, respectively.
2. The method for assigning public consumption to each income decile is described in Finansministeriet (2000*c*).
Source: Ministry of Finance.

Only some of this redistribution is "churning" in the sense that part of an individual's payment of taxes contributes to financing transfers and services he or she expects to receive at some later point in life. A significant amount of redistribution takes place from the working-age group, particularly those aged 40 to 59 years, to both younger and older groups. However, redistribution between individuals over the course of life is considerable as well. It is difficult to establish the exact relative significance of the two components, in the absence of individual-level data. A recent study (Økonomiministeriet, 2001*a*) provides information on the scope of redistribution within and between broad educational groups. It shows that two-thirds of overall redistribution is within each group, while one-third occurs between groups. For example, people without upper secondary education are net receivers over the life cycle, with a quarter of the transfers and services they receive being paid by others. In contrast, people with tertiary education are net contributors, paying the equivalent of one-third of the value of transfers and services they receive.[18]

To a large extent, the above results reflect the fact that equity objectives play an important role in the policy-making process. The Ministry of Finance and the

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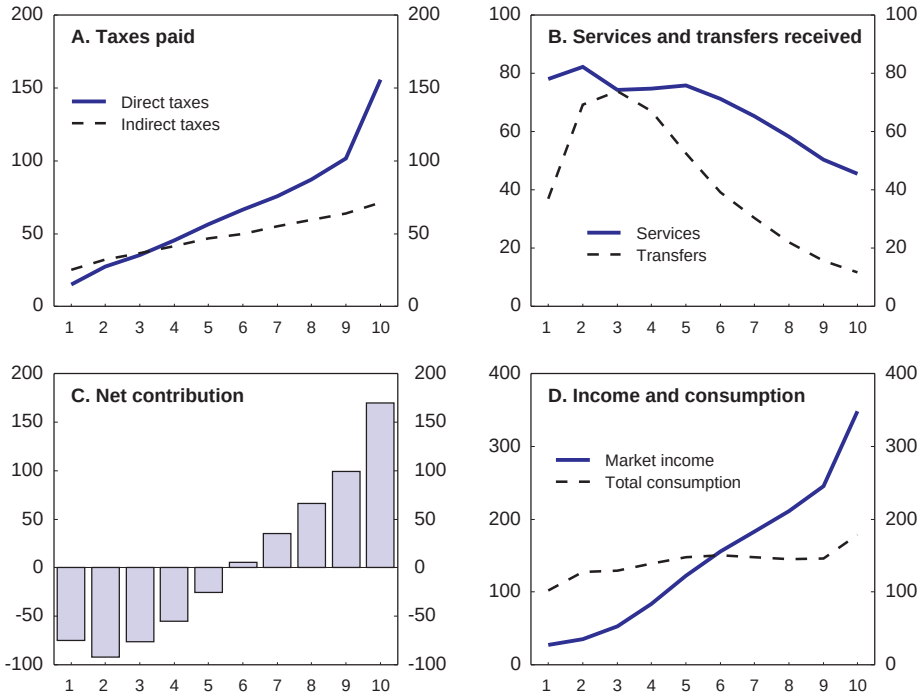
Figure 13. **Distribution of income and total consumption, 1997**

Thousand DKr[1]



1. The population has been ranked according to deciles of disposable income, before payments of indirect taxes and without the value of public services. Income is measured at the family level and adjusted to make it possible to compare families of different size and composition.

Source: Ministry of Finance.

Ministry of Economic Affairs provide regular reviews of the evolution in the distribution of income and consumption (see, for example, Finansministeriet, 2001*a* and Økonomiministeriet, 2001*a*). The so-called Law Model[19] (*Lovmodellen*) facilitates very detailed, albeit static, analysis of the distributional effects of changes in any single component of the tax and transfer system. Hence, it provides a unique tool for carefully tailoring reforms to meet policymakers' distributional objectives. This has been exploited intensively, for example in the design of tax reforms since the mid-1980s, and it is probably a strong contributing factor behind the stability of the income distribution in Denmark.

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However, the availability of very detailed information may also have contributed to an undue bias towards the *status quo* in the policy-making process.[20] Distributional objectives are not stated explicitly. Instead, the focus is on static and marginal distributional effects of discretionary policy changes, which appears to distract attention from the "level" and dynamics of aggregate inequality. Income mobility is very high[21] implying that lifetime inequality is much lower than inequality at any given point in time. If the fundamental objective is to limit the dispersion in lifetime incomes, static measures of inequality provide little guidance for policymakers. The fact that immediate winners and losers are easily identified may reinforce political barriers to reforms or prompt exemptions to be included, even if these often compromise the overall logic of the systems and reduce transparency. This "information trap" may be difficult to overcome altogether, but greater awareness about the risks seems warranted. Hitherto, actual policy-making indicates a reluctance to embark on any reform affecting inequality at the margin. One example is the long-standing tradition of rent controls in the housing market. These are widely recognised as significantly distorting the overall allocation of the housing stock, impeding labour mobility and discouraging maintenance activity. Furthermore, as demonstrated in a recent study by the Economic Council (Det økonomiske Råd, 2001*a*), the resulting artificially low rents are not particularly well targeted at those with low incomes, implying that rent controls cannot be justified on distributional grounds. Nonetheless, it is primarily the visibility of the potential losers and the invisibility of the winners that blocks reform.[22]

Balancing equity and efficiency considerations is a continuous and central task for policymakers. The highly equal distribution of income and consumption is a fundamental strength of the Danish economy in furthering social cohesion. However, with absolute and relative poverty being largely absent, there may be a strong case for undertaking reforms that would help the economy function better, even if these are likely to widen the income distribution somewhat in the short run. This is particularly the case if the greater dispersion results from increasing incomes for middle- and high-income wage earners without negative effects on low-income groups. Reforms that could encourage an increase in average working hours (such as an increase in the threshold for the top tax discussed below) provide an example. Initiatives to encourage a faster pass-through in tertiary education would probably also reduce total redistribution through the implied lower public consumption, but it would nevertheless increase total welfare.

Tax policies

The most recent tax reform (the so-called Whitsun package) was decided in 1998 and will be fully phased-in by 2002. Against this backdrop, considerable public debate on the need for continuing adjustments of the tax system has taken place in the period since the previous *Survey*, and several political parties have

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presented proposals for further reform. Following the change of government in November 2001, a so-called "tax freeze" has been announced. This amounts to a commitment from the new government to avoid raising any tax rates or even adjust excise duties and property taxes in step with general price increases. The only exceptions where a little flexibility is accepted are if adjustments flow from EU requirements or if there are compelling environmental arguments. In these cases, any extra revenue is to be fully offset by cuts elsewhere. According to the government's "working programme", the commitment includes local government taxes. Thus, a general increase in locally determined taxes would be offset by cuts in centrally determined taxes. The nominal freeze of excise duties and property taxes imply a small annual erosion of the bases for these taxes; depending on the exact operational definition of the "tax freeze" this will reduce the budget balance by 0.1 to 0.2 percentage point of GDP per year. While the commitment to abstain from tax increases is laudable, the implied tax-base erosion limits the scope for reducing the more distortionary tax rates on labour income.

This section briefly revisits some of the key tax policy issues in the Danish context.[23] In general, the present system is well designed. But a number of features, such as the low threshold for the top income tax and car taxation, cause considerable distortion and warrant attention, although they do not concern the basic elements of the tax system. The particular tax structure, with high taxes on consumption and high-skilled labour might give rise to some pressure on tax bases in the medium and longer term, but tax-base erosion does not appear to be an imminent concern. The main deviations from neutrality relate to capital income taxation, and these should be addressed as early as possible.

Personal income taxation

In contrast with most other European countries, social-security spending in Denmark is not financed through earmarked contributions from employers and employees.[24] Instead taxes on personal income and goods and services account for a larger share of total tax revenue than elsewhere. The main features of the Danish personal income tax system are depicted in Figure 14. In brief, the system comprises three largely proportional components (labour market and special pension-scheme contributions; local government income taxes; and the ordinary state tax) and two components (the intermediate and top state taxes), which together with a fixed deductible amount (on personal income) make for the progressivity of the system. The complexity of the system arises primarily from the treatment of labour market contributions, the prevalence of multiple income concepts – taxable income, personal income and capital income – and the fact that these enter into the definition of the bases for the individual components in varying combinations (see Box 3).

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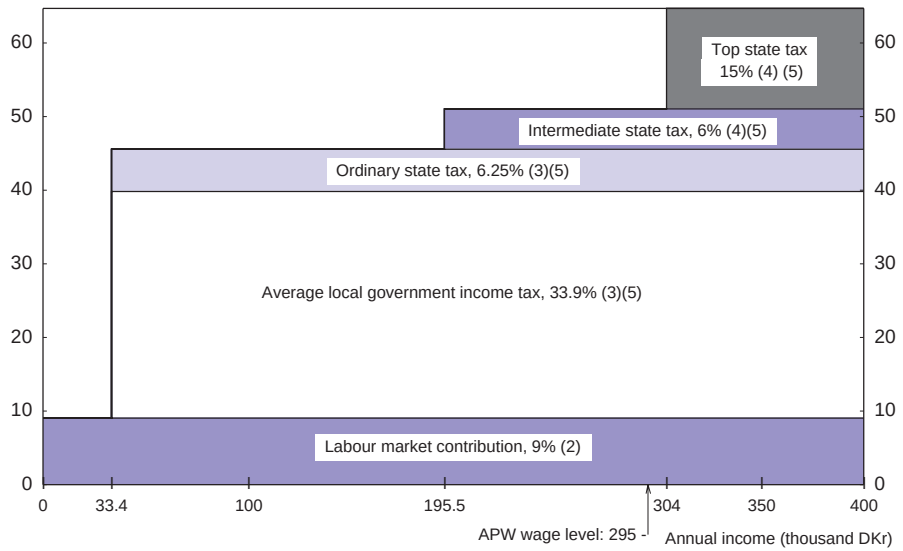
Figure 14. **Main features of the Danish personal income tax system**
2001, per cent[1]



1. For the sake of simplicity some details are omitted, for example those of taxation of self employed. For a more complete description, see Statistics Denmark (2001).
2. Levied on all earned income without any deductions. Includes the special pension scheme savings (1 per cent).
3. The tax base is "taxable income" above DKr 33 400 per year. Interest outlays are not deductible from the state (*i.e.* central government) ordinary tax.
4. The tax base is the total of "personal income" and positive personal capital income. Contributions to capital pension savings are not deductible from the top tax.
5. Labour market contributions (9 per cent) are deductible in all other tax components. To calculate the marginal tax rates, the statutory rates have therefore been multiplied by 0.91.

Source: Statistics Denmark; Ministry of Taxation; OECD.

As discussed in previous *Surveys*, the main changes in the taxation of personal income have been to reduce the generosity of interest deductibility and thereby strengthen private savings; in the Danish context, where households have large net financial liabilities (excluding pension fund assets), this has been accomplished primarily by reducing the tax rate on (negative) personal capital income. The effective broadening of tax bases has paved the way for lowering rates. However, while statutory tax rates have indeed been reduced significantly for the greater part of the income scale, the average marginal tax rate has changed much less because an increasing proportion of taxpayers find themselves in the upper brackets (OECD, 2000*b*). Thus, the reforms have in particular brought about the

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Box 3. **Main income concepts**

A: *Personal income* is defined as the total of wages, salaries and fees; payments in kind; net income from self-employment; and alimonies *less* labour-market contributions (payroll tax); contributions to private pension savings and the special pension scheme.

B: *Capital income* comprises mainly interest received and interest paid; capital yields from the corporation tax scheme for self-employed (in contrast, share income – dividends and capital gains – are taxed according to separate rules). Imputed return on owner-occupied housing was considered capital income until 1999, but was omitted as from 2000 when the imputed rents tax was changed to a property tax. On average, households have negative capital income.

C: *Income deductions* most importantly include contributions to unemployment-insurance funds and trade unions; expenses for transport between home and workplace and alimonies.

D: *Taxable income* is defined as personal income plus capital income minus income deductions.

hoped-for strengthening of private savings, whereas the reduction of tax distortions on work incentives has been partly offset by higher tax rates in local governments. The average marginal tax rate on earned income (for full-time workers) has fallen some 5 percentage points since 1993. Reducing tax distortions on work incentives should be the main objective of any future adjustments to the system of personal income taxation, and it should include measures aimed at bringing the trend increase in local government (average) income tax rates to a halt (see Chapter III).

It might be argued that Denmark has a better starting point for cutting taxes, given its more comfortable overall budgetary position than other EU countries", including Germany and France where tax cuts are underway. The cyclically-adjusted budget surplus of around 2 per cent of GDP (see Chapter I) appears quite robust, while the EU average in 2001 was a structural budget deficit of around 1 per cent of GDP. Furthermore, both gross and net public debt ratios are considerably lower than in most other EU countries. However, the authorities have taken a distinctly prudent view, arguing that maintaining the current fiscal position is necessary to absorb future ageing-related pressures and that the potential for further improvements in employment rates is smaller than in other countries (Ministry of Finance, 2001).

A striking feature of Danish income taxes in a cross-country context is that the highest tax rate, 63 per cent, is paid on all income from around the level of the average production worker (APW) wage or higher (Figure 15).[25] In contrast, those

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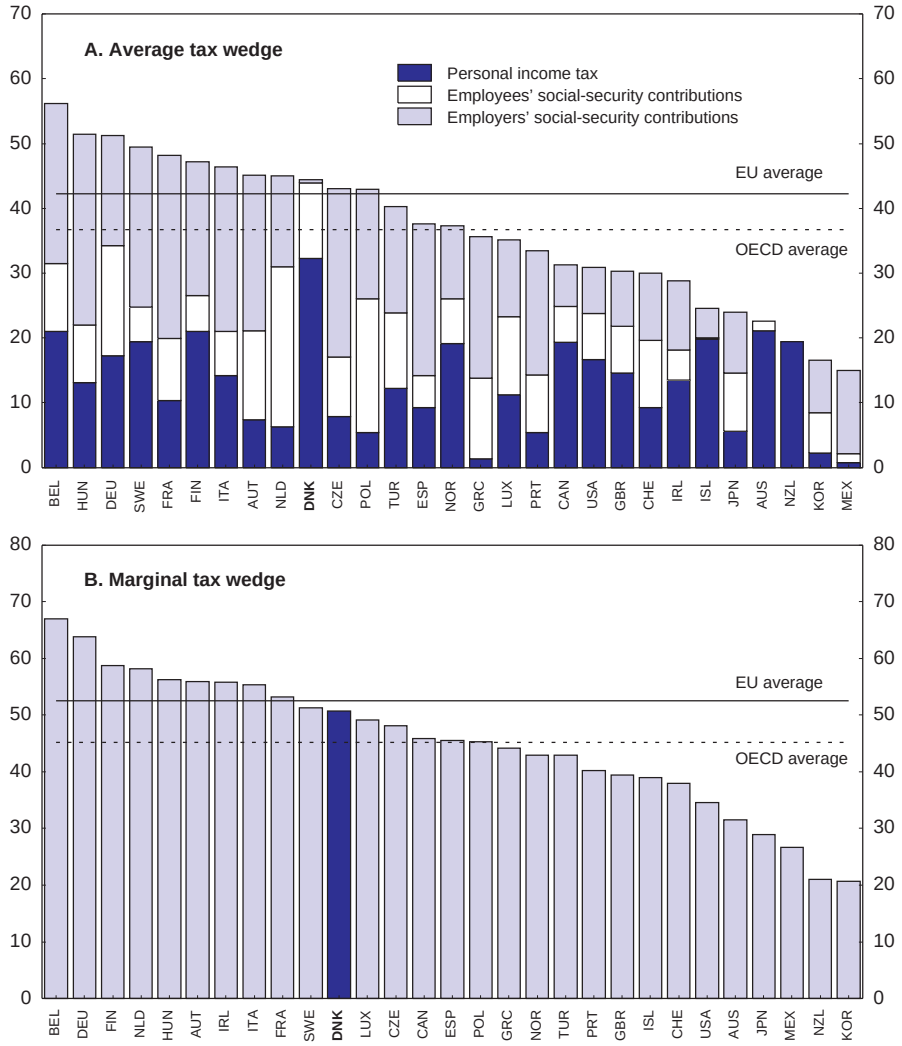
Figure 15. **Tax wedges on labour**
2000, per cent[1, 2]



1. For a single individual at the income level of the average production worker. Tax wedges are calculated by expressing the sum of personal income tax, employee plus employer social-security contributions together with any payroll taxes as a percentage of labour costs (gross wage plus employers' contributions).
2. Data for 2000 are based on estimated wage levels of the average production worker.

Source: OECD.

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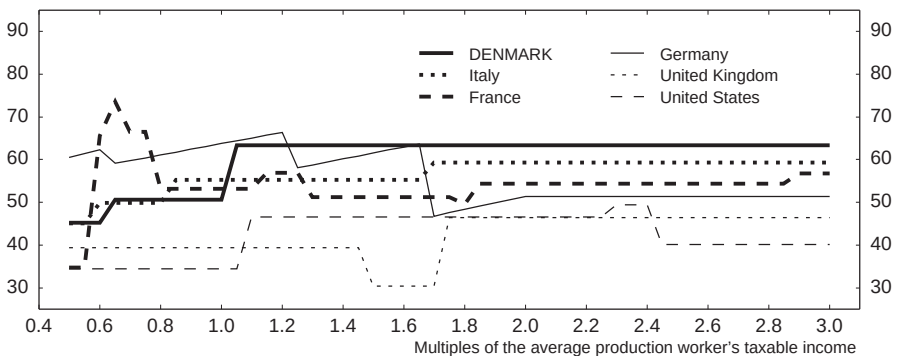
on incomes lower than APW level, face lower tax rates than the EU average (Figure 16). The previous *Survey* singled out a rise in the threshold for the top tax as probably the most efficient tax adjustment possible, *i.e.* the change that would prompt the biggest labour supply response for a given loss of gross revenue. This is because a high proportion of those in the top tax bracket have incomes only marginally above the threshold, implying that an increase could reduce the marginal tax rate for a fairly large group of taxpayers with relatively little revenue foregone. Recent studies (Det økonomiske Råd, 2001*b*; Frederiksen *et al.*, 2001) corroborate this view and conclude that a higher top-tax threshold would be superior to other possible tax adjustments due to greater structural effects. However, estimations for labour supply functions are quite sensitive to the empirical specification and the data used. Several recent estimates have found aggregate substitution elasticities in the range 0.1 to 0.5, varying strongly according to gender and income levels, implying larger elasticities for women and low-income groups. The implied uncertainty surrounding estimates of substitution and income elasticities mean that the exact magnitude of the labour supply response cannot be determined accurately. The speed of any response is uncertain as well. Bearing these caveats in mind, increasing the top-tax threshold may, even applying elasticities at the low end of the aforementioned interval, induce long-term structural effects big enough to significantly compensate the immediate loss of revenue.[26] Gradually

Figure 16. **Marginal tax wedge on labour across incomes**
2000, per cent[1, 2]



1. For a single person with no children.
2. Tax wedges between labour costs to the employer and the corresponding net take-home pay of the employee are calculated by expressing the sum of personal income tax, employees' plus employers' social-security contributions together with any payroll tax, as a percentage of labour costs.

Source: OECD.

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taking steps in this direction in order to provide greater incentives to work is, therefore, clearly warranted.

The case of raising the threshold for the top tax is an exception; structural effects from reducing tax rates could generally not be expected to provide sufficient revenue to compensate for more than a modest faction of the initial loss. Any wider tax reform would therefore have to be fully financed in order to safeguard the sustainability of the public finances. This inevitably implies a trade off between equity and efficiency considerations. While there may be some further room for broadening tax bases through limiting the scope of tax allowances, the character of the potential deductions that could be curbed, most notably the deductibility of interest payments, does not make for easy policy choices. Although the tax relief from interest payments remains generous, the macroeconomic impact of lower house prices from an eventual further limitation of its value for tax purposes would need to be taken into consideration and its implementation carefully managed.

Capital income taxation

Total capital income taxation comprises taxation of property, individuals' capital and share income (dividends and capital gains) as well as corporate taxation and taxes on the return on pension fund assets. While the revenue from total capital income taxation tends to fluctuate more than most other tax components, it amounted to 3¾ per cent of GDP in 1999. An important impetus to changes in capital taxation over the past decade has been the international trend towards reducing corporate tax rates. Denmark followed this course, reducing the corporate tax rate from 40 per cent in the early 1990s to 30 per cent at present. This was largely financed *via* broadening the tax bases by reducing the generosity of tax depreciation rules.

When it was introduced in 1987, the dual system for taxation of personal income was close to the "pure" form, the main deviation being the non-imputation of tax credits to alleviate double taxation of dividends. Capital income was taxed at a flat rate broadly of the same level as those applying to corporate income and the lower-income bracket for earned income and transfer income. However, changes since then have mostly tended to weaken the logic of the personal income tax system and its interplay with other capital taxes. Aside from the reduction of the corporate tax rate, the need to strengthen private savings and discourage tax-induced borrowing has been another source of downward pressure on personal capital income taxation. The tax rate on negative capital income (*i.e.* the tax value of the deductibility of interest payments) was lowered in each of the three tax reforms since 1987, although it is still higher than in most other countries. However, the tax rates applying to positive capital income have not been lowered accordingly, and the basic principle of a flat-rate tax has been abandoned. Thus, in

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the present system, negative net capital income is taxed (*i.e.* deductible) only at the local government income tax rate, while positive net capital income is taxed at the same rate plus the relevant state income tax rate. This implies a flat rate of 33 per cent for negative net capital income but a progressive scale between 40 and 59 per cent for net positive capital income (Table 9).

Table 9. **Main features of capital income taxation**

	Revenue, 1999 DKr billion	Statutory rate		Typical effective taxation of individuals' saving[1]
		1999	2002	
Types of capital income				
Personal capital income[2]	–9.4	–	–	–
Positive	–	40-59	40-59	–
Negative	–	40	33	–
Owner-occupied housing[3]	–	2/6	⅓	–
Share income[4]	4.7	25/40	28/43	–
Return on pension funds assets[5]	14.6	26/5	15	–
Corporate tax	36.6	32	30	–
Capital income taxes, total	46.5	–	–	–
As a percentage of GDP	3.8	–	–	–
Typical investment vehicles				
Pension savings	–	–	–	22.5
Owner-occupied housing				
With carry-over from tax to house prices	–	–	–	25.0
Without carry-over from tax to house prices	–	–	–	33.2
Shares	–	–	–	42.0
Bonds[6]	–	–	–	
Negative net capital income	–	–	–	49.8
Positive net capital income, top tax	–	–	–	74.6

1. Estimated by Det økonomiske Råd (2001). The calculations assume that DKr 500 000 is invested in the respective assets, that the bond yield is 6 per cent and that the inflation rate is 2 per cent. Tax rates from 2002 are applied. The figures are indicative only, as they do not capture all details of the existing rules.
2. Individuals are taxed on their net capital income. Thus all negative capital income is deducted from positive capital income before taxation. If net capital income is negative, a flat rate of 33 per cent (2002) applies. Taxpayers with positive net capital income are taxed according to their marginal personal income tax rate.
3. An imputed 2 per cent return on owner-occupied dwellings entered into positive capital income until 1999 (6 per cent of the assessment value above a certain level). As from 2000, it was replaced by a municipal property tax of 1 per cent of the tax-assessed property value up to around DKr 3 billion (0.8 per cent if the house is bought before 1 July 1998) and 3 per cent of the value above that. Assuming a nominal interest rate of 6 per cent, a 1 per cent tax on the property value roughly corresponds to a 16 per cent tax on the return.
4. The tax treatment of share income is rather complicated. Taxation of dividends follows a separate scheme rather than the normal capital income taxation. Dividends below and above a threshold of DKr 38 500 per year are taxed at 28 and 43 per cent, respectively. Capital gains are tax exempt for small-scale shareholders (with holdings worth less than some DKr 120 000) but are taxed like dividends for other investors. If the shares have been held less than three years, capital gains are treated as positive capital income.
5. The revenue from the ATP and LD pension funds, which are part of the general government, is excluded.
6. The effective tax rate is shown for an investor with negative net capital income and one who has positive net capital income and is paying top tax.

Source: Statistics Denmark (2001); Det økonomiske Råd (2001); OECD.

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Another complication of the personal capital income tax code stems from the fact that income from shares was partially separated out of the tax base in the early 1990s. This was done primarily to alleviate the double taxation of dividends; the typical 40 per cent tax rate on share income from 1993 onwards was lower than the lowest capital income tax rate, which at that time was 52 per cent. However, the current rules for taxation of share income are overly complicated. Different tax rates apply to dividends and capital gains, and, in addition, the tax on the latter depends on the value of individual taxpayers' shareholdings and how long the shares have been held.[27]

It would be considerably simpler to reduce the tax on positive net capital income to the rate applying to negative capital income.[28] This would also weaken the argument for the separate taxation of share income, which could instead be included in the base for capital income taxation. Such an inclusion of share income might eventually be supplemented by an imputation tax credit to alleviate double taxation of dividends, were this considered important. Aligning positive and negative net capital taxation would incur a gross revenue loss of approximately DKr 3 billion (¼ per cent of GDP). The likelihood is that this – all else being equal – would primarily benefit people with high earned income, which is the main political argument for preserving the current rules. Nonetheless, the distributional implications should be balanced against the highly distortionary effects and in particular the incentive for tax avoidance and evasion at present. The validity of the equity argument is also weakened to some extent by the fact that part of the revenue effectively stems from individuals' ignorance of the possibilities for tax arbitrage in the current system. Thus, approximately half of the revenue, *i.e.* some DKr 1½ billion, is paid by persons liable to the highest tax rate on earned income. This group could often reduce their total tax payment by reallocating their funds to lower taxed assets such as real estate, shares or pension savings.

To complete the description of the current patchwork-like design of capital income taxation, both owner-occupied housing and pension saving are taxed at rates that deviate significantly from those of personal capital income taxation. The typical effective tax rates on savings in the form of real estate and, in particular, pensions are considerably lower than for saving in other assets. Thus, with pension savings and positive net capital income (effectively equivalent to net interest receipts) as the extremes, estimated effective tax rates range between 22.5 and 74.6 per cent.[29] The low effective tax rate on owner-occupied housing results mainly from the deductibility of mortgage interest payments, which remain generous by international standards. Moreover, revenues from property taxation are reduced by the exemption of rental housing. The governments' stated intention of nominally freezing tax assessment values introduces a (weak) downward trend in effective property taxation in the future. Given the already low rate this appears to be a step in the wrong direction as it will lead to a wider gap between the effective taxation of owner-occupied housing and other forms of capital income.

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The tax treatment of pension savings generally follows an ETT-model, *i.e.* pension contributions are exempt from tax while the return on assets and the benefits are both taxed. In this respect, Denmark's practice deviates from that of most other countries, where the return on assets typically goes untaxed (Van den Noord and Heady, 2001). An important reason for taxing the return in Denmark has been to limit the possibilities for tax arbitrage. Nevertheless, despite some tightening in recent years, pension saving still enjoys a favourable tax treatment relative to other modes of private saving. Against this background, a higher tax rate on the return of pension funds investment might be warranted, but the scope for such an increase is limited by legal considerations (Box 4).

Taxes on goods and services

Taxes on goods and services account for approximately one-third of total tax revenues, which is somewhat higher than the EU- and OECD-wide averages (Table 10). At 25 per cent, the statutory VAT-rate is high by international standards and, apart from financial services and (imputed) housing services, exemptions are relatively few, implying that the effective VAT rate is also high (Figure 17). Several aspects of taxation on goods and services need attention. High excise duties on a number of products, notably tobacco and alcoholic beverages,[30] encourage tax evasion through cross-border trading. They have been sustainable only because separate agreements with the EU Commission have allowed Denmark to uphold quantitative restrictions on the amounts individuals can import. However, these restrictions must be abolished by the end of 2003 at the latest, which will inevitably necessitate an alignment of Danish tax rates with those of neighbouring countries in order to prevent a collapse of these tax bases.

A notable element of the 1994 and 1998 tax reforms was the increase in environmentally-related taxes, especially on water and electricity, which had the twin objectives of encouraging better environmental outcomes and facilitating some shift in the tax burden away from labour income. Indeed, the revenue from environmentally-related taxes is higher in Denmark, both as a share of total revenues and relative to GDP, than in any other OECD country. This difference is in large part due to vehicle taxes. In particular, the car registration tax (which was introduced in 1924) is extremely high in Denmark; on top of the VAT, a registration tax of 105 per cent is levied on the value up to around DKr 53 000 (€ 7 100) and 180 per cent on the price above that level. Consequently, the market price for cars is around three times the pre-tax price (less for small cars) which obviously distorts consumer behaviour significantly.[31] Car ownership is low by international standards (OECD, 2000), and registrations fluctuate much more over the business cycle than in most other countries (Figure 18), which is difficult to explain by any factors other than the high price level. On average, the revenue from the car registration tax has been 1.1 per cent of GDP, but given the strong cyclical fluctuations

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Box 4. The tax on pension returns and pension funds' interest guarantees

The return on pension funds' investments has been taxable since the introduction in 1984 of the real interest tax (*realrenteafgiften*), which (except for savings before 1983) effectively taxed away any real return on bond assets above 3½ per cent, implying an endogenous variable nominal tax rate. Other assets, notably shares and real estate, enjoyed a more favourable tax treatment (eventually exempt), and fairly strict regulation of the funds' portfolio composition was in place to limit the scope of revenue loss. The tax base has risen vigorously over the past 10 to 15 years; pension funds' balance sheets ballooned over this period, owing to the favourable tax treatment of individuals' pension savings in general and the introduction and expansion of collectively agreed labour-market pension schemes. Pension funds' assets amount at present to some 90 per cent of GDP and are set to increase further.

The endogenous tax rate fell markedly as from the mid-1990s, in line with the decline in real interest rates. From 1999, the variable tax regime was replaced with a fixed nominal rate on pension funds' returns (*pensionsafkastbeskatning*). As in the preceding scheme, share income was initially taxed on much easier terms (5 per cent) than bond and real estate income (26 per cent). In parallel with the change to fixed nominal tax rates, restrictions on the funds' portfolio composition were liberalised.

Meanwhile, the shift to a nominal tax scheme brought a different problem into focus. As in many other countries, pension funds have traditionally issued so-called *interest guarantees* that promise a certain *nominal after-tax* return on deposits. Up to 1994 interest guarantees typically promised a nominal annual after-tax return of 4½ per cent; at that time it was lowered to 3 per cent and in 1999 to 2 per cent. As saving deposits may be held by the funds up to 50 years, such guarantees are notoriously perilous, given the difficulties in predicting interest rate developments over that long a period and the lack of securities with a sufficiently long maturity to allow the funds to hedge themselves. This risk is exacerbated by the fact that the interest guarantees in some of the existing schemes apply to the return *each year*, not just the average return over a longer period.

As had occurred to Japanese insurance companies, the decline in nominal interest rates over the 1990s gradually put an increasing number of pension funds under strain, as the margin narrowed between the market return they could earn and the return they had guaranteed earlier, when nominal interest rates were significantly higher. The Financial Supervision Authority (FSA) sets a reference rate, which the funds must take as the return they can expect to earn on their investment; if the reference rate is below the guaranteed rate, reserves must be set aside accordingly. The FSA supervises that the individual pension funds hold adequate reserves to meet future obligations. The problems were in large part self-imposed in the sense that issuing interest guarantees has been voluntary throughout. Moreover, while the funds in many years earned returns above the guaranteed levels that could have been set aside as reserves to meet shortfalls in later years, an excessive share of such extra returns were often distributed as bonuses instead, partly due to competition among the pension funds. Another

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Box 4. **The tax on pension returns and pension funds' interest guarantees** (*cont.*)

perverse feature of the regulatory framework is that funds under strain have recently been allowed to impose large financial penalties on customers who have wanted to discontinue their engagement thereby effectively jeopardising competition in the sector.

On the other hand, the described situation can also be traced back to inappropriate regulation. With hindsight, it might have been better if policymakers had prohibited the issuance of guarantees. However, through the FSA-set reference rate and the supervision of the funds' reserve capital, the authorities have instead accorded them recognition. As the guarantees specify the after-tax return, policymakers may be liable if changes in the tax rules could be proved to have put pension funds in a situation where they could not meet their obligations. This problem, which obviously does not occur in countries that do not tax the return, became relevant with the shift from a real to a nominal tax; under the old rules, a reduction in the nominal interest rate would endogenously prompt a reduction of the tax rate. With the nominal tax rates, this is no longer the case.

The continued decline in the nominal interest rate after the introduction of the split nominal tax rate in 1999 eventually prompted an adjustment in 2000, whereby a uniform tax rate of 15 per cent was instituted. As an aside, the resulting higher tax rate on equity return (in combination with the increasing funds-to-GDP ratio) is bound to add more volatility to the revenue and thereby to public finances as such. By lowering the tax on the return on bonds, which account for the lion's share of most funds' assets, the uniform tax rate implied a net easing of the funds' financial situation due to the particular methods applied for calculating the required reserve capital. Most recently, a few funds saw their safety margins narrow alarmingly again in the autumn of 2001 as share prices continued to fall, but the subsequent rebound has largely offset this effect. While immediate pressures are likely to be alleviated by advancing an already planned changes in accounting provisions that allow more assets to be included in reserve capital, this will not necessarily prevent problems from re-emerging in the future. In the meantime, policymakers are effectively prevented from raising the tax rate on pension fund returns, even if they might wish to do so.

the proceeds of the tax vary considerably, thereby adding to the volatility of public finances. The typical trough-to-peak difference in the proceeds is around ¾ percentage point of GDP.

The arguments typically advanced in support of high taxation of car ownership are partly distributional and partly fiscal. But while lower taxation would probably show up as higher inequality, it would nevertheless benefit also the

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Table 10. **Features of indirect taxation in Denmark, the EU and the OECD**

Percentage of total tax revenue, 1998

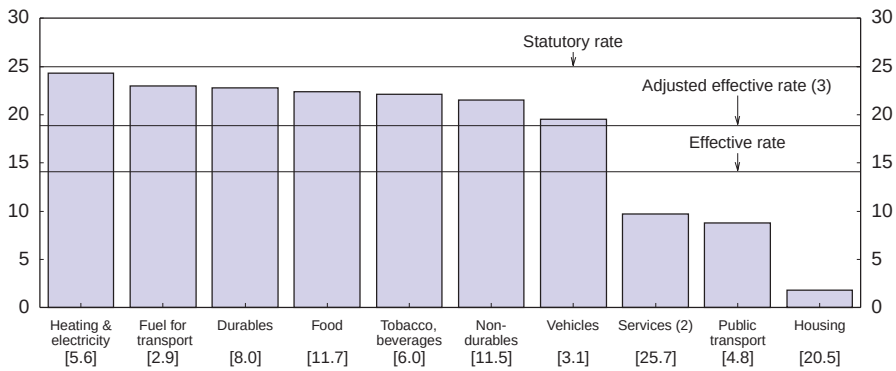
	Denmark	European Union		OECD	
		Unweighted average	Weighted average	Unweighted average	Weighted average
Total taxes on goods and services	**33.2**	30.2	28.8	31.3	23.2
of which:					
General consumption	**19.6**	17.5	17.2	17.9	12.5
Environmental taxes	**10.1**	7.6	–	7.4	–
Non-recurrent vehicle taxes (1995)	**3.0**	1.4[2]	–	1.1[2]	–
Memorandum items:					
Statutory VAT rate[1]	**25.0**	19.4	–	–	–

1. 2001 rates.
2. Average of nine countries for the EU and sixteen countries for the OECD. Only countries with recurrent vehicle taxes are included in the calculation of the averages, which, therefore, are significantly higher than if all countries had been included.

Source: Van den Noord and Heady (2001); Joumard (2001); OECD (2001).

Figure 17. **Effective VAT rates for various consumption groups**

2001, per cent[1]



1. The figures indicated below the labels represent the share of the relevant consumption group in total private consumption.
2. Financial services are exempt from VAT for administrative reasons, but financial institutions are liable for a payroll tax of 9.1 per cent instead.
3. Excluding housing and financial services.

Source: Statistics Denmark; OECD.

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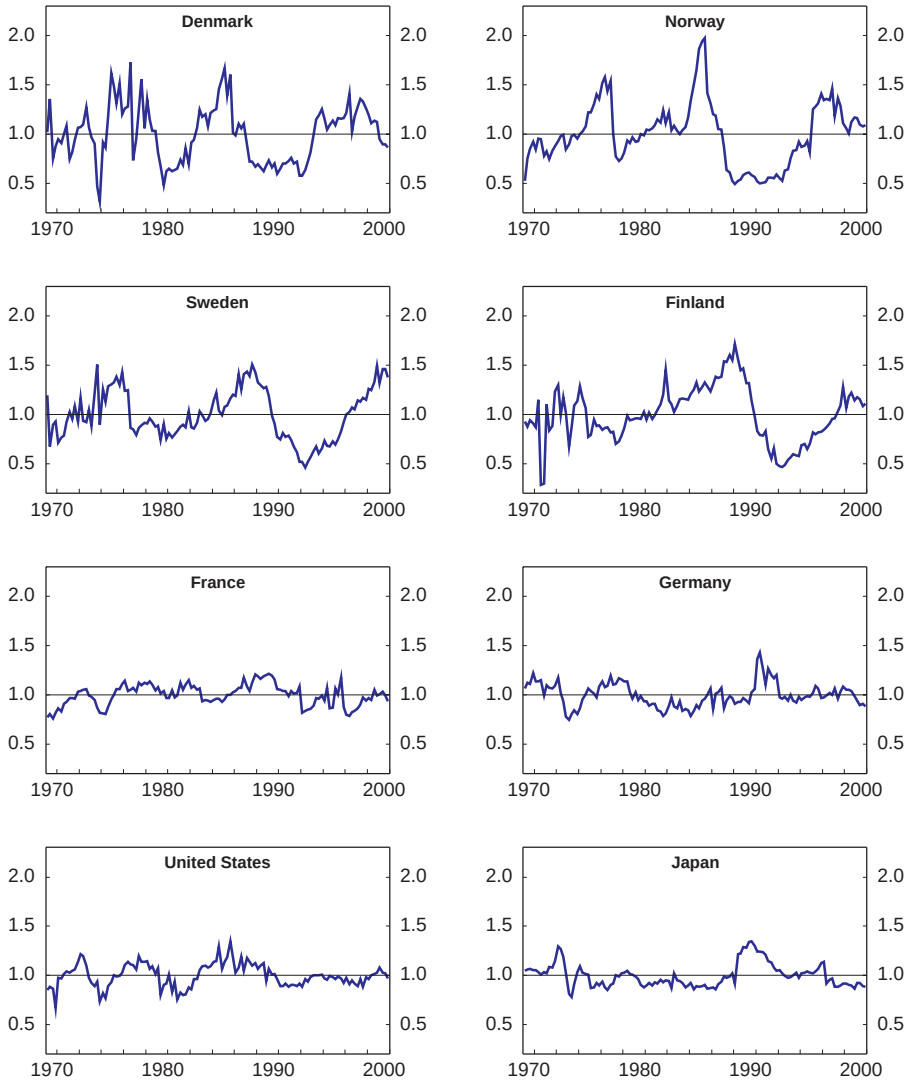
Figure 18. **Car registrations**[1]
Ratio to trend



1. Similar calculations have been undertaken for a number of other countries. In all of these (Austria, Australia, Belgium, Canada, Italy, New Zealand, the Netherlands, Spain and Switzerland) the car registration cycle is much less pronounced than in Denmark.

Source: OECD, *Main Economic Indicators*.

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low-income groups. Thus, notwithstanding the progressivity of the tax scale, those who have opted out (or have effectively been forced out) of car ownership are mainly low- and middle-income households (Table 11). Also, the implications for public finances and income distribution are probably more complex than might appear at first glance. The high market prices almost certainly leads consumers to raise (bigger) loans, for which the resulting interest payments are tax-deductible, to finance their car purchases. Even though this would not result in a revenue loss if the taxation of capital income is symmetric, it is a rather perverse feature of the tax system.

The high car taxation may also implicitly have repercussions on public finances through high public-transport subsidies. At ¾ per cent of GDP public-transport subsidies are higher than in most other countries, probably because services outside peak hours and in sparsely populated areas are extensive. Low utilisation rates imply that the economic costs are high and that environmental arguments for upholding the services are compromised (OECD, 2000*b*). By increasing car penetration rates to average European levels, a less distortionary taxation of cars might reduce the need for subsidies to public transport. On the other hand, it might be argued that the lower incidence of car ownership reduces the need for road infrastructure investments. However, cars are generally used more intensively in Denmark than elsewhere (OECD, 2000*b*), which may be a result of high taxes on ownership. If so, reducing the tax might lead to less intensive use and not necessarily an increase in total distances travelled. A reduction of the registration tax would admittedly involve considerable transition problems, basically because the distortionary effects are very large in the first instance. But, addressing the problems should at least be considered, starting with a thorough analysis of the various issues involved.

Table 11. **Car ownership by household income**
2000

Annual income Thousand DKr	Number of cars per household			Number of households (thousands)
	0	1	2 or more	
	Share of households			
Up to 99	90.4	9.1	0.6	237
100-150	76.1	23.0	0.9	336
150-200	57.5	40.6	1.9	243
200-250	46.5	49.7	3.8	423
300-400	28.2	62.4	9.4	300
4 00+	10.4	62.7	26.9	896
Total	40.4	47.5	12.1	2 434

Source: Statistics Denmark.

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Creating jobs for people at the margin of the labour market

The string of labour-market reforms undertaken over the course of the 1990s has served Denmark well and helped bring unemployment down towards the current 25-year low level of 4¾ per cent (standardised rate). The main emphasis has been on tightening eligibility criteria for unemployment benefits both formally and implicitly by making participation in labour-market programmes compulsory. Previous S*urveys* have provided in-depth analysis of policies to reduce structural unemployment. The thrust of these earlier assessments, as well as the policy recommendations following from them, remain valid. Notwithstanding the achievements so far, further inroads on structural unemployment could be facilitated by, for example, abolishing non-core benefits (such as for holidays) and letting benefits decline with the length of the unemployment spell. The previous S*urvey* also highlighted several impediments to higher labour-force participation: the slow pass-through in the education system, the generosity of paid-leave schemes and the continuing downward trend in average working time. Action in each of these areas remains warranted.

This section provides a discussion of employment opportunities for people at the margin of the labour market and policies to encourage better performance on this count. The theme is of immediate interest in the Danish context, given that overall employment and participation rates are high and unemployment well below the European average. These aspects play an important role in the overall medium-term economic strategy of boosting labour supply. The discussion addresses the following issues in turn: reforming the early retirement (disability) pension scheme; promoting an inclusive labour market; and integration of immigrants into the labour market.

Reforming the early retirement (disability) pension scheme

A comprehensive reform of the early retirement (disability) pension scheme was passed by parliament in late 2000 by a broad majority. The reform, which affects only those entering after 1 January 2003, is ambitious in its objective to improve the employment opportunities for people with less than full working capacity. While the latter is not formally a condition for the pension reform, failing on this count would put considerable strain on the new system.

There were 260 000 early retirement benefit pensioners in 2001, equivalent to 7 per cent of the working-age population, and associated gross public outlays amounted to almost 2¼ per cent of GDP. This makes it the single most significant scheme in the transfer system. However, the annual inflow has declined steadily over the past decade, having abated from above 25 000 in the early 1990s to around 13 000 in more recent years. This decline is partly attributable to changes in the financing of early retirement pensions that have placed a greater share of the associated costs on the individual municipalities. The discretion to

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assess the eligibility of applicants within the legal framework was transferred to the municipalities in 1992. The financial responsibility was partly devolved at the same time, and the Active Social Policy Act from 1998 increased the individual municipalities' share of total costs to around two-thirds.[32] Since then, recipients of early retirement pension have been more expensive for the municipalities than recipients of other transfer schemes except long-term sickness benefit (after 52 weeks). In the short term, this has to some extent resulted in a higher intake to activation and rehabilitation programmes in line with government policies.

There is a complex interdependence between developments in the individual benefit schemes in the social system, notably early retirement pension, sickness benefit, cash benefit and rehabilitation, which have partly overlapping target groups and eligibility criteria. Thus, 75 per cent of the newcomers to the early retirement pension system had been previously receiving either sickness or cash benefit, while less than 10 per cent were in employment (Den Sociale Ankestyrelse, 2001). The distribution of beneficiaries across the system is affected by several factors, including individuals' economic incentives. Moreover, the management of these schemes lies with the municipalities, and there is solid empirical evidence that in their administrative practice they take account of the relative costs. The decline in the intake to the early retirement pension since 1998, induced by increased costs associated therewith, has partly been countered by higher enrolment in neighbouring schemes. Some renewed pickup in the inflow rate to the pension system is generally expected; the Ministry of Social Affairs has estimated that it would have gradually returned to a "normal" level of 16 000 to 17 000 persons if the current pension system had remained in place.

Nevertheless, the reform was not motivated by concern over the economic costs of the system. Rather, the benefit structure of the current system is distinctly complicated with three different types of pension, several potential supplements depending on family status and income, a mix of taxable and non-taxable benefits and a non-transparent interplay with other elements of the transfer system. The three types of pension in the current system differ from one another mainly by the degree of reduced working capacity of the recipients. I*ncreased ordinary pension*[33] can be awarded to those with at least 50 per cent reduced working capacity; to be eligible for *intermediate pension*, working capacity must be reduced by at least two-thirds; while for the *highest pension* it must be negligible. The co-existence of three different types of pension has meant that a high proportion of those receiving the lower-level benefits apply for a reassessment in order to obtain a higher benefit level. This has been reinforced by the fact that eligibility for benefits from private insurance schemes is often linked to receipt of the intermediate or highest benefit level.

A main objective of the reform has been to establish a system with only one pension type; from 2003, applicants whose ability to work is reduced by two-

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thirds or more will be entitled to a pension which varies only according to civil status and total family income. While the reform greatly simplifies the early retirement pension system itself, it makes the interplay with other parts of the benefit system more complex. More than half of those entering the current system are granted the lowest benefit (*i.e.* the one that is removed with the new system). This group would typically have difficulty finding unsubsidised employment. While it is uncertain where these people would end up in the social system in the future, it is important for the success of the reform that as many as possible will move into employment, even if with the help of wage subsidies.[34] At least three factors will affect the outcome:

- The success of the inclusive-labour-market strategy (discussed below).
- The interpretation of the eligibility criteria, since the unmeasurable character of the concept of working capacity means that the administrative practice that will evolve in the future is a critical element.[35] The reduction in the number of annual entrants expected by the Ministry of Social Affairs is less than the current number of entrants on the lowest benefit, implying more generous assessments of eligibility.
- The economic incentives in the new system. While depending partly on the individual's situation, the new system would appear more attractive than the current one for many of those with the best work prospects and *vice versa* for many of those with little hope of finding a job[36] (Table 12). In this sense, the change in the benefit structure may run counter to the objective of boosting employment.

Although decisive assessments are obviously premature overall, the risks would suggest higher costs than assumed.

Promoting an inclusive labour market

The "inclusive labour market" (*det rummelige arbejdsmarked*) is a label that has been employed increasingly in recent years to denote a policy strategy of improving employment opportunities for people at the margin of the labour market. While neither the concept nor the strategy are entirely well defined, the latter may be taken to encompass all relevant policy measures, except those that work directly through strengthening individuals' economic incentives. These are designed to retain jobs for those still in work but in a vulnerable situation and to ameliorate the employment situation for groups that generally have temporary or permanent difficulties in remaining in the ordinary labour market. Broadly defined, the target group comprises some older, mainly unskilled workers near the age of retirement, long-term clients of the social system, would-be early retirement pensioners and immigrants.

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Table 12. **Selected features of the early retirement benefit system before and after reform**[1]

	New system	Current system		
		Increased ordinary	Intermediate	Highest
Eligibility[2]	Two-thirds	Half	Two-thirds	Full
Statutory pension benefit levels	Per cent of maximum unemployment benefit level			
Singles, no children	100	80	94	116
Married and cohabitants				
Maximum (no offsetting of spousal income)	80	62	75	n.a.
Minimum (highest offsetting of spousal income)	64	47	60	n.a.
Average pension benefit levels[3]	100	68	74	101
Disposable income[4]	Per cent of disposable income of a full-time unemployed person			
Singles, no children				
No individual housing benefit	102	80	100	136
Maximum individual housing benefit	102	113	132	158

1. The complexity of the existing system does not allow an exhaustive description. It should be observed that supplements awarded subject to discretionary assessment are not included. These are more important in the new system than in the present one.
2. Indicates the degree of reduced working capacity required to be eligible.
3. Maximum unemployment benefit level = 100.
4. Danish Ministry of Social Affairs' assessment. Excludes supplements and individual housing benefits.

Source: Ministry of Social Affairs.

For all these groups, the only alternative to being employed on special terms is often extensive reliance on public transfers. Hence, a large part of the underlying rationale is to reduce public spending, but there is also a focus on increasing labour supply and the medium-term objective of increasing the workforce by 100 000 is crucially dependent on the success of the strategy (Finansministeriet, 2001*b*). The emphasis on non-economic instruments recognises that it is often lack of employability more than weak economic incentives that prevents employment. The strategy focuses on both job retention and creation of new jobs. It spans a broad range of mostly "soft" measures, typically of an administrative character. These include dialogue with the social partners to incorporate so-called social clauses in collective agreements; compiling local action plans in the municipalities; establishing social partnerships between local authorities and enterprises; and encouraging corporate social responsibility, for example, through guidance and consultancy services on so-called "social and ethical accounting".

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The most concrete element is the so-called *flexi-jobs* (*fleksjob*) scheme, which was introduced in 1998 and has been elaborated upon since then. It entitles individuals to a wage subsidy (which in practice is paid directly to the employers) that compensates for a permanent and significant reduction of their working capacity. The assessment of eligibility is undertaken by the municipalities. The size of the wage subsidy varies (in principle) proportionally with the degree of reduced working capacity, with a ceiling at two-thirds of the minimum wage level. Two-thirds of the subsidy is financed by the central government and the rest by the municipalities. On top of the wage subsidy, employers of people in *flexi-jobs* are exempt from the normal obligation to finance the first two weeks of sickness benefit. People who have been assessed as eligible for a *flexi-job* are entitled to an out-of-work benefit (*ledighedsydelse*) in periods where they are not in work; the benefit level is 91 per cent of normal unemployment benefits, two-thirds of which is financed by the individual municipalities. From the municipalities' perspective there is a strong economic incentive to try and get as many clients as possible out of the social benefit system, where they typically co-finance a greater share of the costs, and into *flexi-jobs*.

It remains to be seen what the potential of this approach is. After a somewhat sluggish start the inflow to the *flexi-jobs* scheme has accelerated of late. By mid-2001, some 12 000 *flexi-jobs* (0.4 per cent of total employment) had been established, of which approximately two-thirds were in the private sector. As the wage subsidy fully compensates the employer for the reduced working capacity, the success of the scheme should, theoretically, be unaffected by the general economic situation. However, it seems likely that it has a much better chance of success in times of prosperity, because it basically relies on the self-interest of enterprises and the goodwill of other employees; it may be vulnerable under less favourable economic conditions. In addition, an important precondition for a successful outcome is that both other employees and their organisations accept that persons holding *flexi-jobs* and other jobs on special terms should be allowed to occupy the same type of functions as everybody else, *i.e.* the wage subsidy should not restrict the range of jobs available.

Challenges facing immigrant integration policies

Immigration policies and the economic integration of immigrants have moved up considerably on the political agenda in recent years, culminating in their pre-eminent position in the 2001 election campaign.[37] A large body of research on immigrants' integration into the labour market has also surfaced. As unemployment has declined, spreading shortages of well-educated people with particular technical skills have prompted calls from employer organisations for increased and more selective immigration. In addition, immigration has been discussed as a potential avenue for alleviating the fiscal burden of a higher old-age dependency ratio.

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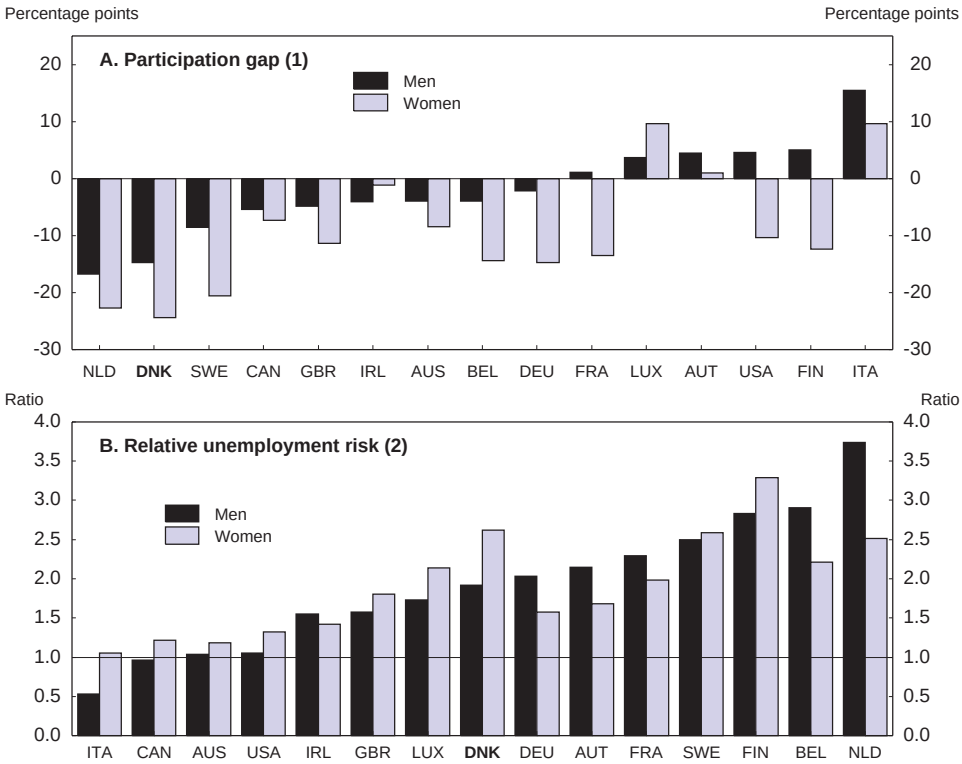
However, labour-market performance of immigrants at present is not particularly promising in this respect. While it is a common feature of most OECD countries that immigrants suffer from above-average unemployment rates, this is most conspicuous in Denmark (as well as in the other Scandinavian and Benelux countries) where the incidence of unemployment amongst immigrants is more than double that for other groups (Figure 19, Panel B).[38] This contrasts sharply with the situation in traditional immigration countries such as the United States, Canada and Australia, but some Southern European countries and Ireland also appear to be more successful. Participation and employment rates basically tell the same story; the employment situation of immigrants in Denmark is rather dismal relative both to the native population (Husted *et al.*, 2001*a*) and by international standards.

Figure 19. **Immigrants' labour force participation and unemployment rates, 1998**



1. Foreigners' minus nationals' participation rate.
2. Foreigners' divided by nationals' unemployment rate.

Source: OECD (2000).

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There is a direct link between the weak labour-market attachment of non-western immigrants and the fact that they are highly over-represented amongst recipients of public income transfers, notably social benefits, such as cash benefit, rehabilitation and social activation (Indenrigsministeriet, 2001*a*). The aggregate figures mask big differences in the labour market status of immigrants from different countries and between men and women. The participation and employment gaps of female immigrants are much bigger than that of male immigrants. In contrast, the gap in unemployment rates is significantly greater for men than for women.

The labour-force participation rate of non-western immigrants has declined significantly over the past 10 to 15 years, and at around 40 per cent it is currently only around half that of native Danes (Mogensen, 2000; Dansk Arbejdsgiverforening, 2000). Moreover, recent research based on register data suggests that immigrant integration has become a steadily more drawn-out process since the early 1970s, even when adjusting for changes in the country-of-origin composition (Schultz-Nielsen, 2001). This is mainly reflected in falling labour-force participation rates after five years in Denmark, from almost 80 per cent for non-western immigrants arriving between 1978 and 1980 to only 50 per cent for those entering between 1993 and 1995. On a more positive note, the participation rate for the more recently arrived immigrants has kept rising more sharply with the length of the stay than it did for earlier cohorts, implying that the gap in the initial participation rates narrows over time. In addition, continued reductions in aggregate unemployment seem to have benefited non-western immigrants considerably in recent years, which are not yet covered by the above studies. However, on balance, these bright spots cannot change the overall rather gloomy impression.

Less is known about the labour-market performance of second-generation immigrants (*i.e.* the children of immigrants), as most of these are still making their way through the education system: of the 80 000 descendants (1½ per cent of the population) only 15 per cent are above 16 years of age. Nevertheless, the information available suggests that descendants do significantly better than their parents, but not as well as youth with native parents. In 1999 descendants aged 16 to 24 years had a labour-force participation rate of 60 per cent, compared with 70 per cent for others in the same age group. Moreover, Husted *et al.* (2001*b*) find that the return on descendants' human capital investment is considerably lower than that of Danes without an immigration background, even with due account taken of differences in various background factors. These, still tentative, findings are corroborated by experience from Sweden (Wadensjö, 2001), where immigration from non-western countries took off earlier than in Denmark.

At least three partly overlapping explanations for the hitherto unsuccessful integration of non-western immigrants have been proposed. The first emphasises immigrants' inadequate endowment of educational and other skills necessary to get a foothold in the labour market. In part this is because immigration policy has not

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been based on trends in labour demand. Refugees, asylum seekers and family reunions have accounted for practically all immigration from non-western countries over the past decades (see Box 5). A relatively large proportion of immigrants has come from countries (Iran, Iraq, Somalia, Pakistan, Lebanon, etc.) that are culturally very different from Denmark. There appears to be a strong relation between immigrants' linguistic skills and their employment situation; knowledge of Danish is

Box 5. Fact sheet on immigration and integration into the labour market

Denmark has administered an almost total freeze on economically motivated immigration from outside the Nordic and EU countries since 1973. Nonetheless, the share of immigrants and their immediate descendants in the total population has risen from 3 per cent in 1980 to more than 7 per cent in 2000, owing largely to an inflow of refugees and family reunions. Excluding descendants, there was an increase from 2.7 to 5.6 per cent.

Despite the freeze on economic immigrants, net immigration rose to post-war high levels in the 1990s. Annual average net immigration in that decade stood at 11 000 persons, equivalent to 0.2 per cent of the population.

The labour-force participation rate of non-western immigrants and descendants was 53 per cent in 1999, compared with 80 per cent for the native population. Unemployment rates for the two groups were 19 and 4 per cent, respectively.

The participation rates of immigrants vary considerably between men and women and depending on the country of origin. For male refugees, the participation rate was 50 per cent in 1999 while it was 31 per cent for female refugees. Among other immigrants, labour force participation is higher for both males and females, but the gender difference is approximately the same as among refugees. Female immigrants' labour force participation rates in Denmark tend to remain close to the levels in their country of origin.

Non-western immigrants and descendants are highly over-represented amongst recipients of public income transfers. 62 per cent of those aged 25 to 66 years received temporary benefits (*i.e.* excluding early retirement transfers) for some period in 1999. The comparable figure for native Danes was 27 per cent, of which a high proportion received unemployment and sickness benefits in a short period only. Thus, if unemployment and sickness benefits are excluded, the proportion of benefits recipients falls to 44 per cent for immigrants and descendants and to 11 per cent for the native adult population.

Immigrants' income mobility is low (and much lower than for native Danes). Sixty per cent were in the bottom three income deciles in 1994; of these two-thirds of these were still in these deciles in 1998 while less than 10 per cent had moved up into the highest 40 per cent.

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found in some studies to be the single most important precondition for successful integration (Schultz-Nielsen, 2001). The language barrier is almost inevitably more difficult to overcome for non-western immigrants coming to Scandinavian countries than for those entering countries with more commonly spoken languages.

A second possible explanation is discrimination in employment opportunities. Empirical evidence suggests that features such as computerisation, job rotation, team-based work organisation and greater involvement of lower-level employees in decision-making processes are particularly widespread in Denmark and Sweden (OECD, 1999*b*). Arguably these characteristics make it more important for employees to possess a high degree of "cultural and linguistic capital" in the sense of well-developed interpersonal communication skills. To the extent that this prompts employers to screen job applicants by their name or country of origin more than by their formal qualifications (educational background and previous work experience) this may disadvantage immigrants and descendants. Using panel-data estimations based on administrative register data, Rosholm *et al.* (2001) find that such statistical discrimination has intensified significantly in both Denmark and Sweden from the mid-1980s to the mid-1990s, and they ascribe it to organisational changes in the workplace.

Third, the dismal integration record could be related to the incentive structures and other features of the welfare system (Smith, 2001; Necef, 2001). An immediate observation is that the countries where the ratio of immigrant-to-average unemployment rates is particularly high – the Scandinavian and the Benelux countries – are all countries with fine-mesh social-security systems. The Danish social security system is characterised by high replacement rates for low-income workers, and it is dominantly universal in its character (*i.e.* benefit entitlements are generally not linked to employment history), implying that immigrants enjoy basically the same rights as others. The implied benefit levels, which, for most non-western immigrants, are generous both relative to conditions in the origin countries and to gainful employment in Denmark, significantly reduce work incentives. At the same time, minimum wages set in collective agreements are high, with the result that low-paid jobs in the services sector are virtually non-existent. While some become self-employed, this is likely to make it difficult for many immigrant groups to price themselves into other types of employment.

For immigration to strengthen public finances, immigrants must be economically self-supporting on average, at least to the same extent as the native population. This is clearly not the situation at present. A study from the Ministry of Economic Affairs (Økonomiministeriet, 1997) estimated the costs of (accumulated) immigration of non-westerners – in terms of below-average tax payments and above-average receipts of income transfers and individual public services – to be around 1 per cent of GDP in 1995. More recently, Petersen (2000) illustrates in a general-equilibrium model simulation that increased immigration would weaken

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rather than strengthen public finances in the long term if its composition remains the same as in the past. Thus, unless the economic integration into the labour market can be significantly improved, policymakers can either accept that immigration has a (modest) fiscal cost or broaden policies by selecting immigrants also on the basis of economic criteria. In its "working programme", the government has signalled that will indeed make possible immigration based on economic criteria (requiring an employment contract and accommodation arrangements) while restricting immigration for other purposes.

The Integration Act that came into force in 1999 requires all immigrants receiving public transfers to participate in a full-time three-year assimilation programme on arrival. These programmes, for which the responsibility has been devolved to the municipalities, include intensive education in Danish language, courses in community understanding and labour market training. The explicit objective behind the assimilation programmes is to enable immigrants to obtain jobs. As the first programmes started only in 1999, a full assessment of their impact is premature. However, preliminary evidence suggests that non-attendance in the language courses remains a concern and that participants do not receive labour market training on the scale that was envisaged in the Integration Act (Indenrigsministeriet, 2001*a*). To safeguard quality, it is important that such problems are addressed, possibly involving more stringent enforcement in cases of unjustified absence. Any scope for further strengthening integration efforts should be pursued. For example, for immigrants with professional qualifications (through formal education or otherwise) from their home country, it would help to develop a more dedicated strategy to make these usable in the Danish labour market. The introduction programmes should be used consistently to test and, if necessary, subsequently upgrade or adapt competences. Some kind of formal certification from the educational authorities might help immigrants to overcome the signalling-type problems that may otherwise expose them to discrimination. Nevertheless, it is doubtful whether better assimilation programmes are sufficient to bring immigrants' average employment rate up to the Danish average, given the aforementioned obstacles and assuming a broadly unchanged immigration pattern in the future.

Reforms to the welfare system that could help overcome problems caused by inadequate employability of immigrants or weak economic incentives are notoriously controversial, not least because lowering benefits for immigrants would imply also scaling back long-established rights of native Danes, to avoid overt discrimination.[39] The new government intends to introduce new eligibility criteria for full social benefits requiring at least seven years of residence in Denmark thereby strengthening economic incentives. Another option would be enhanced active labour market policy that could combine the need for an increase in the labour force with better integration. A number of municipalities have been quite successful in placing immigrants in jobs. A further possible way of strengthening

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economic incentives to search for and accept a job might be to introduce an in-work benefit for these immigrants for a limited number of years upon their arrival to Denmark. Although this also involves a degree of discrimination, the foreign-experts tax scheme already provides a precedent of favourable tax treatment of foreigners working in Denmark. An in-work benefit (such as an earned income tax credit – EITC) would make it more attractive to work by increasing the difference between the take-home pay from work and the alternative of social benefits. Although an EITC would not be suitable as a general instrument in Denmark,[40] some kind of in-work benefit could help alleviate the poverty trap facing many immigrants. It might also induce individuals to accept lower-paid jobs than they would otherwise, although the high minimum wages, set in collective agreements, set a limit to such effects. Thus, it would not directly deal with problems of individuals' productivity being too low to justify employment at the going minimum wage, though it would strengthen individuals' incentive to overcome such employability problems. If such demand side constraints on immigrant employment were thought to be more important than disincentives to seek and accept employment, it might instead be relevant to extend targeted employment subsidies.[41] A comprehensive assessment of the advantages and disadvantages of such targeted schemes would obviously require further analysis, taking into account compliance with international conventions.

Electricity and the environment

Implementation of the electricity reforms ran into difficulty over the green certificates market, and that part of the reforms has been put on hold for the present time. This instrument was designed to oblige all electricity users to purchase 20 per cent of their electricity from renewable resources and had been scheduled to be introduced from the beginning of 2000. Wind energy, the most significant renewable source in Denmark, would then shift from a fixed tariff of 60 øre per kWh, to a market where the price would be set between a minimum of 10 øre and a maximum of 27 øre per kWh, on top of a fixed 33 øre per kWh component. However, after repeated delays and extensive opposition from a wide range of groups, there is currently no firm date for implementation and the fixed tariff regime remains in place for the meantime.

The delays and uncertainties have had some unfortunate consequences for investment patterns in the industry. In 1999 the government announced a transitional scheme whereby all wind turbines in operation or approved for installation by 1 January 2000 would continue to receive the fixed tariff for a transition period of five to six years before being required to participate in the market. Any new turbines approved after that cut-off date would receive a mixed payment of 33 øre per kWh plus a green certificate of uncertain value, until the market was established. This produced a surge of new installations profiting from the old

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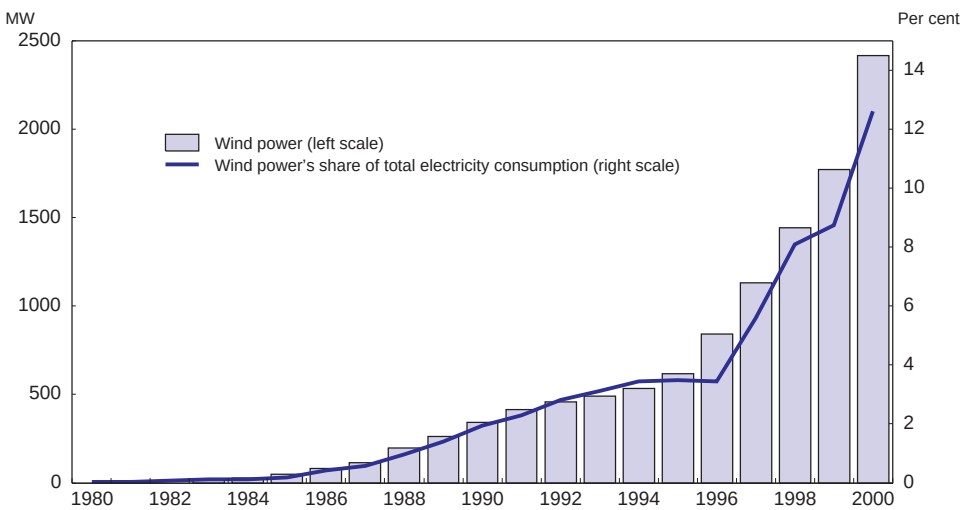
rules,[42] and coming on stream in 2000, adding more than 30 per cent to generation capacity. It was sufficient to lift overall renewable generation (including bio-mass) to about 20 per cent of the electricity market (Figure 20). But with the future framework still unclear and no market yet established, investment has more or less stopped.

The alacrity with which investors took advantage of the announced closing of the old scheme confirms that the previous system guaranteed excessive rates of return for wind-turbine investment given the present state of technology, otherwise investors could have been confident of receiving this degree of subsidy *via* the green certificates system. Action is now needed to unwind the excessive tariffs that are imposed on users and generate significant rents to owners and to implement a more sensible approach to any further increases in renewable generation capacity. At the present time, wind generation installed before 2000 receives a fixed tariff of 60 øre per kWh for a certain amount of production, depending on the size of the turbine. Capacity installed between 1 January 2000 and 1 January 2003 receives 43 øre per kWh for production equivalent to 22 000 full load hours. These tariffs are high compared with total costs in new Nordic turbines of around 34 to 40 øre. Costs have fallen sharply in recent years, and wind is among the cheapest sources overall (Figure 21), even without the full internalisation of the costs of CO_2 and acid rain precursor emissions. Operating costs for wind

Figure 20. **Wind power capacity and share of electricity consumption**



Source: Ministry of Energy and Environment.

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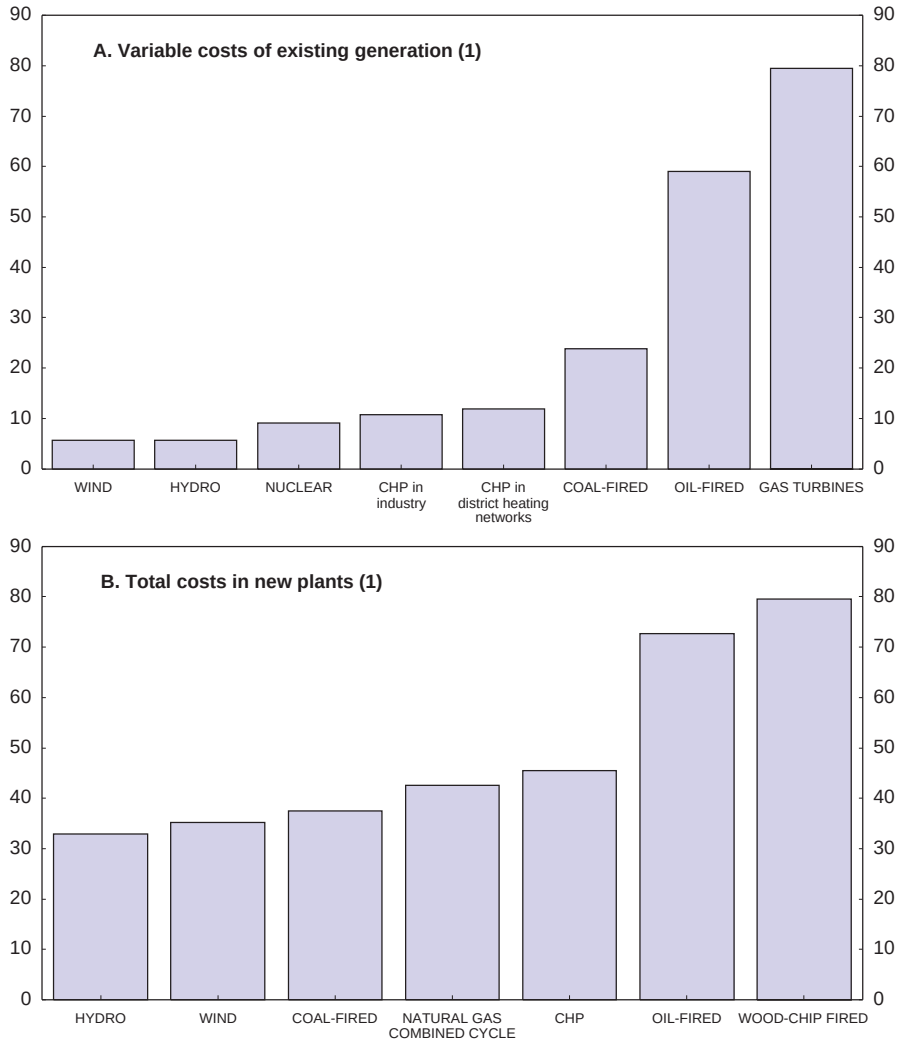
Figure 21. **Electricity generation costs**

Danish øre



1. Average for all type of process in 2000. Total costs assume 4 per cent real interest rate and a depreciation period of 20 years, except for hydro where 50 years is used. Although these costs are calculated for Swedish generation, they can be considered broadly representative for all Nordic countries.

Source: Swedish National Energy Administration.

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power are dramatically lower than Nordpool prices, which have climbed sharply in 2001 to around 20 øre per kWh. Users nevertheless pay high prices compared with elsewhere in the Nordic area, in part because of the "Public Service Obligation" (PSO) requirement (Table 13), which obliges them to purchase a certain proportion of "priority" electricity (renewables and output from certain combined heat and power plants) and pay the higher price. As a result, owners of newer turbines are able to earn high returns at the expense of the rest of the economy.

The Danish objective of increasing the proportion of renewable electricity supply as an end in itself complicates the policy framework. If the main objective were to internalise the external costs associated with emissions of greenhouse gases and acid rain precursors, then the CO_2 cap-and-trade system, now up and running for electricity (see Annex I), along with taxation of SO_x and NO_x emissions provide a sensible framework for encouraging the least-cost and most efficient generation techniques *via* market signals. Within this framework, renewables energy use would become relatively more attractive if the cap on emissions were lowered. The PSO could be removed, and demand and supply would both determine the market price and ensure that electricity is provided by the most efficient producers. This strategy would probably not deliver support to wind energy to

Table 13. **Electricity prices in Nordic markets**
1 July 2001, Euros/100 kWh

	Domestic customer			
	3 500 kWh		20 000 kWh	
	Excluding taxes	Including taxes	Excluding taxes	Including taxes
Denmark	**8.13**	**21.06**	**6.54**	**18.21**
Finland	6.66	8.98	3.81	5.51
Norway	8.75	12.60	5.26	8.28
Sweden	6.84	10.88	5.41	9.09
Germany (Western Zone)	11.21	14.79	6.60	8.76
	Industry consumers			
	16 GWh		2 GWh	
	Excluding taxes	Including taxes	Excluding taxes	Including taxes
Denmark	**6.13**	**10.52**	**5.64**	**9.90**
Finland	5.06	6.70	3.80	5.17
Norway	6.08	7.54	4.15	5.14
Sweden	3.88	4.85	3.75	4.68
Germany (Western Zone)	10.16	12.15	6.65	8.04

Source: Eurostat.

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match the current level; instead the cost of meeting commitments already made to turbine owners would have to be shifted onto the taxpayer, making the cost of government policy more transparent and the associated economic distortions easier to identify, being spread across taxpayers instead of electricity consumers. A revived green certificates scheme could serve as a direct instrument for promoting renewable energy, but it would need to be redesigned so that the price of the certificate be allowed to fall to zero if supply of renewable energy exceeded the target and the electricity itself be purchased at market prices. However, with an effective cap on emissions, a separate target for renewables may add complexity without providing much additional benefit. To complicate the situation, the instruments available in this area will depend on developments in the European Union and the Nordic countries. As Denmark and other national authorities develop their own programmes for addressing environmental concerns, the challenge of future integration into a common system grows larger.

In any event, despite progress made towards liberalisation of energy markets, a number of short-comings that were already identified in the previous *Survey* still require attention. In particular, the ownership structure in the sector leaves much to be desired. Although production, distribution and network companies are now legally separated, there is nothing to prevent the same owners for all these companies. In practice, municipalities remain the dominant owners, with vertical integration leading to risks of continued cross-subsidisation, and political considerations overriding economic criteria. Furthermore, the municipalities have no incentive to sell their stakes, since any realised revenue would be deducted from their block grant. Greater emphasis on benchmarking would at least help promote transparency and make it easier for Danish consumers to assess how their local government is performing in this respect.[43] In addition, although liberalisation of the natural gas market is now underway, in accordance with the EU gas directive, the pace is slow: from 2003, the 15 largest consumers will have the right to choose their supplier, with only a political agreement for further review at that point. Although the Danish National Oil and Gas state-owned enterprise (DONG) has been required to separate the operation of the transmission and storage network from the supply to customers, *de facto* monopoly conditions still prevail, as there has been no entry of alternative suppliers.

Actions to improve medium term performance

The analysis in this chapter has pointed to a number of areas where reforms would improve the way the Danish economy functions. A number of recommendations were also made in earlier *Surveys* and progress on these is summarised in Table 14 along with new recommendations arising from this assessment. Although Denmark has already made many changes in recent years, the momentum for reform has dissipated and needs reinvigorating. The new government now has an

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Table 14. **Recommendations for Structural Reforms**

The table follows up on recommendations from the 2000 *Survey* (Table 16) and summarises action taken since then. New recommendations, resulting from the analysis in this *Survey* are included as well, except those on public expenditure policies, which are summarised in Table 22.

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Reducing structural unemployment		
Reduce benefit generosity.	No action.	Consider reducing the highest statutory replacement rate and allow benefits to gradually decline with the length of the spell, for example by implementing the so-called annual income model for benefit calculation (see the previous *Survey*).
Extend the waiting period for benefits.	No action.	As a first step, extend the number of employer-paid days (currently two days).
Reduce UI benefit duration.	The 1999 labour market reform has been fully phased in as planned (restricting the benefit period to four years and imposing longer qualification periods), but no further action has been taken.	Shorten duration further and abolish benefits for non-core purposes such as holidays.
Improve the efficiency of ALMPs.	The requirement that the unemployed participate in programmes during at least 75 per cent of their unemployment period has been difficult to administer in practice and has been criticised for leading to "assembly line activation" of poor quality. A more flexible operational definition of what constitutes full-time activation is being considered with a view to preventing low-quality activation. In addition, modest administrative measures have been taken, aiming to strengthen incentives on the PES to focus more clearly on the programme quality.	Closely monitor that the greater flexibility does not in practice weaken the testing of individuals availability. Introduce competition (to the PES) for placement services and promote a more competitive environment for providing educational activation activities.
	The new government has proposed to extend the early activation requirement (after six months) currently applying to unemployed under 25 years of age to those aged 25 to 29 years.	Implement the government's proposal.

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Table 14. **Recommendations for Structural Reforms** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Increasing labour supply		
Reduce scope for early withdrawal from the labour force.	The early retirement benefit reform implemented in 1999 (see previous *Survey*) seems, up to now, to have brought about the hoped-for modest reduction of the inflow to that scheme. However, there is, as yet, very limited information on the reform's impact on the employment rate for those in the age group affected by the reform (60 to 64 years). Unemployment for this age group has not declined in parallel with overall unemployment, which might reflect, in part, "benefit arbitrage". The higher annual contributions from 35 years of age now required for earning eligibility at age 60, has discouraged around 20 per cent of those aged 35 to 39 years from paying contributions.	Continue to carefully monitor the inflow to the scheme and the labour market status of those aged 60 to 64 years. Take further steps to strengthen incentives to postpone retirement if the effects of the reform fall short of expectations or if labour market pressures intensify more than expected.
	A reform of the early retirement (disability) pension scheme has been adopted and is to come into force in 2003. The reform tightens eligibility criteria by requiring that working capacity be reduced by two-thirds (currently one-half). Those affected by this tightening are to be offered jobs on special terms instead. The reform also greatly simplifies the benefit structure.	The outcome is difficult to predict, but the success of the reform is closely linked to boosting employment opportunities for people with reduced working capacity, *i.e.* to the creation of a more *inclusive labour market*. In this context, it may prove counterproductive that the benefit level in the new system will be higher than the lowest rate in the current system.
Promote a more inclusive labour market.	As part of the early retirement pension reform, a number of administrative measures were taken aiming to boost the so-called *flexi-jobs* scheme. Furthermore, an agreement has been made with the municipalities who have agreed to reinforce their efforts to encourage the business sector to create more *flexi-jobs*. Social benefits for people made "unemployed" from *flexi-jobs* have been made more generous.	Strengthen efforts to remove barriers for establishing jobs on special terms for those with reduced working capacity. In particular, other employees need to understand and accept that, in this context, a wage subsidy does not inflict unfair competition but is a precondition for overcoming employability problems. On the other hand, it is crucial to ensure that stringent eligibility assessments for the *flexi-jobs* scheme are maintained.
Adjust the sickness benefit scheme.	The statutory 12-month limit to sickness benefit, already compromised by numerous exceptions, was further eroded by the agreements on the budget for 2001. The municipalities' economic incentives to curtail sickness benefit expenditures have been strengthened. They now pay the full costs for sickness benefit beyond 12 months of duration and 50 per cent of the costs already after four weeks (previously: eight weeks).	Introduce a waiting period and enforce the current 12-month time limit.

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Table 14. **Recommendations for Structural Reforms** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Reduce excessive time spent in the education system.	No discretionary policy changes have been implemented.	Consider introducing tuition fees for tertiary education. Reduce the period for which students can receive educational grants, and set stricter time limits for completion of higher education. Consider abolishing the voluntary tenth grade of lower secondary education.
Improve immigrant integration into the labour market.	An Integration Act came into force during 1999 and 2000. Responsibility for implementing of policies was devolved to the municipalities, and participation in three-year introduction courses was made compulsory for recipients of social transfers. The new government that came into office in November 2001 placed immigration and integration policies in a new separate ministry.	Monitor closely the municipalities' implementation of integration policies. Consider introducing a temporary in-work benefit targeted on immigrant groups facing the weakest economic incentives to seek work.
Distribution policies		
State equity objectives more clearly.	Political objectives of distribution policies are not, at present, stated explicitly. The current focus on marginal changes in the income distribution narrows the structural reform agenda without necessarily pursuing equity goals efficiently.	Welfare gains from structural reforms should be allowed, more often, to take precedence over keeping static measures of inequality at current low levels. This is particularly relevant where reforms might benefit high-income groups the most but with little, or no, effect on low-income groups.
Tax policies		
Reduce marginal personal income tax rates.	The 1999 tax reform has been implemented as planned, containing a ¾ percentage point reduction in the ordinary state tax rate in 2001 and 2002. Local governments' income tax rates rose 0.4 percentage point in 2001 and 0.1 percentage point in 2002.	Increase the income threshold for paying the top tax rate. Consider the scope for further broadening of the tax base to allow rate cuts. For example, the tax-value of deductible interest outlays remains high by international standards.
Remove asymmetry in capital income taxation.	The spread between tax rates on capital income and interest outlays has increased with the 1999 tax reform.	Align the tax treatment of positive net capital income with that of interest outlays. Further to this, reconsider the need for having a separate tax scheme for share income (dividends and capital gains) and in any case simplify the taxation of share income.
Reduce the corporate tax rate.	The corporate tax rate was reduced 2 percentage points in 2001 (to 30 per cent) following a similar cut in 2000.	The corporate tax rate is well aligned with neighbouring countries. However, there may still be some leeway for reducing tax expenditures further, thereby making additional modest rate cuts possible.

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Table 14. **Recommendations for Structural Reforms** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Safeguard effective taxation of home-ownership.	The tax-value of the deductibility of interest payments was reduced in each of the years 1999, 2000 and 2001 as part of the 1999 tax reform. The tax freeze instituted by the new government implies a small but steady annual erosion of property tax bases, as nominal tax payments are to be frozen on their 2001 levels for individual house owners.	As a minimum, increase tax assessment values in line with house price developments. Introduce a property tax on rental housing.
Housing policies		
Dismantle rent controls and extend property rights in social housing.	The new government has proposed allowing tenants in social housing to buy their apartments.	Remove the rent controls applying to private rental housing. Any induced economic rent to landlords resulting from higher property prices should be taxed away. Rent regulation in social housing should also be modified to eliminate unwarranted rent differentials within and across vintages.
Clarify governance structures in the social housing sector.	No action.	A strategy to deal with the expected wealth accumulation in the sector should be developed in order to avoid lock-in effects of capital and over-investment in housing. Increasing property rights might be one element.
Improve incentives for cost reductions in urban renewal.	No action.	Reconsider the current rules so as to avoid the perverse incentives in the present system. As a first step, implement policies that would increase cost awareness.
Reconsider the housing benefit scheme.	No action.	Benefits available to pensioners should be aligned with those for other groups.
Education policies		
Secure basic skills, in particular at compulsory level.	The number of pupils' instruction hours has increased in many municipalities with the emphasis on enhancing reading skills. This seems to have paid off in terms of better outcomes. Teaching of English has been advanced one year. The OECD's recently published PISA study indicated that, in an international comparison, Danish students have average reading skills, above-average skills in mathematics and below-average skills in natural science.	Consider making the introductory school year prior to primary school (*børnehaveklasse*) compulsory and strengthen its educational content.
	The working time agreement from 1999, which aimed to increase classroom hours and introduce performance-related pay on a small scale, has proved difficult to implement at the school level. Uncertainty about its extension has emerged due to discontent among teachers.	Ensure an increase in teachers' classroom hours towards the OECD average to reverse the trend increase in unit costs. In addition, make sure that resources are gradually shifted from the area of pre-school childcare to primary and lower secondary education in step with the entry into school-age of the big cohorts born in the first half of the 1990s.

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Table 14. **Recommendations for Structural Reforms** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Strengthen quality and cost-effectiveness in adult education and training.	No action.	Introduce sizeable user charges on adult education and training for the employed, thereby shifting the main financing responsibility to the employers. Further narrow the range of courses receiving subsidies, in particular by eliminating those with a firm-specific content.
Competition policies		
Strengthen the Competition Act.	An adjustment of the Competition Act was made in 2000, introducing merger control and modest fines for first-time violations.	The potential fines for violating the Competition Act remain modest, and an increase seems warranted.
Strengthen competition in electricity generation.	The electricity reforms approved in 1999 are being progressively implemented except for the planned green certificates for renewables energy. As a result, the Public Service Obligation (PSO) remains in place, keeping the electricity market heavily segmented and prices artificially high.	Remove the PSO and allow consumers to purchase electricity from the cheapest supplier. Convert the fixed tariff commitments given to wind-turbine owners into a diminishing fixed-term government subsidy.
Further increase vertical separation in energy distribution sector.	No action.	Further tighten the rules to reduce the extent to which municipalities can in practice continue to own production, network and distribution facilities and remove the current financial disincentives for divestment.
Go further than EU minimum requirements for deregulating the natural gas sector.	Present liberalisation opens up competition for 15 large consumers from 2003. State-owned Danish National Oil and Gas (DONG) remains a *de facto* monopoly, although it has been required to separate transmission and storage from supply.	Liberalise the market for all consumers as rapidly as possible, and expose DONG to further competition.
Utilise competition to improve resource use in road and rail transport more consistently. In particular, long-distance bus services should be deregulated.	Tendering on five specific routes has done little to promote competition, given the particular routes chosen.	Develop a comprehensive strategy for increasing competition between different modes of transport that allows effective competition on all routes and involves converting current protection granted to rail into subsidies paid for the provision of social services, where these are deemed necessary.

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Table 14. **Recommendations for Structural Reforms** (*cont.*)

Previous or new proposal	Action taken	Current OECD assessment/recommendations
Verify whether competition in the telecommunications sector strengthens further and access to the network becomes cheaper.	Consumer prices have fallen and Danish prices are among the lowest in the OECD. Interconnection agreements must be made public. The National Telecommunications Agency is developing a LRAIC model for interconnection tariffs that is planned for implementation by end of 2002. Full number portability will be achieved by 1 April 2002.	Ensure that implementation takes place.
Environment policies		
Water charges should be revised to treat users equally, and agricultural discharges should be taxed or traded under a permits scheme.	No action.	Use the environmental component of current water charges to equalise abatement costs, removing the major rebates given to industrial users. Shift to a tax on net nitrogen discharges from farms or establish a tradable permits scheme.
Carbon tax and permit trading schemes should be rationalised to reduce costs of meeting targets and develop a more integrated approach.	CO_2 cap-and-trade system is now operating for electricity producers.	Extend cap-and-trade system to cover all carbon emissions with the aim of encouraging reductions to be made where abatement costs are lowest across all sectors.
Make transport policies more consistent in encouraging emissions reductions.	Road pricing study is underway with conclusion expected in 2003.	Re-examine the taxation of car transport in conjunction with road pricing project.
Allow economic instruments to function in waste management.	No action, except for revising packaging tax to shift from being based on weight to its environmental impact.	Abolish national targets, which are based on a rigid interpretation of the waste hierarchy, and instead allow economic incentives to achieve the optimal mix of recycling, incineration and landfill.

Source: OECD.

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ideal opportunity to address the outstanding issues and put in place policies that will help to raise potential output as well as improve the efficiency with which resources are used. Increasing total hours worked, whether by expanding the number of labour market participants or by arresting or even reversing the current trend towards shorter hours per employee should be the top priority. To help set out an agenda for reform, the key specific recommendations in each area discussed are outlined below:

- To increase the *labour supply* in line with the government's medium term ambitions would require further scaling back the eligibility and/or generosity of public benefits. Discontinuing paid leave schemes and instituting measures speeding up passage through the education system would make a positive contribution. Proposed extensions to the maternity leave scheme should be resisted as they compromise the commitment to the stated objectives.

- The *tax system* could be improved in several ways. Reducing the average marginal tax rate by raising the threshold for the top income tax rate would stimulate a labour supply response with less overall budget costs than other tax reductions. The tax rate on positive capital income could be brought into line with the taxation of other capital income, to remove current distortions on the allocation of savings as well as reducing current incentives for avoidance and evasion. The currently very high registration tax on cars should be reduced and the environmental damage addressed instead by higher charges on the actual use of vehicles.

- The objective of promoting a more *inclusive labour market*, *i.e.* encouraging higher participation and employment for those on the edge of the labour market, should be pursued, but not at any cost. Boosting the *flexi-jobs* scheme further and strengthening immigrant integration into the labour market could make a valuable contribution to increasing overall labour supply. However, stated objectives are highly ambitious and considerable barriers need to be overcome. The attempts to make the *flexi-jobs* scheme attractive to individuals have prompted policymakers to establish a system with out-of-work compensation and early retirement benefit in the social system, which in many respects is parallel to the ordinary unemployment benefit and early retirement schemes. This significantly reduces individuals economic incentives and may prove to be counterproductive.

- W*ind energy* needs to be integrated into the electricity market in a way that allows the electricity market to function properly. This means removing the Public Service Obligation and allowing users to purchase electricity from the cheapest suppliers. The present level of support to

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existing turbines should be scaled back as far as possible and shown transparently as a subsidy in the government budget. Further integration of international markets should be facilitated and environmental measures should be consistent with and designed to be easy to incorporate in future European Union schemes. As a start, the CO_2 cap-and-trade scheme should be extended to cover all producers so as to minimise abatement costs.

To spur further reform, the current bias towards maintaining the *status quo* on distribution grounds needs to be overcome. To help this process, the policy debate could usefully devote more attention to assessments of the full costs and benefits of present policies and alternatives and require analyses to be made, where they are not already available. This may require some further investment in strengthening the techniques available to provide a sounder basis for advice to policymakers than is currently offered.

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III. Enhancing expenditure control with a decentralised public sector

Overview

A central feature of the Scandinavian welfare state such as Denmark's is its fine-mesh social-security safety net and the provision of a broad range of free or almost free public services. Such a system almost inevitably implies a high level of public outlays and considerable redistribution of income and consumption. The analysis of Danish expenditure policies in this chapter takes as a point of departure that the current system enjoys broad support, at least in its fundamental design. Nonetheless, an assessment of expenditure policies cannot escape dealing with issues seen by many as basic features of the welfare state. However, the principal issue faced here is not whether the level of public spending is appropriate, but whether adequate policies are in place to encourage a high level of efficiency and ensure overall spending control.

The chapter focuses on the key policy issues, whereas most factual background information is provided in subsequent annexes. However, the analysis starts out by describing the level and structure of current public expenditure in a historical and cross-country context. The higher level of total expenditure in Denmark than in most other EU countries is partially attributable to technical factors, notably the higher taxation of income transfers in Denmark. However, public consumption constitutes a larger proportion of GDP than in any other OECD country except Sweden, and the scope of redistribution through public finances is greater than elsewhere. Next, the chapter outlines existing institutional settings and policy processes, and considerable attention is devoted to intergovernmental relations. These are particularly important in the Danish context, inasmuch as local governments – counties and municipalities – have been assigned greater responsibilities and taxing powers than in almost any other OECD country.

A number of factors may combine to put upward pressure on public spending and to challenge the extensive reliance on tax financing in the future. On the revenue side, these would comprise most importantly a continued downward drift in average working hours and a demographically induced contraction of labour supply, thereby reducing the income and expenditure tax base (see Chapter II). On

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the expenditure side, the prospective rise in the old-age dependency ratio will also put upward pressure on public spending. Furthermore, the relative costs of public-sector production are likely to trend upwards, because such production is necessarily more labour intensive than its private sector counterpart (due to the so-called "Baumol disease"). An increase in relative costs could be further reinforced if the already mounting evidence of shortages of skilled labour in the public sector prompts an increase in relative wages. Finally, depending primarily on the income elasticity of demand, pressures for higher-standard services and increased coverage may prove to be considerable in many areas.

The possible quantitative significance of these pressures, let alone their timing, is difficult to assess with any accuracy, and offsetting factors may mitigate them. Nonetheless, it is reasonably safe to assume that, overall, they will slowly but steadily strain public finances. Policy measures that aim at restraining public expenditure or enhancing the efficiency of provision and administration are warranted in their own right, but they will be even more important if the underlying upward pressure on public outlays proves to be significant. Among the policy issues raised in the chapter, the two most important are reforming the framework for managing local governments and moving towards a greater reliance on competition and market signals in the provision of public services. A more consistent use of contracting out, user charges and client choice could promote greater efficiency and help contain existing and future expenditure pressures. Considering the current sound stance of public finances, failing to deal with these challenges will not risk prompting an imminent fiscal imbalance. However, to safeguard the favourable starting point and facilitate better outcomes in terms of value for money, they nonetheless deserve attention sooner rather than later. Social transfers account for almost half of gross public spending, but the main policy issues that arise in this area are dealt with in Chapter II in the context of increasing labour supply. Therefore, they are not given further specific attention in this chapter.

Public expenditure in perspective

Real public consumption has risen relatively rapidly over the past six to eight years. This has not been driven by demographic factors but reflects considerable increases in standards and coverage rates of public services (OECD, 2000*b*). The fastest-growing areas have been secondary health care, pharmaceuticals, education and childcare (Finansministeriet, 2001*c*). Nonetheless, the overall expenditure-to-GDP ratio declined significantly over the second half of the 1990s (Figure 22), as a result of dwindling transfer payments and interest outlays, the latter caused by a declining debt ratios and lower interest rates. Finally, according to the national accounts, wages have risen less than in the private sector and the deflator for non-wage public consumption has increased less than the GDP deflator (Annex II).[44] To the

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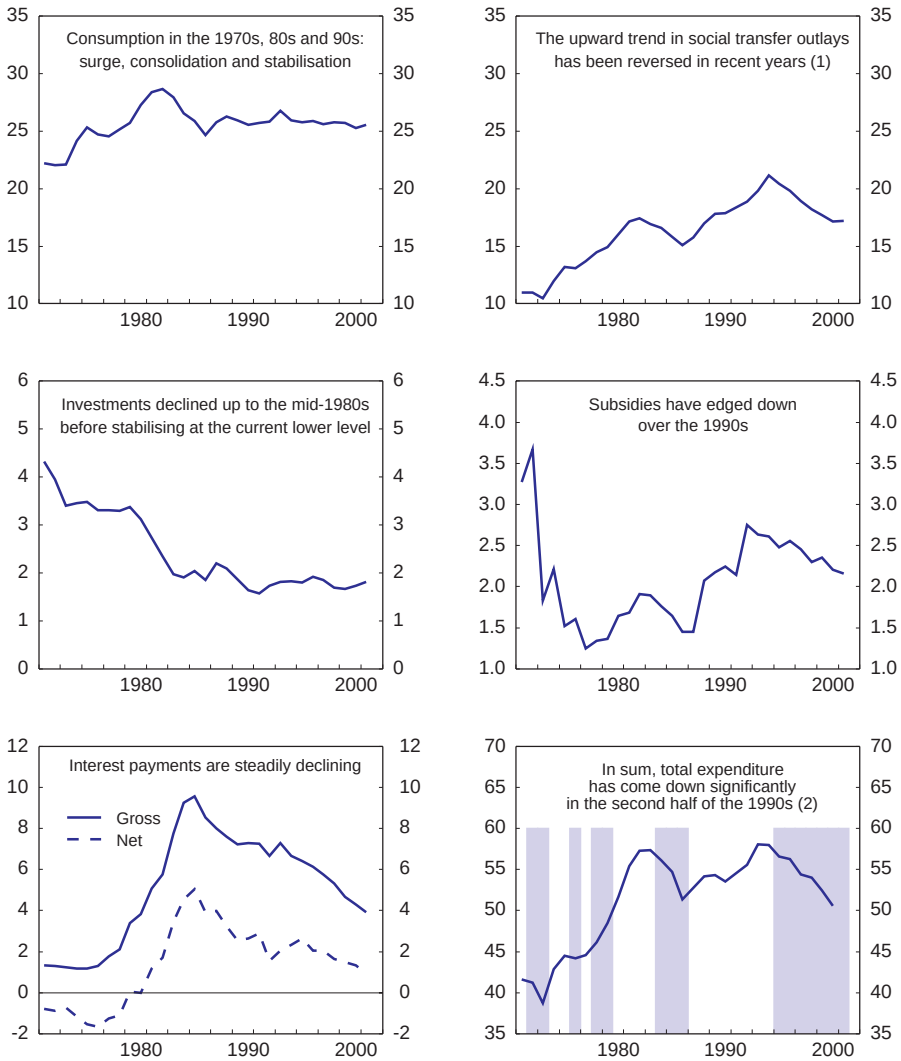
Figure 22. **Trends in public expenditure, 1971-2001**

Per cent of GDP



1. Pensions and social assistance benefits were increased but made taxable as from 1994 with little impact on net transfers. This implied a 1.9 percentage-point increase in the transfers-to-GDP ratio.
2. The shaded areas designate years of economic recovery as represented by an increasing employment rate.

Source: Statistics Denmark; OECD.

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extent that this reflects a temporary favourable development in relative prices and costs in the public sector, the lower potential for productivity increases in the provision of labour-intensive public services could cause the deflator for public consumption to increase faster than the GDP deflator in the future.

At 50 per cent, Denmark's public-expenditure-to-GDP ratio was 6 percentage points higher than the EU average in 2000 and 12 percentage points higher than the average level in the OECD area as a whole. However, accounting for the taxation of public transfers narrows these differences. Nevertheless, public consumption is higher than in any other OECD country except Sweden; at 25 per cent in 2000 its share of GDP was some 10 percentage points higher than in Ireland, the United States and Japan, while it was six percentage points above the EU average, a figure that has been fairly stable over the most recent decade (Figure 23). In contrast, social security outlays (transfer payments) are lower in Denmark than in a number of other EU countries once adjusted for the tax treatment (see Annex II for a further discussion). In particular, public pension outlays are relatively small in Denmark, reflecting their predominantly flat rate and means-tested nature. In contrast, the earnings-related elements, which are a feature of the public pension systems of most countries, are almost entirely organised in private pension funds in Denmark. At some 90 per cent of GDP, private pension funds are bigger in Denmark than in most other countries. The relatively low level of net social transfers outlays also reflects the high labour force participation rate and the below-average unemployment rate. While the level is fairly modest compared with the EU average, the Danish transfer system tends to involve more redistribution than those of other countries. This is due to the fact that whereas most EU countries' transfer systems link contributions and benefits to individuals' earnings, the Danish system is predominantly a flat rate system with significant elements of means testing (see Chapter II).

The institutional set-up and policy-making processes

Two particular features underpin the way expenditure policies are formed in Denmark. The first is the fact that coalition minority governments are the norm. The second is the extensive division of labour between the state (*i.e.* central government), counties and municipalities. Both of these elements contribute to a distinctly co-operative political climate. This section highlights the main features of the institutional settings that frame expenditure policies, while Annex III provides additional information.

The budget process

The budget cycle is determined mainly by the deadlines specified in the constitution. The budget proposal is prepared by the government between January

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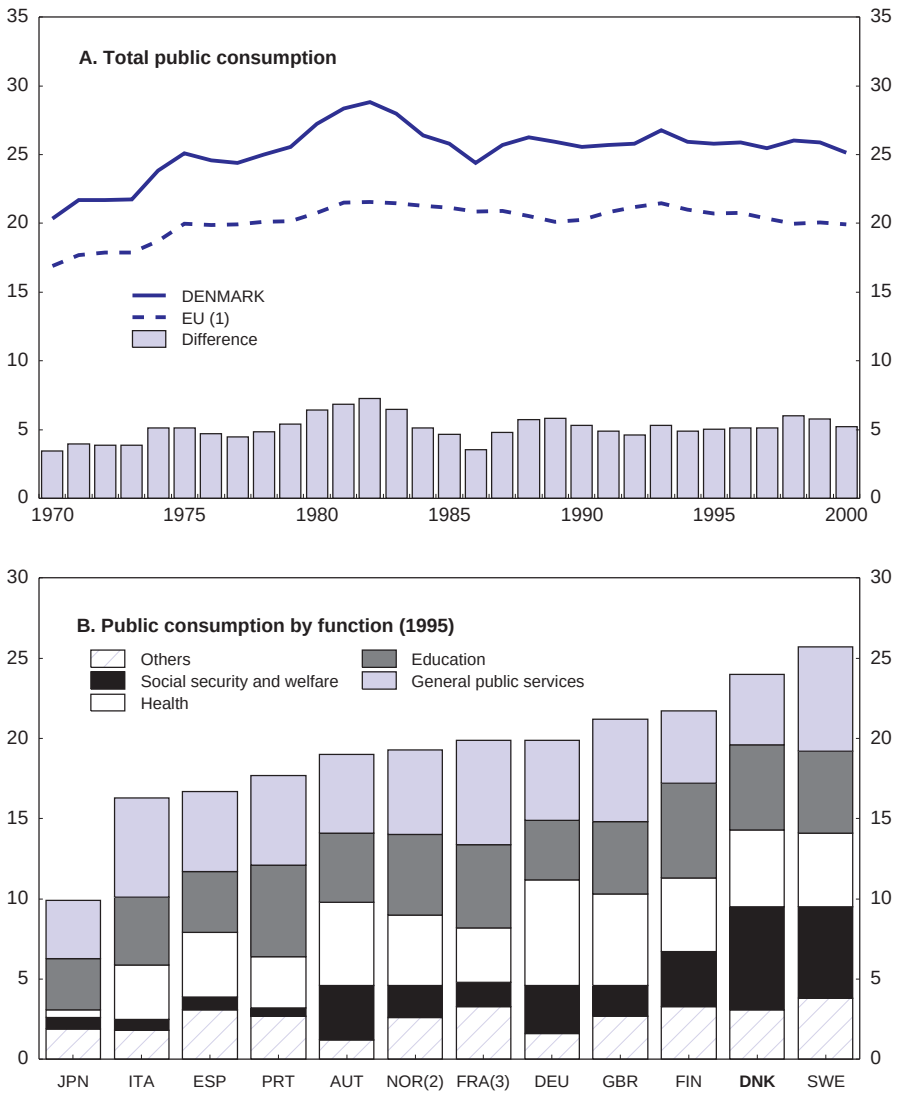
Figure 23. **Public consumption in selected OECD countries**
Per cent of GDP



1. Excluding Luxembourg and Greece.
2. 1991.
3. 1993.

Source: OECD.

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and June and presented to the parliament in late August. Discussions with opposition parties designed to build majority support for the budget usually take place in October/November. As in other countries, the adoption of the budget is also a vote of confidence. This creates a momentum for policy-making that, in the Danish tradition of coalition minority governments, is not present during the rest of the year. The August budget proposal should be seen not only as a statement of the government's priorities, but also to some extent as the strategic starting point for the subsequent political discussions, in which "new" topics may be raised by the government or opposition parties. Some of the measures required to finance new initiatives are not necessarily detailed in the budget proposal but may appear as unspecified savings or revenues to be decided upon during the negotiations. Likewise, measures may occasionally be included in the budget proposal mainly for tactical reasons and which the government is willing to "trade away". In addition, structural policy issues, which may have only limited consequences for next year's budget, are often brought up in the budget negotiation process, rather than being launched as separate discussions during the rest of the parliamentary session.[45]

While the budget proposal may not always be complete in stating the government's policy intentions it does serve an important purpose in specifying the government's intended overall fiscal stance. Although the fiscal stance in the final agreement may deviate from the budget proposal, such deviations have most often been relatively small. The specific fiscal stance measure on which the negotiations focus may vary from year to year depending primarily upon what the government considers a suitable target at the time. Since the early 1990s the so-called "fiscal effect" has become a key indicator along with the overall budget figures and indicators of the long-term sustainability of public finances taking into account the ageing of the population. The fiscal effect is based on model simulation, it measures the imputed first-year impact on GDP of discretionary fiscal policy initiatives, whether on the expenditure or the revenue side of the budget. In contrast the political commitment to keep a lid on total spending was stronger during most of the 1980s and nominal ceilings for total, non-cyclical, central-government outlays occupied a more central role in the process. In this sense, the focus in the budget process has shifted considerably from restraining total expenditures in the 1980s to a more pragmatic, macroeconomic-management approach since the mid-1990s. In this regime, slippage on the expenditure side relative to the budget proposal has frequently been tolerated if it is compensated by additional revenue so that the fiscal-effect target has been achieved. The "tax freeze" declared by the new government (see Chapter II) is likely to imply a significant change to this approach to fiscal policy making. Thus, with the "tax freeze", greater attention is needed throughout the budget process to keep spending within the originally set target, as any slippage would otherwise weaken the budget balance.

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Intergovernmental relations

The government and parliament have devolved a very wide range of responsibilities to local governments (*i.e.* counties and municipalities). Annex IV provides information on the main responsibilities of these lower levels of government as well as on their revenue structure. Local governments provide the lion's share of public-sector services and account for 70 per cent of total public consumption (Figure 24). The extensive reliance on local governments in the conduct of public-sector activities is clearly evident in an international comparison of local-government outlays relative to GDP and to total government expenditure. Denmark is second to none on both these measures. Local-government spending was 30 per cent of GDP in 1995,[46] followed by Sweden at 22 per cent, and Norway and Finland at 16 to 18 per cent. In other countries, local-government spending (excluding spending at the state level in federal countries) is often below 10 per cent of GDP. While (*ad hoc*) decisions have also played a part, the decentralisation strategy has been based on a blend of pragmatic political arguments, such as the strengthening of local democracy, and efficiency considerations. There has been broad consensus that central governing of local service provision is inferior to local governing both in administrative efficiency and in targeting services at the right people. Danish counties and municipalities have also been assigned taxing powers that are possibly greater than in any other OECD country with the exception of Sweden (Table 15). The share of own revenues in financing local governments' expenditure is around 80 per cent, with taxes making up almost 60 per cent. The right to set income tax rates is widely seen as the cornerstone of local government autonomy, although it is not considered a constitutional right as it is in Sweden.[47]

Having devolved the responsibility for most public services to counties and municipalities, the government and the parliament cannot exert direct control over total public-sector activity. While this problem is equally relevant in Norway and Sweden, those countries have put legal constraints on counties' and municipalities' right to set their own tax rates, although they were recently lifted in Sweden's case. Danish governments have largely abstained from such direct intervention. Instead, Denmark has opted for a system of formalised budget co-operation between the central government and local government associations. The roots of this system go back to the early 1970s and developed into formal annual negotiations as from 1980.

As their core elements the resulting agreements stipulate aggregate expenditure levels and tax rates, and – subject to subsequent parliamentary support – set the size of untied (block) grants. At the initiative of the central government, they have also come to play an important role in stating objectives and establishing new initiatives in individual policy areas. Thus, treatment of specific policy areas have become more prevalent in the agreements in recent years.

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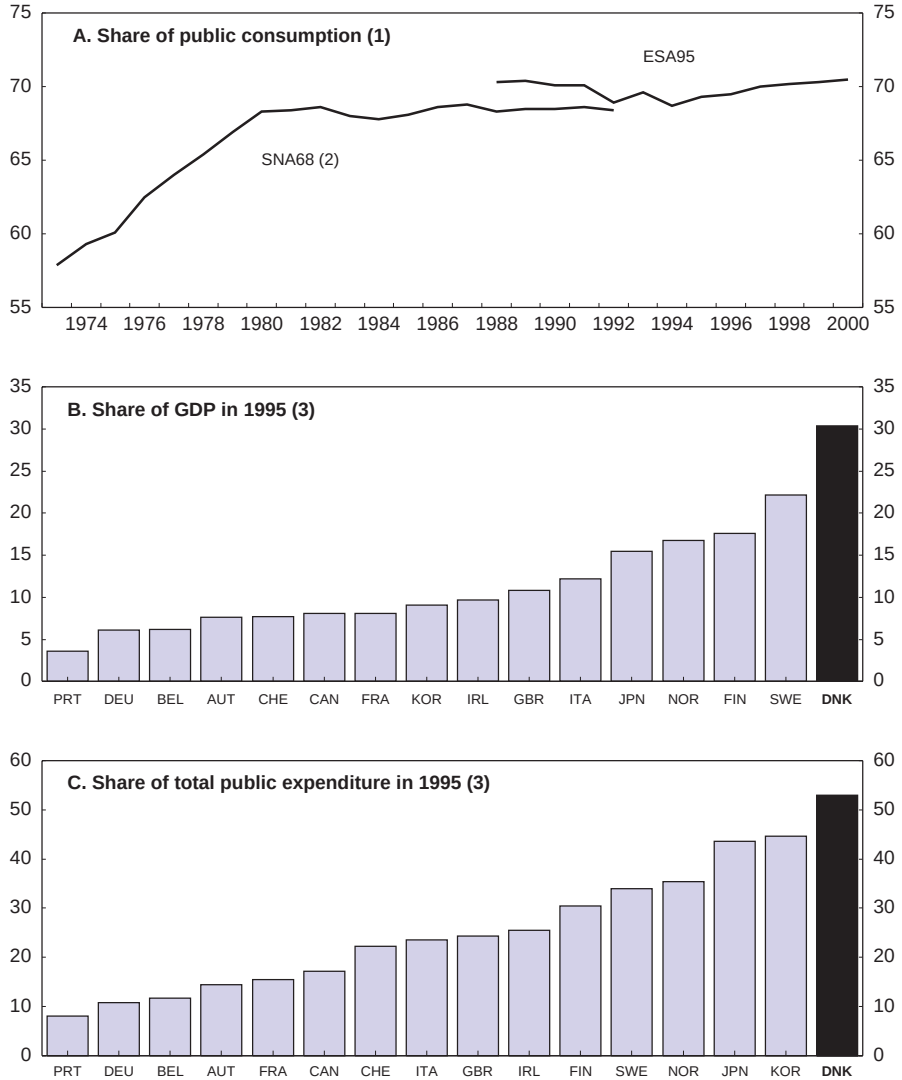
Figure 24. **Local government expenditure**

Per cent



1. Consumption by local governments as a percentage of total public consumption.
2. Data for 1977 to 1979 are interpolated.
3. The figures do not include spending by state and provincial governments in federal countries.

Source: OECD, *National Accounts*.

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Table 15. **Local governments' taxing powers in selected OECD countries, 1995**[1]

	Local government taxes relative to:		Discretion to set taxes[2]	Summary indicator of taxing powers[3]
	Total taxes	GDP		
Sweden	32.0	15.4	100.0	15.4
Denmark	**31.0**	**15.5**	**95.1**	**14.7**
Finland	22.0	10.1	89.0	9.0
Belgium	28.0	12.5	57.9	7.2
Spain	13.0	4.3	66.6	2.9
Germany	29.0	12.0	12.8	1.5
United Kingdom	4.0	1.4	100.0	1.4
Netherlands	3.0	1.2	100.0	1.2
Austria	19.0	8.3	10.0	0.8
Portugal	6.0	2.1	31.5	0.7
Norway	20.0	8.5	3.3	0.3

1. The countries are ranked in descending order according to the value of the summary indicator of taxing powers.
2. The figures show the percentage of their total taxes for which local governments hold full discretion over the tax rate or both the tax rate and the tax base. A value of 100 designates full discretion.
3. The summary indicator is the product of the ratio of local-government taxes to GDP and the degree of discretion to set taxes independently. Thus, it measures local-government taxes with full discretion as a percentage of GDP.

Source: OECD (1999*c*).

These vary in their level of detail and the degree of commitment imposed on the parties. In some cases, specific targets have been stated, for example for the development of additional childcare places and hospital operations undertaken. In other cases, they have been limited to declarations of intent and highlighting of priority areas.

The timing of the negotiations is dictated by the legal requirement that the Ministry of the Interior must inform individual counties and municipalities about the size of their block grant for the coming year by 1 July at the latest. This is to allow local governments enough time to determine their budgets by 15 October. Consequently, the agreements are typically reached in early June and the voting in the parliament's finance committee takes place one or two weeks later. The precedence has been established whereby the proposed block grant is routinely passed by a big majority in the committee if it originates from an agreement between the central government and local government associations.

As the local government associations cannot act legally on behalf of their members, the agreements are not binding for individual counties and municipalities. The strength of the agreement framework therefore relies crucially on the associations' ability (despite a lack of efficient instruments) to ensure aggregate compliance. In practice, the compliance has been far from complete (see below). The average tax rate is revealed only when the local governments submit their budgets in mid-October. Nevertheless, and despite these shortcomings, the cen-

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tral government has invested more time and prestige in the budget co-operation process in recent years.

Assessing public expenditure outcomes in selected areas

Assessing how well existing frameworks and policies function in ensuring or encouraging good outcomes is a very complex task, particularly in a cross-country context. The scarcity of unambiguous indicators of good performance as well as problems of properly accounting for differences in stated policy objectives would often render decisive judgements difficult, and an overall rating would hardly be meaningful. Instead, the following sections provide some selective analysis and assessments of performance in four areas, namely childcare, elderly care, education and health. These spending areas, which accounted for two-thirds of public consumption in 2000, have risen more rapidly over the 1990s than have other spending components, and are high by international standards. In large part, this is the result of a deliberate policy to facilitate high female labour force participation (Figure 25). Thus, care and support functions that were earlier attended to within the families have increasingly been taken over by the public sector. While this is a trend that has been visible in many other countries as well, it manifested itself earlier and more clearly in the Scandinavian countries than elsewhere.

Childcare spending surged in the 1990s

In principle, childcare services are not rationed; the objective is that all parents, regardless of their work status, have a right to day care places on demand. Public-sector outlays on day care are high by international standards, owing both to high staff-ratios (and, therefore, unit costs) and broad coverage. At 69 per cent, the coverage rate, *i.e.* the provision of day-care places as a percentage of the number of pre-school children, was significantly above the levels in other European countries already in 1996 (Table 16), and it has risen since then to 75 per cent in 2001. In particular, the coverage rates for children aged six months to three years are higher in Denmark than elsewhere. Consequently, gross public expenditure for formal out-of-home day care was also higher than elsewhere. However, it should be observed that the total leave period and/or tax credits for at-home care are less generous than in many other European countries, including Sweden, Finland, France and Germany.

While a gradual shift to cheaper modes of care has facilitated a decline in real unit costs over the 1990s, real public spending has nonetheless increased as much as 5 per cent per year on average over the past decade. Demographic needs explain some of the increase, but, in addition, coverage has broadened in two respects. F*irst*, the proportion of children aged six months to nine years attending childcare facilities has risen from less than 60 per cent in the beginning of the decade to more than 80 per cent in 2000, owing in part to a rapid expansion of

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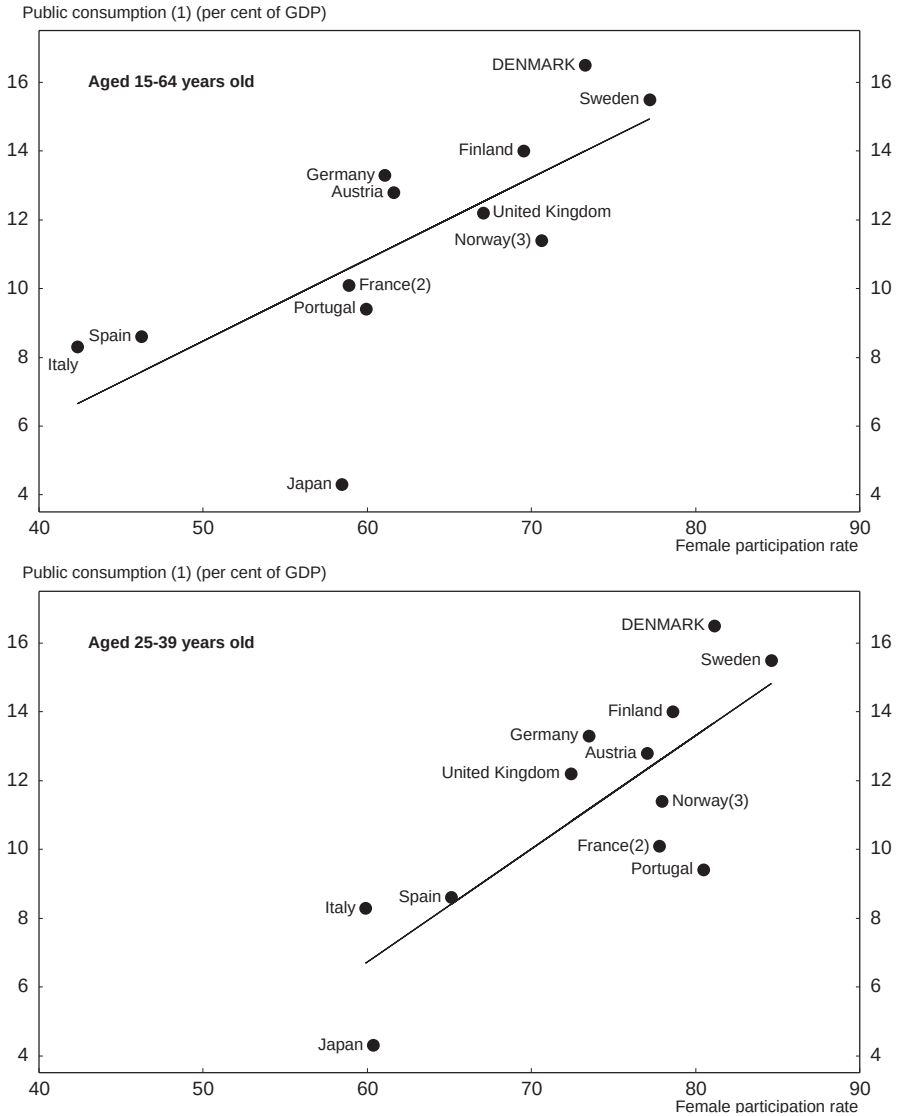
Figure 25. **Public consumption on education, health and social services and female participation rate, 1995**



1. On education, health and social services.
2. 1993.
3. 1991.

Source: OECD.

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Table 16. **Out-of-home childcare in selected European countries, 1996**

	Denmark	Sweden	Finland	Germany	France	Netherland	United Kingdom
Gross public expenditure[1]	**2.0**	1.3	1.1	0.4	0.9	–	1.3
Provision of full-time places[2]	**69**	61	42	38	60	28	41
Recommended staff ratio[3]							
0-2 years	**1:3**	-	1:4	1:5-7	1:5-8	1:4	1:3
3-6 years	**1:6**	1:5	1:7	1:10-14	1:27	1:8	1:8
Unit costs[4]	**5 664**	3 210	3 665	4 648	3 487	3 630	4 910

1. Expenditure for formal day care. per cent of GDP.
2. Per cent of pre-school children.
3. Staff ratios in day care centres, except for France (3 to 6 years) where data refer to nursery school, which almost all children in this age group attend. In a few cases, the staff ratios reported in the table do not correspond exactly to the age group. Full details are available in Rostgaard and Fridberg (1998).
4. Expenditure per child in pre-primary centre-based care on public and private institutions in (US dollars converted using PPPs). Based on full-time equivalents.

Source: OECD Social Expenditure Data Base, OECD (2001*a*) and Rostgaard and Fridberg (1998).

after-school services for children between six and nine years of age. A broadly similar trend has been observed in most other countries for pre-school care but not for after-school facilities (OECD, 2001*b*).[48] *Second*, part-time day care places for children of pre-school age have largely been closed down and replaced by more expensive full-time places, thereby implying a considerable standardisation of services. The surge in demand and provision of childcare is partly a response to a reduction of user charges in the early 1990s[49] and the fact that these do not increase in accordance with the number of hours per week children actually attend the facilities. While childcare subsidies are generally economically justifiable on grounds of significant externalities (see, for example, Blomquist, 1999), the surge in childcare spending over the past decade has not been accompanied by higher labour force participation. On the contrary, participation rates among men and – in particular – women aged 25 to 39 years have declined in the same period (OECD, 2000*b*). In part, this is attributable to the introduction in 1994, of a parental leave scheme in addition to maternity leave. But in itself, the parental leave should have reduced the demand for out-of-home childcare. In sum, the current extensive provision is likely to result to some extent from excessive demand. This could probably be reduced by redesigning user charges to reflect on the number of hours of attendance, although marginal prices should not be set so high that they risk encouraging full-time secondary earners to reduce working hours.

Elderly care would benefit from introducing demand signals consistently

Care for the elderly is a municipal responsibility and mainly comprises nursing homes and personal and practical home help (the latter being mainly cleaning, doing the laundry and shopping). Denmark has successfully pursued a

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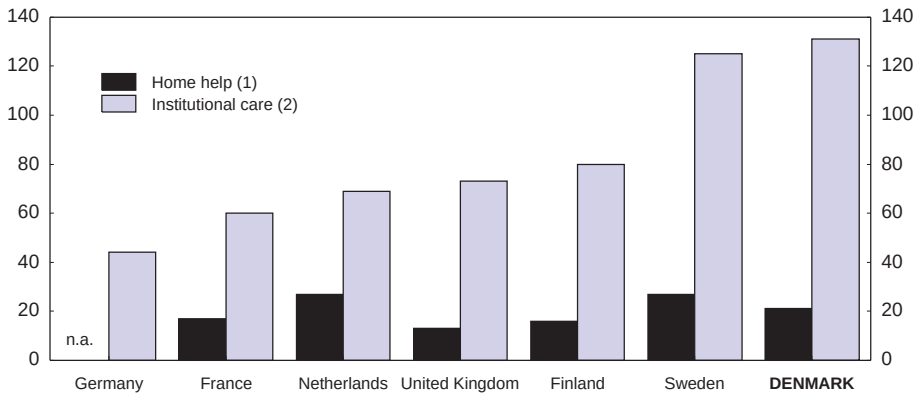
policy of making it feasible for the elderly to stay as long as possible in their own homes. The number of places in nursing homes has been reduced over the past 20 years, while sheltered accommodation and home help services have increased in scope. Meals aside, all services are provided free of charge. Although client-choice models have been introduced in a few places, the vast majority of municipalities provides the services themselves. Spending control is exercised exclusively through staffing policies and discretionary assessment of need. The move from nursing homes to cheaper services has contained unit costs, and the eligibility criteria implicitly embodied in the assessment of need have generally been tightened in recent years. Total spending has remained stable as a percentage of GDP over the past decade, but at a high level by international standards, in large part because the public sector had already assumed much of the responsibility for elderly care in the 1960s and 1970s. Moreover, nursing home staff ratios are much higher than elsewhere in Europe, except Sweden (Figure 26).

While spending control *per se* is thus far less of a concern for elderly care, it may increasingly become so in the future. The news media have reported concerns that legitimate demand is being suppressed and that quality is suffering. While a thorough assessment of such claims is beyond the scope of this chapter,[50] it seems safe to conclude that clients do not benefit fully from the resources allocated, as absenteeism is high. Moreover, it is clear that the sector is extremely

Figure 26. **Full-time staff of home help and residents in institutional care**

For 100 recipients, latest available years



1. Finland, France and the Netherlands: 1996; United Kingdom: 1995; Germany and Sweden: 1994; Denmark: 1992; figures for institutional care only include municipal nursing homes.
2. Finland: 1996; France: home nursing 1996; England and the Netherlands: 1995; Denmark: 1992; Sweden: 1989.

Source: Joint Committee on Taxation (2001*b*).

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supply-side controlled, with market signals and user choice being negligible. Competition is weak or non-existent, and political resistance against user charges is widespread and profound, not only for care services but also for practical help. Moreover, while the municipalities have a legal obligation to provide care and support to those in need, the assessment of need itself is undertaken by the individual municipalities with few restrictions.

It may prove difficult to sustain the strict supply-side rationing in the longer term, and policymakers should allow demand signals to play a greater role. At the same time, abandoning the municipalities' obligation to provide practical help could free resources for personal care. While assessment of need is a necessary tool for containing public spending, the case for using it as a means to curb total demand is weak. More scope for client choice both of providers and service standards, would in itself further welfare, but to limit a surge in demand it could be accompanied by private co-payment either as individual user charges or *via* insurance schemes.[51] Indeed, elderly care lends itself well to private insurance solutions. Such insurance schemes could finance supplementary services topping up the basic tax-financed provision but would need to be under public regulation. One possible advantage of insurance schemes could be greater transparency by stating entitlement criteria more explicitly. At present, effective eligibility criteria may vary considerably across, or even within, municipalities. From the users' viewpoint, this is both arbitrary and non-transparent.

The new government has outlined a reform of elderly care policies in its "working programme". The main thread is to significantly widen the scope of client choice once the eligibility for care or practical help has been positively assessed. This comprises free choice of nursing home or sheltered accommodation (also outside the municipality of residence) and the right to be accompanied by one's spouse; free choice – both for personal care and practical help – between municipal services and private providers (persons or companies); and greater scope for individuals to determine the content of practical help. These are laudable intentions. However, the more extensive rights may also lead to greater demand pressure and user charges should be introduced to prevent a rise in spending. Moreover, to make client choice a reality, it is important that adequate private supply is encouraged, one precondition being that the procedures for certifying private companies are kept as simple as possible.

Evolution of health care policies

From a level of 8 per cent of GDP in 1980, public expenditure on health had fallen to 6¾ per cent of GDP by 1998. While the 1980 level was distinctly high by international standards, the current level is comparable with those in most other OECD countries (Table 17). It should be observed that differences in countries' demarcation of health expenditure might to some extent distort the compar-

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Table 17. **Public expenditure on health, 1998**
Per cent of GDP

	Public expenditure on health[1]	National accounts consumption of health services			
		Health related transfers to households[2]	Public sector[3]	Private[4]	Total
Australia	5.9	2.1	3.0	5.1	8.1
Austria	5.8	0.1	5.5	2.2	7.7
Belgium	7.9	–	–	7.7	–
Canada	6.6	–	6.6	2.4	9.0
Denmark	**6.8**	**0.0**	**5.0**	**1.5**	**6.5**
Finland	5.3	1.0	4.3	2.8	7.1
France	7.3	6.6	3.4	6.2	9.6
Germany	7.9	–	6.4	2.1	8.5
Ireland	4.8	0.5	4.4	2.0	6.4
Italy	5.7	1.9	3.4	4.1	7.5
Japan	6.0	4.9	0.4	6.3	6.7
Netherlands	6.0	6.9	–	5.7	5.7
New Zealand	6.2	2.2	2.8	5.1	7.9
Norway	7.4	–	5.7	1.3	7.0
Portugal	5.2	–	3.3	3.4	6.7
Spain	5.4	1.3	4.3	3.2	7.5
Sweden	7.0	–	4.6	2.0	6.6
Switzerland	7.7	0.6	1.5	7.8	9.3
United Kingdom	5.6	0.1	5.3	1.0	6.3
United States	6.1	4.7	1.3	12.4	13.7
Country mean	**6.3**	**2.4**	**4.0**	**4.0**	**7.8**

1. As defined in the OECD health data base.
2. Finland, Ireland and the Netherlands (1997); Spain and Japan (1996); Italy (1995); France (1993).
3. Finland and Ireland (1997); Australia, Japan and Norway (1996); Canada, France, Italy and Sweden (1995); Spain (1994); France and Portugal (1993).
4. Australia, Canada, Finland, Ireland and Spain (1997); Belgium, France, Germany, Italy, Japan, Norway and Sweden (1996); Portugal (1995).

Source: OECD Health Data Base.

ison. Two features of the Danish health care sector have been particularly important in restraining expenditure.

- *First*, the gatekeeper function served by general practitioners prohibits or significantly restricts direct access to other medical services.
- *Second*, the sector's vertical integration is stronger than in most other OECD countries. The full responsibility for planning, operating, financing and economic management of all essential parts of the sector is concentrated at one level of government – the counties (Pedersen, 1999). For example, the agreements on general practitioners' remuneration is negotiated by the association of county councils, who also own, operate and finance hospitals.

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This strong vertical integration has helped facilitate expenditure control, for example through budget envelopes for individual hospitals. No individual user charges apply for primary and secondary health services, and user choice in the latter is limited. Patients have had the right to choose hospital services outside their own county since 1994, but such factors as insufficient information and strong preferences for treatment at the local hospital have limited the exercise of that privilege. The new government has proposed to extend the patients' choice to include private hospitals, which should help to encourage more competition and efficiency. However, at the same time it weakens the vertically integrated structure (because the counties would no longer own and operate all hospitals) and it could prove to be difficult to reconcile with the budget-envelope approach (see below).

Meanwhile, the supply-side-governed hospital sector needs to respond, if only indirectly, to demand signals in order to contain strains that might otherwise arise from a mismatch between demand and supply. Increasing average waiting times in the first half of the 1990s have led policymakers to channel additional resources to the hospital sector in recent years, and the decline in total expenditure since 1980 has bottomed out. However, despite rapid increase in the number of hospital treatments, average waiting times have not yet started to shorten. In 2000, the average waiting time for elective surgical overnight-stay patients was around 100 days (Ministry of Health, 2001). Maximum waiting periods have been guaranteed for an increasing number of treatments.

Most recently, the new government has proposed allocating an additional DKr 1½ billion to the health-care sector in 2002 and giving all patients the right, as from 1 July 2002, to seek treatment at a private or foreign hospital, if treatment has not been offered by the public health-care system within a period of two months. Considering the highly strained capacity situation in the sector, patients will probably have to be willing to accept treatment abroad if effective waiting times are to be reduced significantly in the near term. Even if productivity is improved, the capacity of the sector needs to be expanded to gradually achieve a significant shortening of effective waiting periods. While the size of the implied increase in spending is uncertain, the basic point is that a commitment to a general maximum waiting period would imply *de facto* a big step in the direction of linking, more formally, appropriations to "demand" with accordingly higher budget costs. However, it is a question whether the capacity can realistically be expanded without an increase in real unit costs. In particular, it has been difficult for a number of years to attract new staff on the scale required just to compensate the annual outflow into retirement and other activities. While this has probably multiple explanations, relative wage levels could easily be one of them.

A sign of potential strain in the hospital sector is the apparently growing interest in private health-insurance schemes financed primarily by employers. Such medical insurance has hitherto been discouraged by insufficient private sec-

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tor capacity and the absence of any public subsidies to treatment in private hospitals. While these remain important barriers,[52] evidence is spreading that private funds are increasingly available to pay the costs of overcoming them, at least for certain groups of employees or citizens. It is premature to assess how big a market could evolve in the future, but policy makers could face a difficult choice. On the one hand, maintaining strict expenditure control may encourage the emergence of a supplementary private insurance market and a growing private hospital sector and thus compromise the traditional objective of "free and equal access". On the other hand, adding enough resources to the public hospital sector to satisfy the demand of those groups who are most willing to pay, thereby discouraging supplementary private solutions, may prove to be expensive.

Education is costly compared with the outcome

At 8.3 per cent of GDP in 1998 Denmark's total spending on education is slightly above the levels in Sweden and Norway and significantly higher than in any other OECD country (Table 18). The higher level of public educational expenditure can be put down to four main factors. *First*, unit costs are very high by international standards in primary and lower secondary education. *Second*, outlays for adult and vocational training are distinctly high, reflecting the priority given to active labour market policies as well as remarkably generous subsidies to leisure time education and employers' training of their staff. *Third*, while private-sector co-payments in tertiary education are significant in a number of countries, they are negligible in Denmark. Hence, education expenditure is largely publicly provided and tax financed. *Finally*, as in the other Scandinavian countries, the student grants scheme is very generous, thereby contributing to the high spending level, although this does restore incentives to invest in human capital, which are depressed by the compressed wage structure.

A notable feature of Danish education policies is the relatively high priority given to primary and lower secondary education relative to tertiary education:

- Primary and lower secondary education are costly (relative to GDP) by international standards, but the greater resources are not matched by correspondingly better schooling outcomes.[53] The high spending level is mainly due to high unit costs, as Denmark has the lowest student-to-staff ratio in the OECD area. However, this is not reflected in higher annual instruction hours and only partly in smaller class sizes. Instead, it is because teachers teach fewer classroom hours than elsewhere – this has been recognised by policy makers as unwarranted. Faced with the prospect that the parliament would otherwise impose legislation, the teachers' union agreed to new work time rules as part of the 1998 collective agreement. But the new rules intended to increase the average number of instruction hours do not yet seem to have delivered on this count.

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Table 18. **Spending and unit costs at various levels of education**

	Public spending				Total spending	Unit costs		
	Primary and Secondary[1]	Tertiary	*of which*: Student grants	All levels of education[2]	All levels of education[2]	Primary	Secondary	Tertiary
	Per cent of GDP					Expenditure per full-time student relative to GDP per capita[3]		
Australia	3.5	1.2	0.1	4.8	5.9	16.0	24.0	48.0
Austria	4.0	1.6	0.2	6.3	6.5	26.0	35.0	48.0
Belgium	3.5	1.1	0.2	5.2	–	–	–	–
Canada	3.7	1.8	0.3	5.7	6.3	–	–	–
Denmark	**4.9**	**2.2**	**0.7**	**8.3**	**8.4**	**26.0**	**28.0**	**37.0**
Finland	3.8	2.0	0.3	6.2	–	21.0	23.0	34.0
France	4.2	1.0	0.0	6.0	6.3	18.0	31.0	34.0
Germany	3.0	1.1	0.1	4.6	5.6	15.0	27.0	41.0
Greece	2.3	1.1	0.1	3.5	4.8	17.0	23.0	29.0
Ireland	3.3	1.1	0.0	4.5	4.9	12.0	17.0	38.0
Italy	3.5	0.8	0.1	4.9	5.1	26.0	29.0	28.0
Japan	2.8	0.4	0.0	3.5	4.4	21.0	24.0	41.0
Netherlands	3.1	1.4	0.2	4.9	5.0	15.0	21.0	44.0
New Zealand	4.9	1.8	0.7	7.2	–	–	–	–
Norway	4.6	2.0	0.6	7.7	7.8	22.0	28.0	42.0
Portugal	4.3	1.0	0.0	5.7	5.8	21.0	29.0	–
Spain	3.3	0.9	0.1	4.5	5.2	19.0	27.0	30.0
Sweden	5.3	2.1	0.6	8.0	8.2	26.0	26.0	61.0
Switzerland	4.1	1.1	0.0	5.5	6.0	24.0	34.0	61.0
United Kingdom	3.4	1.1	0.3	4.9	5.2	15.0	24.0	45.0
United States	3.4	1.3	0.2	5.1	6.7	19.0	24.0	61.0
Country mean	**3.8**	**1.3**	**0.2**	**5.6**	**6.0**	**19.9**	**26.3**	**40.1**

1. Includes post-secondary non-tertiary education.
2. Includes undistributed outlays and advanced research programmes. Undistributed outlays for programmes that do not conform to international classifications are above average in Denmark. This reflects high spending on leisure-time education as well as adult and vocational education partly in the context of active labour market policies.
3. Multiplied by 100. Includes both public and private institutions except for Austria, Norway, Italy, Portugal and Switzerland where data refer to public institutions only and Greece and United Kingdom for which they concern public and government-dependent institutions.

Source: OECD (2001*b*).

Moreover, the teacher's union has indicated that it may discontinue the agreement when it comes up for renewal in 2002.

– For tertiary education, total expenditure relative to GDP is close to the unweighted average level in the countries reported in Table 18 but lower than the weighted OECD total, which is boosted considerably by the high spending level in the United States. This reflects above-average public spending in Denmark and below-average private funding.

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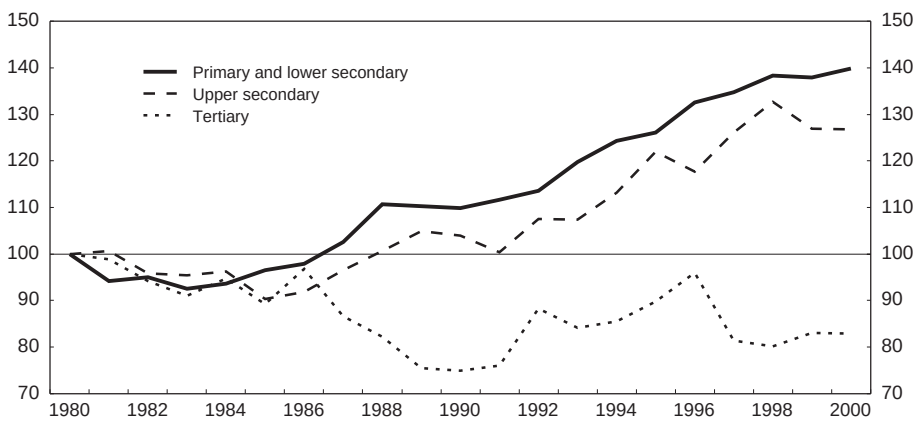
While enrolment rates in tertiary education are fairly high, unit costs are relatively low.

The current composition of education expenditure results, in part, from an upward trend in unit costs within primary and lower secondary education since 1980, combined with a fall for tertiary education (Figure 27). This suggests asymmetric behaviour, at least for primary and secondary education: total spending in this sector tends to adjust upwards when the number of students rises, but fails to adjust downwards when numbers decline (Finansministeriet, 1999*a* and Heinesen, 2000). This difference is likely to be the combined result of several factors, two of which are mentioned here.

– *First*, most staff in the primary and lower secondary sectors are on permanent contracts, and the degree of unionisation is very high. In contrast, a large part of the teaching in the tertiary sector is undertaken by external part-time teachers, typically on temporary contracts. These differences may reduce flexibility and the variability of total costs in the primary and lower secondary sector (at least in the short run) because of union resistance against spending cuts.

– *Second*, the responsibility for primary education lies with the municipalities, upper secondary education with the counties, and tertiary education

Figure 27. **Real unit costs at various levels of education**[1]
1980 = 100



1. Unit costs deflated by the index used for constant-price calculations by the Ministry of Finance in the Budget. The main input to the index is the evolution of public-sector wages.

Source: Ministry of Finance.

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with the state. Unit costs have increased most at the most decentralised levels of government and least at the central level. It seems plausible that client involvement and awareness is greater at the local level, which tends to strengthen the position of spending advocates.

A separate issue is adult and vocational education, which comprises courses and formal education for the unemployed and training of employees, as well as leisure time education. At more than one per cent of GDP, public expenditure is significantly above that in other countries.[54] This reflects political preferences and has not been justified on economic arguments, for example from cost-benefit analyses. While public spending in this sub-sector is prevalent in most countries, the Danish system is distinctly generous on two counts.

– *First*, subsidies for leisure-time education are significant both in coverage and on a per-student basis. While unit costs vary significantly between the specific types of education, they are generally very high compared with other types of education. In "folk high schools",[55] for example, the annual public unit cost of a full-time student is 25 per cent above the average level for tertiary education, and only 10 per cent below the most expensive types of tertiary education.

– *Second*, Denmark stands out as virtually the only country in which recurrent training of the employed enjoys non-negligible public subsidies (OECD, 2001*c*). In addition to often paying the full costs of providing vocational training courses even for people in employment, the government also pays a benefit to the participants in such courses (thereby freeing the employer from the wage costs) as well as compensation for transportation costs. In some cases, economic externalities may be large enough to justify some subsidies, but in general these payments should be cut back significantly and, as a minimum, sizeable user charges should be levied.

Challenges and scope for changes in policy frameworks

This section provides a discussion of the scope for adjusting current policy frameworks in ways that may further the broad and general objective of optimising the use of public funds, *i.e.* providing high-quality services and income support to the right people at the lowest possible cost, thereby mitigating, as much as possible, the distortionary effects from the tax financing. The focus is largely on institutional adjustments and cross-sector instruments (in contrast to the above sector-specific policies). The discussion will proceed in four stages: reforming the budget process; managing local governments; encouraging competition and extending market signals; and enhancing administrative efficiency.

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Reforming the budget process

The main deficiency of the budget process (which is described in more detail in Annex III) is the absence of a binding expenditure ceiling and the resulting risk of discretionary slippage in the political discussions that take place between the presentation and the adoption of the budget. The fiscal balance, as such, has not been a big concern, as additional spending has typically been financed by higher revenues. But no firm mechanisms have acted to prevent a ratcheting-up of both expenditure and taxes. The multi-annual budget framework has so far been of little use in this respect, as amendments to the budget are typically added to the ministerial expenditure ceilings of subsequent years. On several counts, in particular for entitlement outlays, the three budget forecast years are largely a mechanical prolongation. The multi-year budgeting framework should be improved to facilitate early identification of problems and better integration with the overall medium-term economic strategy.

Another concern is that the state budget is adopted after those of local governments, but nonetheless it has frequently contained last-minute amendments that impact on their economic situation. This weakens the transparency of the budget process for the general government sector as a whole. One way of addressing both problems would be to have the parliament vote on the level of total expenditure for the coming fiscal year early in the process.[56] In addition to furthering transparency and parliament involvement in the budget process this would strengthen its top-down character and weaken the current "patchwork-approach" to policy making.[57] Such a vote should be taken before the commencement of the negotiations between the central and local governments, and the ceiling should include central-government grants to local governments. While the details would obviously have to be analysed further, a stronger political commitment is the prime precondition for an expenditure target to be efficient. The new government has stated its intention to restrain public expenditure, and the declared "tax freeze" is intended to reinforce that commitment. Indeed, the pledge to not raise any taxes would put the fiscal balance at greater risk unless expenditure is also reined in more tightly. Nevertheless, instituting an overall expenditure ceiling would further underpin a tight spending stance and safeguard the fiscal surplus.

Managing local governments

The merits of a highly decentralised public sector may be compromised if governance structures and the regulatory framework are not well aligned. One challenge is that the central government cannot directly control the implementation of many sectoral policies, nor total public spending, but must act indirectly through regulation of local governments' activities. Moreover, if economies of scale are significant, it would be important for overall efficiency to match the

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Box 6. Micro-managing local governments

The annual agreements with local governments have become steadily more detailed over the past decade, both specifying desired policies and outcomes in various sectors and in nailing down technical details in the text itself. This could be interpreted as indicating a fundamental conflict between local government autonomy and the central government's ambition to control policy developments even in areas that have been devolved.

In the health care sector, the government has increasingly taken over formulating policies and determining priorities. In particular, this is manifest in the national programmes in specific areas such as cardio-vascular disease, cancer and psychiatric illness. On the central government's initiative, the 2002 Agreement with the counties stated the objective of reducing maximum waiting periods for all hospital treatment to three months over a three-year period. A few very specific programmes, for example, screening for breast cancer, have been imposed on all counties. The commitment to reducing waiting periods has been reinforced by the new government's client choice programme described above.

The fiscal bill in recent years has typically included initiatives with consequences for the lower levels of government, even in allocating additional funds to services for which local governments are responsible. As agreements on the fiscal bill are reached after local governments' budgets are adopted, this affects the transparency of the whole budget process.

Legal regulation is perceived to have increased in scope and detail. One of the most obvious examples is the Social Services Act of 1998, which strengthened appeal possibilities, introduced user councils in new areas and obliged local governments to draw up service declarations to a greater extent than before. The aim of this line of policy, which has been taken furthest within elderly care, is to clarify and ensure individuals' legal rights, contain expectations at realistic levels and strengthen transparency in general. While these are laudable objectives, they may unintentionally add to demand-side pressures rather than curb them.

devolution of responsibilities with the size structure of local governments. The following sections provide a discussion of some of the issues in intergovernmental relationships where policy reforms might be warranted.

Unclear governance structure weakens accountability

Following almost three decades of continuous decentralisation, some reversal is apparent, although only a few responsibilities have formally been moved back to the central government. The government and parliament are increasingly engaging in setting standards and micro-managing local governments' implementation of sectoral policies (Box 6). This reflects, *de facto*, less willingness

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by central-government politicians to accept variations in service levels and standards across municipalities. Such differences have indeed narrowed.

Despite the much greater involvement of the central government in many spending areas, local governments' taxing powers have remained untouched, and the framework for economic management of counties and municipalities have been left largely unchanged for more than a decade. More direct and thorough central government control risks compromising the logic of the decentralised system in two important ways.

- *First*, some of the potential benefits – richer local democracy and adjustment to local needs and preferences – are obviously weakened the more the central government interferes, and the case for local-government tax financing is weakened. The example of secondary health care may serve to illustrate this; effectuating the government's client choice programme basically implies that the level of activity will be determined by what is required to comply with a general maximum waiting period of two months. This leaves little scope for the counties to prioritise resources and it is not obvious, therefore, that the funding is best placed with the counties.
- *Second*, accountability is likely to suffer. Repeatedly, local governments have blamed overruns on agreed spending levels and tax rates on centrally imposed regulation and objectives; and equally often parliamentarians have criticised local governments for not meeting centrally initiated objectives in certain spending areas as well as for not complying with the agreed tax rates. By increasingly micro-managing on the expenditure side but leaving taxing powers unchanged, the central government has effectively separated the discretion to decide policies and service levels from the responsibility to finance spending.

There is a strong need for realigning these two powers and for clarifying the responsibilities between the various levels of government more generally. In those areas where the central government specifies standards and imposes detailed regulation, it might as well also take over the financial responsibility and compensate local governments through direct reimbursements. Conversely, where local-government tax financing is deemed warranted, the central government should abstain from excessive micro-management.

Overall expenditure control needs to be enhanced

As local governments' spending accounts for more than half of total public sector outlays, controlling their aggregate expenditure is of paramount importance. However, the formalised budget co-operation between the central and local governments has failed to deliver on this count. Local governments have typically responded to negative deviations from their budget targets by raising revenues

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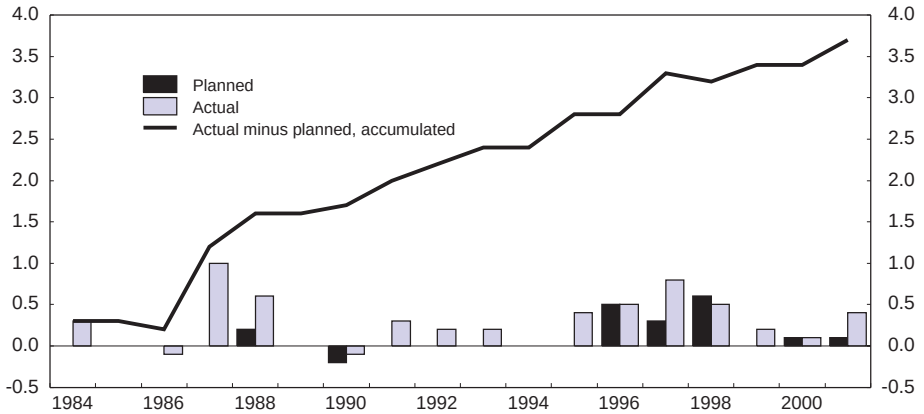
Figure 28. **Planned and actual income-tax rate increases by local governments**[1]

Percentage points



1. "Planned" refers to the annual agreements between the central and local governments, except in 1987 and 1988, where targets were set unilaterally by the central government.

Source: Ministry of the Interior; Statistics Denmark; Blom-Hansen (1996).

more than cutting expenditures, and to positive surprises by increasing spending more than reducing taxes (Rattsø and Tovmo, 1997). Practically all annual agreements between the various levels of government since 1984 have included targets for the aggregate tax rate. Although some increases were accepted in the second half of the 1990s (primarily at the county level), the agreed levels of block grants from the central government have typically assumed unchanged tax rates. However, the actual outcome has been a steady upward trajectory for both counties and municipalities. The actual changes in tax rates have exceeded agreed changes in 11 out of the past 18 years (Figure 28). The cumulative increase in local governments' income tax rates in this period has been 5 percentage points of which only 1¼ percentage points were foreseen in the agreements themselves.[58] While local governments obviously bear a substantial part of the responsibility for this outcome, the unclear governance structure, which falls back on the central government, also contributed.

An innovation in the process of budget co-operation took place in 1998 with the four-year agreement between the central government and the National Association of Local Governments. The agreement was an attempt to focus attention more directly on expenditure developments, both on the side of the municipalities and amongst central government spending ministers. It set a 1 per cent target for the acceptable annual rise in real operating costs, *i.e.* mainly public con-

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sumption, and specified the corresponding expenditure levels for each of the years 1999 to 2002.[59] In addition, quarterly follow-up meetings between the parties were instituted to enable early action in case of deviations from the set course. However, the average increase in the years 1999 to 2001 was 1½ per cent and, against the backdrop of a generous agreement, the municipalities' budgets suggest a further real increase of around 1¾ per cent in 2002.

At least two problems arise from the unplanned upward drift in local governments' income tax rates. *First*, it is fundamentally unsound to maintain a system that is not complied with; this weakens the central government's credibility and it raises moral hazard problems. *Second*, individual local governments are unlikely to take into account the national externalities resulting from higher tax rates, *i.e.* distortion of incentives to work and educate.

By not acting in response, the government and the parliament have *de facto* given their tacit consent to non-compliance. It is the combination of an unclear governance structure, an atomised decision process and extensive taxing powers for individual local governments that has led to weak aggregate expenditure control. While the right for local governments to set their own income tax rates freely is seen by many as the epitome of local-government autonomy, restricting this right could, nonetheless, be necessary because of the negative effects on the overall economy of rising tax rates.[60] The new government has stated that the announced "tax freeze" will also apply to local government taxes, but it has not yet specified any changes to the budget co-operation framework that could ensure this.[61] One possibility is to impose economic sanctions on those counties and municipalities that raise taxes. Another more flexible approach would be to cap or freeze the income tax rate but allow local governments to fund themselves in other ways, for example by increasing the scope for raising property taxes, lowering taxpayers' basic allowances or reducing the constraints on user charges.[62] This would safeguard local government autonomy and, at the same time, prevent further detrimental effects on labour supply from the continued upward drift in tax rates.

Are there too many small municipalities?

It is questionable whether the structure that was created three decades ago remains well aligned with the current range of local government responsibilities. The average number of inhabitants per municipality is approximately 19 000, with the 20 smallest municipalities having less than 5 000 inhabitants. While this is much smaller than in, for example, Ireland or the United Kingdom, it is considerably bigger than in France, Italy or Spain (Mouritzen, 1999). Thus, the main point is not that Danish municipalities are small by international standards but that the list of their responsibilities is much longer. One possible drawback of the current municipal structure is that it may impede reforms that could encourage competition and user choice (see below).

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A number of studies indicate that economies of scale in the provision of services are prevalent, even if rather modest (Mouritzen, 1999 and Indenrigsministeriet, 2001*b*).[63] However, those studies include only the efficiency in the provision of core services and not whether small municipalities have the ability to carry out professional administration of complex tasks such as, for example, assessment of eligibility for early retirement pension; provision of labour market programmes for cash benefit recipients; integration of refugees into the labour market; or environmental supervision of businesses. While empirical evidence is lacking, anecdotal information indicates that this might be a relevant concern, and public debate on the need for a reform of the municipal structure has surfaced. Experience from the 1970 reform suggests that building support for such a change can be a politically cumbersome and time-consuming process.[64] So far, central government policymakers have been reluctant to address the issue of a new reform, but it may become inescapable in the near future. The devolution of the above-mentioned responsibilities, which have mostly taken place over the past decade, has added a new dimension to municipalities' portfolios. On top of the traditional obligations to provide core personal services, they also have the job of ensuring as far as possible that social clients receive "active offers" instead of "passive transfers". It seems rather obvious that authorities with 5 000 to 10 000 residents have very different conditions for managing this than do those of 50 000 to 100 000 inhabitants.

Encouraging competition and extending market signals

This section considers ways in which extending market mechanisms, such as contracting out, user charges and client choice, can improve allocative efficiency. If such instruments are not applied, containing expenditures will depend solely on supply-side rationing such as queuing or assessment of need. Both of these instruments have their own inherent weaknesses; a main drawback of queuing is the absence of prioritisation in the allocation process, while assessment of need tends to be a very non-transparent mechanism from the client's point of view. The division of political competence between different levels of government may be an important barrier to strengthening competition and introducing market signals. For example, municipalities have significant discretion in deciding on contracting out, but little freedom to set user charges.

Contracting out

A certain political consensus has emerged in recent years, at least in principle, that exposing public services to competition may be a means of making their provision more efficient and that this can be reconciled with public financing and regulation. Policies governing contracting out are twofold. *First*, central government authorities are required, by virtue of a Ministry of Finance circular, to call for

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Table 19. **Local governments' purchases of external services, 1991-99**
Per cent of operating costs

	1991	1993	1995	1997	1999	Change 1991-99
Technical services and environment	43.2	44.5	47.8	47.5	49.0	5.8
Utilities	10.6	16.2	17.2	17.6	16.9	6.3
Roads	39.2	40.8	42.8	38.6	38.7	–0.5
Education and culture	5.8	5.4	5.4	5.7	5.9	0.1
Hospital services	4.0	4.1	3.8	4.1	4.9	0.9
Social security and health	3.9	3.3	3.8	4.1	4.4	0.5
Administration	16.0	17.0	17.4	17.8	18.6	2.6
Total	9.1	9.3	9.8	9.8	10.1	1.0

Source: Konkurrencestyrelsen (2001).

tenders on all services "regularly". Hitherto, adherence to this requirement has been rather mediocre (Udliciteringsrådet, 2000), and its enforcement needs to be enhanced, for example through the contracts with central government authorities and their directors. *Second*, individual local governments have considerable freedom to decide their own policies; core administrative responsibilities and primary education aside, there are only few restrictions.

A recent study by the Competition Authority on trends in the use of tendering and outsourcing in counties and municipalities suggests that the value of services contracted out has increased only marginally over the past decade (Konkurrencestyrelsen, 2001).[65] Exposure to competition is highest for technical services, road building and, to a lesser extent, utilities and administration (Table 19). In contrast, contracting out is very limited – around five per cent – in the big spending areas like education, childcare, elderly care and health. In these areas the Danish "level" of contracting out is only half that in Sweden and largely confined to auxiliary services such as cleaning, transport or "meals-on-wheels" services. In sum, the diffusion of tendering and outsourcing has progressed only slowly despite extensive public attention and political debate on the issue.

Where contracting out has been used, experience on changes in quality and price effects has been mixed. Available information suggest that cost reductions of some 15 to 25 per cent have often been achieved, although with experience spanning a greater interval. These effects are not extendable to all public service provision, as the most promising projects are likely to have been the ones first outsourced. Moreover, the possibilities of establishing a competitive private market may be limited in some areas, thereby reducing or eliminating the potential gains. Also, there are some examples that the quality of services outsourced has not met the expectations. The existing disparity across municipalities may serve as the starting point for assessing the magnitude of the macro-level savings

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potential from a consistent use of outsourcing. If all municipalities contracted out at the same scale as the most active already do, services worth an additional DKr 6 billion would be exposed to competition (Konkurrencestyrelsen, 2001). Assuming, hypothetically, a 20 per cent average cost reduction, this might imply modest total savings of some DKr 1¼ billion (0.1 per cent of GDP). This illustrates that if more significant reductions of public spending are to be achieved, contracting out needs to be applied much more widely than at present. This should both involve a more intensive use of the instrument within the existing legal framework and allowing private provision for a broader range of services.

A number of economic, political and institutional barriers need to be overcome if this is to take place. For example, the existing municipal structure is likely to impede the diffusion of outsourcing due to diseconomies of scale and the inherent fragmentation of the decision process. The latter is likely to discourage potential suppliers, as it takes a certain market size and some minimum security for the investment and development costs to pay off. This, in turn, could reduce the first-mover advantages for those municipalities that consider scaling up the use of outsourcing, thereby delaying the build-up of competitive markets. While this would seem to warrant adjustments of the municipal structure, this is not likely to come about by itself, at least in the near term. Therefore, political attention to address the problem is called for. One way of moving ahead might be to try to find ways of encouraging greater inter-municipal co-operation, for example, by extending the use of inter-municipal purchasing agencies and joint administrative systems. Imposing regular tendering for some services might also be considered.

User charges

The term *user charges* is employed here in a broad sense. Most importantly, it designates situations where individual consumers or companies contribute directly with payment for services provided or funded by the public sector. However, it is also taken to include collective, private co-payment arrangements – such as private insurance schemes – even if these only directly link individuals' consumption and payment when a part-charge is levied for services.[66] As a general policy instrument, user charges have a number of advantages.

- They may curtail demand and, therefore, public expenditure. User charges provide information to consumers and officials about the demand for, and costs of, providing services. The provision of public services free of charge obviously risks prompting excessive demand as the social costs of supplying them are largely irrelevant for the individual consumer's behaviour.[67]
- They result in an allocation of funds determined directly by the demand of consumers instead of being expressed indirectly through the political process and by the electorate as a whole.

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– They provide an alternative to tax financing or mandatory social security contributions. Unlike user charges, tax financing redistributes income and consumption possibilities. Likewise, taxes reduce individuals' returns to work, save and invest in their human capital while user charges do not.

– As consumers are likely to pay more attention to the quality of services for which they pay, user charges may well prompt demand-side pressure for better outcomes, thereby furthering supply-side efficiency. This effect is probably reinforced if, at the same time, consumers have some scope for choosing between different providers.

In recent decades, there has been virtually no introduction of co-payment in new areas or increased co-payment in areas where it already existed. On the contrary, existing user charges have generally been lowered, for example for childcare and practical home help. Nevertheless, the share of private co-payment of publicly provided services increased from 5.9 per cent of total costs in 1977 to 13 per cent in 1997. The most distinct increase – particularly in the latter half of the period – was for education, health and social services (Finansministeriet, 2000*a*). The increase is attributable to the fact that spending rose most on services that had above-average levels of user charges in the 1970s, notably childcare and pharmaceuticals.

Danish policymakers have been distinctly reluctant to deal with the issue of user charges, let alone develop a coherent strategy for their application. In particular, introducing user charges in new areas has been controversial, not least on distributional grounds (see Box 7). The current mixture may best be seen as the accidental inheritance of the past. No attempts have been made to assess whether private co-payment is used where it makes most sense economically or whether the split between user financing and collective financing is appropriate. Randomness, more than absence of user financing, characterises current policies. Thus, sizeable user charges exist for some services such as childcare, dental care and pharmaceuticals. In the last two areas, total private co-payments are in practice high by international standards, as there is no ceiling on individuals' total payment.[68] On the other hand, Denmark is distinctive for having virtually no private co-payment for primary and secondary health care, elderly care and tertiary education (Table 20).[69] A range of non-core services for which demand is likely to be sensitive to the price, for example general adult education, vocational training and public libraries, are also provided with little or no user payment.

Developing a coherent strategy for the application of user charges would most likely include a broadening of their use and allowing local governments more discretion in setting their levels. It is difficult to judge *a priori* the extent to which an increase in the overall scope of user charges is warranted, although the higher share of public finances applied to overall consumption than elsewhere indicates a potential for a wider scope of user payments. Equity considerations,

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Box 7. Distributional effects of user and tax financing of additional spending

An increase in user charges for *existing services* would typically widen the dispersion of total consumption possibilities, although in principle this depends on how the extra revenue is used. Per capita public consumption declines with rising income (Finansministeriet, 2000*c*). Therefore, increasing user charges on the "average public consumption bundle" would be relatively more burdensome for low income households and even if the revenue is channelled back as a lump sum tax cut they would have a net reduction of consumption, while the opposite would be the case for high income households. Even so, the "loss of equality" should be balanced against the mitigation of overall static distortions and the potential favourable dynamic effects.

Moreover, the relevant weighting is different at the margin, when considering the mode of financing for *new or higher quality services*. In this situation user charges would not cause less redistribution but merely an increase in the "price" of private consumption. In contrast, tax financing would increase redistribution but at the cost of greater distortion of incentives to work, save and acquire human capital. Finally, even if discretionary measures are not taken to broaden user charges, this may evolve endogenously if private insurance schemes spread in areas such as hospital treatment or elderly care. As much as politically decided charges, this could challenge the principle of "free and equal access" in the future.

distortionary effects associated with the alternative of tax financing, and a careful evaluation of price sensitivities would be three of the elements entering this assessment. While a more consistent overall approach to user payment is paramount, examples of more specific changes for consideration would comprise the following (see also Annex V):

- While parental payments for *childcare* are not especially low in a cross-country context, they are inappropriate inasmuch as they are not linked to the individual child's actual attendance. Redesigning user charges by relating them to marginal consumption could possibly facilitate a reduction of excessive demand but would need to be balanced against potential negative labour supply effects.
- Denmark is distinct in not having any user charges for *elderly care*. Current trends suggest that the pressure on supply-side rationing is considerable as "demand" for free services is booming. Introducing partial user payment, in combination with more extensive client choice and competition,

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Table 20. **User charges for various personal services**

Private co-payment percentage except as noted

	Elderly care		Childcare		Healthcare services				Tertiary education[4]
	Home help	Institutional accommodation	Aged 0-3[1]	Aged 3-6[2]	In-patient acute care	Out-patient medical care	Pharma-ceuticals	General practitioner office visit[3]	
Denmark	**0**	**5**	**33**	**33**	**0**	**5**	**43**	**0**	**3**
Sweden	6	8	20	0	2	27	29	15	11
Norway	–	–	28-45	28-5	0	–	42	14	6
Finland	14	20	10-15	10-15	8	23	52	–	–
Germany	–	< 25	16-20	–	2	10	52	–	8
Belgium	–	–	17-25	0	–109	–	–	–	–
France	–	–	23	0	8	43	46	–	10
United Kingdom	7	27	30-60	0	1	12	10	–	25
Italy	–	–	36	0	15	28	35	–	19
Netherlands	10	10-39[5]	44	0	19	37	36	–	4
Austria	–	–	–	–	10	20	50	–	1
Portugal	–	–	11	0	–	–	–	–	8
Canada	–	–	–	–	10	41	57	26	17
United States	–	–	60	60	45	–	75	–	48
Australia	–	–	9	0	27	40	50	17	32
New Zealand	–	–	–	–	8	53	35	–	–

1. In most countries, user charges vary with family income. Where a range is not specified, the figures reported in the table are maximum charges for Denmark and Sweden and average charges for other countries except Australia and Portugal where they refer to the percentage of disposable income. In Finland, charges are subject to a ceiling.
2. In countries with no charges, the services in question are part of the education system and are typically part-time. In Australia, the Netherlands and the United Kingdom pre-primary education is free from 4 years and in Sweden from 5 years.
3. US dollars at 1998 exchange rates.
4. The figures show the share of private funds in financing educational institutions. Data refer to 1998 and are net of public subsidies.
5. 10 per cent in nursing homes and 39 per cent in residential homes.

Source: OECD Health Data Base 2000, OECD (2001*a*); Rostgaard and Fridberg (1998).

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is warranted in this area, and it is worth considering abandoning public subsidies for some practical services, such as cleaning.

– The substantial subsidies for adult and vocational education appear difficult to justify on grounds of economic externalities or equity concerns. Sizeable user charges should be introduced in this area, thereby shifting the costs back to the employers.

An issue that has been debated recently is whether it would be fruitful to distinguish between basic public services financed through taxes and supplementary services, which the public sector could provide against full user payment.[70] This has been suggested as a way of forestalling demand pressures from eventually manifesting themselves in higher tax rates. While this might indeed alleviate pressures in some areas, in general publicly provided supplementary services should be used with caution. The implied increase in public sector production could in practice compromise policies of encouraging contracting out and might easily risk crowding out private sector production. A better way would be to expand outsourcing and client choice, in which case supplementary services would be purchased from a private provider.

Client choice and vouchers

While consumer choice is a fundamental characteristic of private consumption, there is typically more limited or no scope for users of public services to choose between providers or to influence their content; obviously, this reflects the zero or very low payments involved. The extent to which the limited choice impinges on perceived service quality is likely to vary across consumers and with the nature of the services in question; if proximity to the provider is considered of dominant importance, local monopoly-like supply conditions may prevail in any circumstance. Nonetheless, the authorities have long recognised that having a choice is often valuable in itself and assumed that users will increasingly demand tailored services and greater possibility to choose in the future (Kommunernes Landsforening *et al.*, 1999 and Indenrigsministeriet, 2001*b*). Indeed, client choice and influence has increased in scope over the past decade, for example *via* a legal right to choose hospital treatment outside the county of residence. Limited geographical choice has also been introduced for childcare, and a few municipalities have introduced voucher models for home help services. As discussed above, the new government has stated its intention to greatly widen the scope for client choice in the areas of elderly care and secondary health care. Within primary education there is a long-standing tradition for private schools. These are entitled to a per student subsidy of 85 per cent of the municipalities' costs and have increased their market share slightly over the 1990s to some 10 per cent.

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While a comprehensive review of the current extent of user choice and the scope for increasing it within existing framework is beyond the remit of this chapter, the concept itself has some important bearings on expenditure policies. On the one hand, client choice, for example in the form of vouchers, is likely to have some desirable effects. Surveys corroborate the view that user satisfaction and perceived quality tend to increase with the scope of choice. In addition, unless capacity constraints severely restrict client choice, it may facilitate competition among providers, thereby furthering efficient resource allocation. Client choice and vouchers may increase transparency in the decision making process by making the costs and trade-offs more visible to clients and policy makers. Finally, vouchers could make it easier for consumers to purchase supplementary services from their provider, thereby potentially helping to mitigate demand pressures at the margin. On the other hand, unrestricted client choice would often be difficult to reconcile with overall expenditure control, especially for services provided free of charge. The experience in 1997 and 1998 in the area of adult education was a forceful demonstration of these shortcomings; demand and public spending surged in response to abandoning budget envelopes and introducing extensive client choice accompanied by only negligible user co-payment. Thus, budget envelopes and rationing through assessment of need are often unavoidable instruments for safeguarding the control over total spending, but this does not preclude the above advantages from manifesting themselves.

The two areas in which the new government is extending client choice – elderly care and secondary health care – are both characterised by not having any (imminent) upper limit to demand. The budgetary risk of client choice in those two sectors could, therefore, be large; while the assessment-of-need rationing limits the risk in the case of elderly care there would be no efficient constraints on health spending if the budget envelopes are not maintained. In contrast, for services with virtually unrestricted access, such as childcare and ordinary education, there is typically little risk that client choice would prompt an unsustainable demand response, and the instrument would be particularly suitable in these areas. Indeed, the use of vouchers in OECD countries is most common in the education sector (Cave, 2001). While more widespread but careful use of the instrument is warranted, the fragmented municipal structure may be an impediment to initiating such a process.

Enhancing administrative efficiency

Well-designed public management policies are likely to be important in furthering good performance, *i.e.* encouraging high productivity and ensuring that stated policy objectives are reached efficiently. However, measuring public sector productivity and goal achievement at the aggregate level is notoriously difficult. Therefore, while this section provides a brief overview of current trends in public

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management policies, it contains no attempt to quantify their effects on macroeconomic performance.

Promoting good performance

Danish public management policies have changed considerably over the most recent decade, notably at the national level. The main thread has been a wish to link appropriations, jobs and pay more closely to performance, thereby enhancing overall productivity. For example, company-style accounting by state enterprises and agencies spread considerably over the 1990s. The same is true for performance-related contracts with public authorities and their directors, which virtually did not exist 10 years ago. Contracts are used at two levels. Performance-related contracts are signed within individual ministries between the department and its agencies. Also, performance-pay contracts can be signed with directors of individual government agencies. In 2000, around half of all government agencies had signed contracts, while the equivalent share for directors of government agencies was around a quarter. Both types of contracts have increased significantly in number in recent years (Table 21). The performance criteria in the contracts vary. For directors' contracts *ex post* performance is quantified on a scale from 0 to 100. The average score in 1999 was 67 per cent.

The use of fixed-term employment contracts has been scaled up significantly, while "life-time" civil-service jobs are used less frequently, thereby increasing overall flexibility. At present, some 15 per cent of public-sector managers are employed on a fixed-term basis; the typical duration is three to five years with a common pay supplement of 15 to 25 per cent. Finally, a wage reform was introduced in the late 1990s that will gradually weaken seniority-based wage elements and make wage-setting more individualised.

Developing a consistent policy for evaluation of government activities should be a priority. Ideally, this should include overall goal-achievement within policy areas, the quality of services provided and the efficiency of individual government institutions or entities. Such evaluation can help identifying non-performing units as well as best practices. Applying a broad definition, evaluation currently takes

Table 21. **The prevalence of performance-related contracts in central government**

Number of contracts

	1992	1993	1994	1995	1996	1997	1998	1999	2000
Contracts at the institutional level	9	15	20	27	31	38	56	81	83
Managers'contracts	–	–	–	5	12	21	35	40	48

Source: Ministry of Finance.

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place at several levels; it is conducted by numerous actors; and the nature of the evaluation can take many different forms. Thus, it covers a wide field from appropriation control conducted by the National Audit Agency through outcome and efficiency analysis by sector-specific evaluation centres to scientific research and analysis by private consulting companies ordered by public-sector entities. A few defining characteristics and tendencies may be worth mentioning:

- Total evaluation activity seems to be increasing. This is likely to reflect, in part, the aforementioned sharper focus on public management policies with the Ministry of Finance as the main driver. In addition, the traditional conflict of interest between spending advocates and budget advocates has increasingly developed into a competition between analytically based arguments, which has prompted an increase in the use of analysis of efficiency, policy effects and outcomes on both sides.
- The central government has involved itself more directly in policy areas where counties and municipalities are formally responsible. This manifests itself both in the outlining of policy objectives and in regulation (see above). In addition, the acceptance of local or regional variations in service standards has decreased amongst central-level policy makers. In continuing of these tendencies, policymakers at the central level have looked for ways to ensure that intended policies are pursued at lower levels of government and that resources are used efficiently. Evaluation lends itself well to this end. Independent evaluation agencies were established for the secondary health-care sector and education sector in 1999 under the Ministries of Health and Education, respectively.

Despite the increasing level of activity, there is still considerable scope for anchoring evaluations more closely and systematically in the budget process. In fact, current activities are not guided by an overall evaluation policy. Much of the ongoing effort is rather fragmented and detached from the political process. Without compromising the current diversity, developing a strategy for systematic use of cost-benefit analysis and evaluations as an integral part of the budget process is warranted. This could involve concentrating more resources under the auspices of a single independent agency that would have the capacity to strengthen the quality as well as the legitimacy of the analysis. A stronger attachment to the budget process could also be facilitated by a wider use of sunset clauses, making eventual extension of appropriations dependent in part on the result of evaluation.

Digital administration

Denmark is well placed to take advantage of ICT technology as a means to strengthen administrative efficiency in the public sector. Personal computer and internet penetration rates are amongst the highest in the world, and access to broadband is likely to increase rapidly in the near term. In addition, surveys suggest

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that there is considerable openness and interest in the population for using electronic self-service facilities through the internet. One obvious advantage from the consumers' point of view is the effective extension of opening hours. Self-service facilities are prevalent in many areas already, including, for example, submission of tax declarations, reservation and loan services in municipal libraries, booking services and access to all central government forms. A government portal for all business services is currently being developed.

While better supply of electronic self-service facilities is clearly to the advantage of citizens and firms, in themselves such facilities do not necessarily reduce the costs of public-sector administration and thereby promote efficiency. To achieve this end, changes in working procedures within individual public-sector entities and across organisational structures may be required. The government has recently launched a strategy for furthering digital administration in the public sector with the explicit objective of reducing administrative costs as much as possible (Kommunernes Landsforening *et al.*, 2001). A task force has been set up to co-ordinate and lead the development of concrete initiatives.[71] The focus on digital administration is motivated in large part by the fact that reaping the savings potential in this area will help make more resources available for providing personal services. The potential size of the savings will depend in part on the technologies that will become available, and no estimate thereof has been made. Instead, the strategy focuses on furthering changes in working procedures and improving the possibilities of electronic data exchange within and across government entities and between the public sector and its users. This is bound to be an ongoing process for an extended period of time, but with the existing Central Population Register (see Box 8), Denmark probably has a better starting point than many other countries.

Summing up

The primary policy issues that arise from the analysis in this chapter concern overall expenditure control in counties and municipalities and strengthening the role of competition and market signals in the provision of a wide range of public services. As most such services are provided by local governments, these issues are strongly interrelated. The fundamental question is whether the traditional policies of widespread local-government autonomy and "free and equal access" can be reconciled, in their present form, with effective expenditure control in the future.

The framework of formalised inter-governmental budget co-operation has not prevented an unplanned increase in local governments' income tax rates. The principal deficiency of the co-operative approach as it has worked out in practice is that it embodies no efficient mechanism to ensure that actions taken by individual counties and municipalities, when taken together, conform to the aggregate spending and tax levels envisaged in the agreements. In this sense, local govern-

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Box 8. The Central Population Register (CPR)

The CPR register was established already in the late 1960s as a precondition for the introduction of the withholding tax in 1970. Similar registers were established around the same time in the other Scandinavian countries. The register identifies all persons resident in the country or holding Danish citizenship by a civil registration number.

This number is used as identification in virtually all public sector administrative systems and for all contacts between individuals and public authorities. Thus, when contacting a general practitioner, being hospitalised, enrolling children in school, applying for a social transfer, submitting a tax declaration or receiving elderly care, the civil registration number is always used in the individual's dossier. The number must also be stated to employers for their wage administration, and its presentation is a precondition for opening a bank account. Subject to the restrictions concerning privacy set in the CPR Act, this facilitates extensive exchange of data between public-sector authorities.

ment autonomy has prevailed at the expense of aggregate spending control. Adjustments to the budget co-operation framework should be a top priority and would need to involve restrictions on local governments' taxing powers. At the same time, there is a strong need to clarify responsibilities between the various levels of government. In some cases this would involve less micro-management by the central government.

More efficient resource allocation could be achieved by enhancing competitive structures in the provision of services as well as allowing a greater role for demand signals such as user charges and client choice. This is particularly relevant in the case of Denmark inasmuch as public consumption is a larger share of spending than elsewhere. One barrier impeding such reforms is the widespread perception amongst policymakers that these instruments conflict with distributional objectives. However, as has been argued in this chapter, this perception is at best partial and at worst misleading. Institutional barriers are important too. One factor is the fragmented policy decision process inherent in the decentralised government structure itself. Another is the fact that potential synergies from the simultaneous use of several instruments are difficult to reap because the authority to change policies does not always lie with the same level of government. While some of these barriers are difficult to overcome and involve trade-offs, they nevertheless deserve attention.

A summary of the recommendations that flow from the analysis in this chapter is contained in Table 22.

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Table 22. **Summary of recommendations**

Recommendation	Present situation	Elaboration
Budget process		
Introduce politically binding expenditure ceiling.	The fiscal-stance measures currently guiding fiscal policies focus on the budget balance but do little to restrain expenditure growth. The existing ministerial expenditure ceilings put forward in the budget proposal in August are not binding in the following political negotiation process.	A ceiling for total central-government expenditure, which is binding throughout the budget process, should be introduced. While determining its precise demarcation would warrant further analysis, to be effective it would need to be set in nominal terms and include grants to local governments. Achieving political agreement on the overall spending level prior to the budget negotiations with the local governments would strengthen top-down control.
Strengthen multi-annual budgeting procedures.	The current procedure of budgeting outlays in the three budget forecast years following the fiscal year are not fully consistent with medium term projections.	Improve the consistency with medium-term projections and use forecasting techniques to provide best possible projections of future expenditure.
Link evaluations closer to the budget process.	No overall evaluation policy is in place consistently assessing the performance and productivity of government agencies or the goal-achievement of government activities. Expenditure analyses are carried out within the ministries as an integrated part of the budget process. However, while evaluation activity in independent publicly-funded institutes is increasing, its impact on the budget process is limited.	Evaluation of performance and productivity developments in individual spending areas should be linked closer to the budget process itself, possibly in tandem with a wider use of sunset clauses. Consideration should be given to concentrate more responsibility for evaluations with an independent body.
Local governments		
Improve spending control and compliance with agreements.	Local governments have repeatedly exceeded agreed spending levels, and their income-tax rates have drifted up as a result. This compromises central government tax policies and has adverse impacts on labour supply.	Local governments' entitlement to set income-tax rates should be restricted, either through a centrally determined tax freeze or by imposing financial penalties on tax increases. To reconcile this with local-government autonomy, counties and municipalities could be allowed more discretion in setting property tax rates, the basic tax allowance and user charges.
Clarify responsibilities between different levels of government and match with taxing powers.	By setting objectives and imposing regulation the central government has assumed *de facto* control in areas that have been devolved to local governments. This makes clear accountability more difficult.	In policy areas where the central government wants to play a major role in setting service standards, it should take back the financing responsibility accordingly. On the other hand, where the central government does not want to reduce local governments' taxing powers, it should abstain from detailed regulation.

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Table 22. **Summary of recommendations** (*cont.*)

Recommendation	Present situation	Elaboration
Encourage greater efficiency in local governments.	Economies of scale warrant a certain minimum size of municipalities. Moreover, the fragmented political decision process risks impeding the spread of competitive structures.	Efforts to develop instruments encouraging inter-municipal co-operation, voluntary mergers and enhancing the role of market signals should be strengthened.
Strengthen competition and market signals		
Encourage competition in the provision of publicly financed services.	Compliance with the Ministry of Finance circular on tendering and outsourcing has been mediocre, and the trend rise in municipalities' use of these instruments has stalled.	Encouraging competitive framework conditions is warranted for a wider range of services. Enforcement of existing provisions regarding central-government agencies' contracting out should be strengthened, and local governments should have more freedom and stronger incentives to establish competitive production structures. More effort should be devoted to identifying and removing barriers to outsourcing.
Develop a coherent strategy for user charges.	User charges are at present a reflection of past traditions and are not, necessarily aligned with the price elasticities of demand. As a result they are not providing appropriate market signals.	User charges should in general be designed so as to optimise the demand response. Higher or more well-designed charges seem appropriate in areas such as home help, childcare and adult and tertiary education. Local governments should be allowed more discretion in setting charges.
Strengthen client choice to exploit synergies.	Clients' choice has been furthered in some areas, but inadequacy of information, capacity constraints and political reluctance continue to restrain its impact. The new government has outlined a more comprehensive clients' choice programme.	Clients' choice may often in itself raise the perceived quality of services. In addition, synergy effects may arise, as allowing clients to determine the prevailing providers could also foster greater competition. However, for services with no imminent level of satiation, a necessary balance needs to be struck between clients' choice and expenditure control. More client choice would often need to be accompanied by user charges.
Administrative efficiency		
Broaden and deepen the use of performance-improving instruments.	Policies to promote efficiency such as performance contracting and performance-related pay has strengthened significantly.	There is still scope for improving implementation of such policies, particularly at lower levels of government.
Strengthen digital administration.	Denmark is well placed to take advantage of ICT to improve the efficiency of public administration. A task force has been set up in the Ministry of Finance to push this ahead.	Continue developing the strategy for e-government and ensure that savings are collected for centrally determined re-allocation.

Source: OECD.

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Notes

1. The Danish authorities estimate that hurricane repairs added around ¾ percentage points to measured activity in 2000.
2. Danish exports are traditionally less sensitive to the business cycle than exports from other EU countries.
3. There is an expanding cluster of pharmaceutical and biotech companies in the Øresund area connecting Denmark and Sweden know as "Medicon Valley".
4. However, the number of residential buildings started in 2001 actually rose because of composition shifts between apartment buildings and single-family houses.
5. Although the split between households and corporations shows marked differences, the distinction is mainly in timing since the disposable income of corporations finally accrues to the household sector.
6. Denmark meets all the requirements for entry into the third stage of European Monetary Union.
7. Ministry of Finance estimates of a Taylor rule for Denmark and the euro area in June 2001, indicated that the anticipated euro area interest rates were only slightly below those required for Denmark, and converging (Ministry of Finance, 2001).
8. Newly recruited staff who have not earned rights to paid vacation due to previous unemployment are entitled to "unemployment benefits" while on vacation, if they belonged to an unemployment insurance fund.
9. The discussion in the previous *Survey* was related to an earlier government scenario covering only the period up to 2005. In the authorities' updated projection, most of the increase in labour supply is still assumed to come about before 2005.
10. In the short term, sickness benefit, cash benefit and rehabilitation benefits all serve as close substitutes. However, none of these schemes are supposed to compensate individuals over longer periods of time. It seems most likely that a large accumulated increase in the number of recipients of these benefits in the short term will eventually lead to renewed pick-up in the inflow to early retirement pension.
11. The different assessments by the Ministry of Finance and the Economic Council are to some extent due to the fact that the former includes a larger contribution from administrative measures including new, but not yet announced measures, different assumptions with respect to the number of *flexi-jobs*, and the extent to which a continuation of the recent reduction in the outflow to early retirement will be reflected in a larger labour force.
12. The government has proposed an extension of the maternity leave scheme from the current 28 weeks to 52 weeks, while at the same time abolishing the existing paid leave scheme.

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13. The OECD's growth project found evidence of a positive impact of R&D on growth (see, for example, Bassanini *et al.*, 2001).
14. An introduction of tax credits for R&D could be fully financed, for example by lowering the depreciation rate on investment in capital with less externalities and a long lifetime such as construction.
15. The results reported in Panel A of Figure 12 are based on studies of income distribution in 13 OECD Member countries using national-source data with the potential *caveat* that differences in methodology and definitions may distort the cross-country comparison. However, a recent study by Heady *et al.* (2001) using a consistent data set for 13 EU countries (Finland and Sweden are not included) confirms the impression that the inequality of disposable incomes was lower in Denmark in the mid-1990s than elsewhere.
16. These are short-run figures and the amount of redistribution through public consumption is likely to be significantly lower in a lifetime perspective.
17. Private consumption is here assumed to be equal to disposable income.
18. These are average figures. Within-group variations are likely to be significant, with all groups comprising both net receivers and net contributors.
19. The Law Model, which was established around 1980, is basically a set of very detailed household accounts matched with a wide range of socio-economic background variables. The database is an annually updated representative sample of some 10 000 adults. The model has been established by merging personal data from different administrative systems into one comprehensive database, thereby making use of the fact that the civil registration number is used in all such systems providing a common "identifier".
20. This is obviously not an argument for suppressing such information. Also, on a more positive note, the Law Model has been instrumental in identifying sources of high effective marginal tax rates resulting from means-tested benefits. This has facilitated adjustments aimed at limiting the detrimental impact of poverty traps.
21. For example, analysis by the Ministry of Economic Affairs (Økonomiministeriet, 2000) shows that 50 per cent of the population change income decile from one year to the next, while half of those in the bottom two income deciles in 1994 had moved into higher deciles four years later. While the compressed wage structure in itself makes changes between deciles from one year to the next more likely, it is nevertheless remarkable that only ¼ per cent of the adult population had disposable incomes below the relative poverty line (50 per cent of the median level) throughout the five-year period 1994 to 1998.
22. Because the ultimate beneficiaries of any reform can only be identified once all the effects have fully worked through the economy, the winners may themselves not realise what they stand to gain.
23. The discussion is deliberately "topical". See Det økonomiske Råd (2001*a*) for a recent comprehensive overview of Danish tax policies, Van den Noord and Heady (2001) for a synthesis discussion of tax policies in the OECD area at large and Joumard (2001) for a review of policy trends in the EU area.
24. Employees' contributions to the unemployment-insurance and early-retirement benefit schemes constitute the main exceptions from the general rule, but even these finance only a fraction of total outlays under these schemes.
25. Liability to pay the top state tax typically requires gross income to be slightly higher than the APW wages level, as contributions to pension saving are deductible.

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26. The Economic Council has estimated that increasing the threshold from 1.0 to 1.3 times the average production worker's wage could be 90 per cent self financing in the long term. This is based on a labour supply function with an aggregate substitution elasticity of 0.1 and an effective elasticity in the particular experiment of around 0.15. According to a Ministry of Finance estimate assuming an aggregate substitution elasticity of 0.1 and a decreasing income-related profile, such a measure could be 50 to 60 per cent self-financing, before taking into account the debt servicing costs arising from the delay between the loss of revenue and the potential supply response.
27. Small-scale investors' capital gains are tax exempt if they have held the shares for three years or more. Investors with holdings worth more than some DKr 120 000 are liable to tax on capital gains in all circumstances, but at a rate that may vary between 28 and 59 per cent.
28. It should be noted, however, that this would induce significant asymmetries between the marginal tax on earnings and capital income, respectively. This would provide a stronger incentive (for taxpayers with positive net capital income) to convert earned income into capital income when possible.
29. The calculation underestimates the effective tax on pension savings for some groups as it does not take into account the fact that private pension benefits are partly offset against the public old-age pension.
30. Some reduction in the taxes on wine and beer has been implemented in recent years, but the revenue from alcoholic beverages and tobacco still amounted to 1 per cent of GDP in 2000.
31. It also impacts on producers' behaviour. The pre-tax prices on new cars are amongst the lowest in Europe, while, on the other hand, prices on spare parts are amongst the highest (Konkurrencestyrelsen, 2001).
32. The central government finances the full costs for early retirement pension, but only 35 per cent are reimbursements to individual municipalities corresponding to their actual outlays. The remaining 65 per cent of total outlays are *de facto* channelled back as untied grants, as they are allocated to the municipalities in accordance with the distribution of taxable income.
33. A fourth type of pension, labelled *ordinary pension*, is identical in its eligibility criteria to increased ordinary pension but applies to applicants aged 60 and above. It is awarded to relatively few people, and for simplicity, it is left out in the following description.
34. The great uncertainty about the effects of the reform is recognised in the explanatory memorandum to the bill, which requires a stocktaking report be submitted to parliament during its 2006/07 session.
35. Considerable emphasis is put in the bill on the introduction of a new administrative criterion for assessing the eligibility of future applicants (*arbejdsevnekriteriet*) and on establishing new working methods. The significance of these innovations cannot be assessed as yet.
36. The greater generosity of the new system is much less pronounced if individual housing benefits are taken into account, but for those who do not receive them, the incentives to work are weaker in the new system than in the old one.
37. In mid-January 2002, the new government presented a new immigration policy as an input to subsequent political discussions. Among its elements, the programme proposes that full social benefit rights be made contingent on having lived in Denmark in seven out of the eight most recent years. Also, stricter criteria for family reunions as

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well as for obtaining asylum and permanent residency are proposed. A more comprehensive revision of integration policies is scheduled for the autumn of 2002. The analysis in this section addresses the general economic aspects of immigration only, and in particular it does not include any assessment of the government's immigration programme or its economic impacts.

38. For non-western immigrants, the unemployment rate is approximately four times that of native Danes.

39. The so-called introduction benefit given to non-employed immigrants in their first three years in Denmark was originally set at a lower level than the normal cash benefit level when the Integration Act was adopted in 1998. However, the two were soon aligned as the lower level of the introduction benefit was considered to probably conflict with international conventions.

40. The structural characteristics of the Danish economy, notably a compressed wage structure and already high marginal tax rates, do not lend themselves well for introducing a general EITC scheme (Erhvervsministeriet *et al.*, 1996; Finansministeriet, 1995 and OECD, 1997). The implied higher marginal tax rates in the tax credit's phase-out income interval would affect a large proportion of the workforce, thereby probably more than offsetting the positive effects of better incentives at lower income levels. The target immigrant population of such a scheme, on the other hand, is generally characterised by low earnings capacity and the disincentive problems in the phase-out range may, therefore, affect relatively few.

41. Wage subsidies are offered to employers recruiting immigrants in temporary labour-market programmes. However, these are no more generous than for other unemployed and the type of job functions allowed is considerably restricted by the fact they are not allowed to crowd out unsubsidised jobs. A separate scheme is in place for highly educated immigrants, whereby employers can receive a subsidy of 50 per cent of the wage for the first six months of employment.

42. The rules on wind turbine ownership were also reformed to make large corporate investment possible.

43. Such transparency, allowing users to compare electricity charges across the country, is already a requirement in Sweden.

44. Davidsen and Hanus (2001) gives a detailed account of the evolution in the public consumption deflator from 1992 to 2000. According to their analysis, methodological weaknesses and shifts in the national accounts have tended to underestimate price increases and overestimate volume increases over this period.

45. Recent examples of structural reforms that were presented only during the political process include the labour market reforms in 1995 and 1998, the reform of the early retirement benefit scheme in 1998, the abolition of the wealth tax in 1995 and the adult education reform in 1999.

46. More recent comparable information is not yet available due to the change of national accounts standards.

47. Thus, legal constraints on the taxing powers have been applied on a few occasions in the past. The tax rates for both counties and municipalities were determined by law in the fiscal year 1971/72 and – for the counties only – were capped in 1988. In addition, legal sanctions in case of non-compliance with the agreements could be used during a few years in the mid-1980s.

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48. The supply of after-school facilities has tended to create its own demand through a sort of network externality: parents may have felt forced to send their children to an after-school facility because most other children go there, so there is nobody to play with at home.
49. A 50 per cent "brothers and sisters" discount was introduced in 1991. Thus, parents with more than one child in public day care pay full user charges for the most expensive place and 50 per cent of the normal charge for subsequent places.
50. Finansministeriet (2000*b*) reports results from a recent survey of satisfaction with elderly care, which suggest that the perceived quality is lower for practical help than for other services. Moreover, non-users are found to be distinctly less satisfied than users, a result also seen in similar surveys in other areas.
51. The political resistance against user charges is founded on distribution concerns, but it is both illogical (see Box 6 below) and inconsistent. For example, it is not obvious why differences in housing standards for elderly receiving help in nursing homes are deemed unacceptable while they are not for those receiving help in their own homes.
52. The right to choose private hospitals implies that "the money goes with the patient". However, according to the new government's "working programme", private hospitals are not allowed to charge for services provided under the imminent client choice programme.
53. Results from the 2000 survey of the OECD's Programme for International Student Assessment (PISA) indicate that the reading skills of Danish 15 year olds are close to the OECD average, while skills in mathematics are above average and scientific literacy below average (OECD, 2001*d*).
54. International standards for classification and demarcation of adult and vocational education do not exist, impeding cross-country comparisons. Nonetheless, the higher figure for "undistributed" educational spending in Table 18 above suggests that expenditure in this area is higher than in most other countries.
55. "Folk high schools" (*folkehøjskoler*) have a tradition of more than 150 years. They are independent, private, non-examination boarding schools providing courses over shorter or longer periods. Subject to the approval by the Minister of Education of their foundations, they are eligible for government subsidies on a per-student basis. The schools are open to all and determine their curricula themselves.
56. This should be seen as a complement to further developing the framework for sound micro-budgeting (see below).
57. In recent years, new elements in the budget have been supported by varying majority coalitions on a case-by-case basis – thereby giving rise to the patchwork characterisation. The budget bill itself is voted upon separately and is typically passed by a broad majority.
58. This updrift in tax rates was a continuation of the trend of the 1970s and the 1980s; thus, the average income tax rate of local governments rose more than 7 percentage points between 1970 and 1983. Even though the budget co-operation was less formalised and the agreements less explicit during this period, it is reasonable to assume that a large part of the tax increases even prior to 1984 was considered unwelcome *ex ante*. Thus, when the PPII-report (P*erspektivplan* II) from 1973 recommended formal budget co-operation, it was precisely out of a concern that tax rates would otherwise rise. The temporary freeze of local-government income tax rates in 1971/72 originated from the same concern. It was proposed by the conservative-liberal government then in office as a temporary arrangement until budget co-operation could be established. In a parlia-

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mentary reading the social-democratic speaker endorsed the proposal and gave the following statement, which in hindsight is rather thought-provoking: "My party recognises that taxation has reached such a level that the situation would be serious to many taxpayers should further tax increases occur. Therefore, we would like to contribute to reducing taxation ... we recognise that intervention is now necessary" (quoted by Blom-Hansen, 1998).

59. The multi-year character of the agreement applied to the spending level only, as the level of untied grants was still to be subject to annual negotiations. Moreover, in the subsequent years' agreements, explicit expenditure targets have not been set for the years after 2002.

60. If local differences in (the evolution of) tax rates and expenditure levels fully reflected varying local preferences they might be considered logical and even warranted effects of local democracy. In this case, the Tiebout effect, *i.e.* that people "vote with their feet", would ensure that actual policies were in keeping with voters' preferences. However, several factors limit the relevance of the Tiebout effect in practice, at least in the Danish context. F*irst*, local-level policy makers and their electorates are unlikely to take into account second-round effects on labour supply from raising taxes. S*econd*, moving costs are considerable, and the advantages sought in another jurisdiction may be transient. F*inally*, the initial premise is not valid in practice. Besides differences in the efficiency of providing services, variation in the trends and levels of local tax rates also reflect incomplete equalisation and the less than full reimbursement of transfer outlays. These factors are beyond local control, which adds to the overall complexity.

61. The government has signalled that it would cut central government taxes accordingly if local government taxes are raised.

62. Some flexibility might also be maintained by putting in place a trading scheme in taxing rights on top of a tax freeze thereby making it possible for counties and municipalities with a preference for increasing spending to buy permits to raise taxes from other counties and municipalities lowering their rates. It is a complicating factor, though, that both the levels and changes in tax rates across local governments reflect much else than local preferences.

63. The economically optimal size is estimated by the Ministry of the Interior to be around 20 000 to 30 000 inhabitants; beyond this size, there is a weak tendency for costs to rise. Some 200 municipalities (70 per cent) are smaller than this, and the potential annual efficiency gain if, hypothetically, all municipalities were of optimal size (and assuming the same population density in all municipalities) has been estimated to be roughly DKr 1¾ billion, *i.e.* around 1¾ per cent of total local government outlays.

64. That reform required more than a decade of preparation.

65. On average, services worth 11 per cent of the municipalities' operating costs were contracted out in 1999. A small group of municipalities purchased more than 15 per cent of their services from external providers, while in the large majority of municipalities, the figure was between 7 and 10 per cent. Cross-country comparisons of the use of tendering and outsourcing are impeded by the scarcity of data. Using the proportion of non-wage outlays in total public consumption as an indirect indicator, the share of services produced "in-house" by the Danish public sector appears to be above average – and contracting out thus below average (OECD, 2000*b*).

66. For private co-payment to impinge on demand it should be linked to individuals' marginal consumption. Payment arrangements already apply to number of services (for example, waste disposal, wastewater treatment and radio and television broadcasting)

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where users collectively finance (a part of) the costs for providing the services, but where the individual user's payment is independent of actual consumption.

67. Winter and Moutizen (2001) demonstrate that just giving information to consumers about the true costs of providing public services prompts a reduction of demand.

68. OECD's health database suggests that the share of private co-payment in total health expenditure is slightly above the average level in other EU countries (but lower than in, for example, the United States and Canada), despite the fact that there are charges are in place only for pharmaceuticals and dental care.

69. Cross-country comparisons of the levels of user charges, especially aggregate payments, should be interpreted with caution. For example, the level and the marginal properties of private co-payment arrangements may be blurred if they are offset by reimbursement arrangements or if private insurance schemes enjoy preferential tax treatment.

70. Supplementary services might include longer opening hours in childcare institutions, single rooms in hospitals, better standard accommodation in nursing homes or additional home help services.

71. The task force has its own board of top civil servants. While located in the Ministry of Finance, officials from all tiers of government, including local governments, are present in the task force.

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Annex I

CO_2 emissions

Denmark has well-established CO_2 taxes in other sectors and introduced an emissions trading system for electricity from 2000. Total quotas for emissions from electricity production have been set and emissions allowances assigned to eight individual power companies and allows for trading and "banking" or carrying forward of unused quota to subsequent years. There are some 500 carbon-emitting electricity producers, but these are combined heat and power companies and are exempted as they produce less than 100 000 tonnes of CO_2 emissions each. In total they emit around 1.9 million tonnes.

These permits are given for free, but to provide an incentive for emissions reductions and for emissions-free technologies to develop, they need to carry a positive price. Emissions in 2000 were just under 20 million tonnes, 3 million tonnes less than the planned ceiling for total emissions in that year.* With emissions allocations of 22 million tonnes in 2001, 21 million tonnes in 2002 and 20 million tonnes in 2003, it would seem unlikely that these CO_2 quotas will demand a high price in the near term or that any trading will take place, unless there is a strong upward market price movement. It will also not have any effect on emissions in the near future, beyond hypothetically reducing investment in new fossil fuel technology.

Nevertheless the cap-and-trade system provides a useful framework for reducing emissions and one that could readily be integrated with international systems as they develop. However, to ensure that reductions in emissions are undertaken by those with the lowest abatement cost across the country as a whole, it would make more sense to have a single integrated scheme that applies to all sources of emissions in Denmark, instead of maintaining the present separate sectoral CO_2 taxation arrangements. This would also provide a more compelling model for other countries to emulate.

* The necessary EC approval was only received in April 2000.

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Annex II
Public expenditure in perspective

With a view to identifying the key factual characteristics of Danish public expenditure, this Annex provides a brief overview of historical trends in public expenditure and a cross-country comparison of the current level and structure of public spending.

The historical evolution

The unsuccessful economic policy in the decade following the 1973/74 oil crisis caused total public expenditure to surge from just over 40 per cent of GDP to almost 60 per cent between 1973 and 1982. With some cyclical fluctuation, this high level persisted until the mid-1990s, before declining considerably over the most recent seven years (see Figure 22 in the main text).

The expansion of the public sector and of social welfare spending was not fully planned, but resulted, in part, from a process where new or better-standard services and transfer schemes turned out to be more expensive *ex post* than had been envisaged.[1] Public consumption rose by an average of 3.4 per cent annually in constant prices from 1973 to 1982 (compared with GDP growth of 1.3 per cent) and jumped from 23 per cent of GDP to 29 over this period. At the same time, social transfers increased from 10 to 17 per cent of GDP; this was the combined result of rising unemployment, easier eligibility criteria for early retirement pension in particular and more generous benefit levels in general. The interrelated declines in public investment activity and subsidies offset only a small part of the surge in public consumption and social transfers. Moreover, as taxes were not raised in line with the higher levels of primary spending, a large fiscal deficit occurred at a time when real and nominal interest rates rose to post-ar peak levels. The upshot was a marked jump in interest payments, which rose by some 6 to 7 percentage points of GDP.

The difficult experience of the 1970s helped, eventually, to bring about greater political consciousness of the importance of overall expenditure control and management. The years from 1982 to 1986 were a period of considerable fiscal consolidation. Real public consumption was restrained, benefit levels were held nominally unchanged, and public-sector investment continued to shrink. At the same time, output grew vigorously, supported primarily by rapidly declining interest rates (following the introduction of the fixed-exchange-rate regime in 1982) and deregulation of capital markets and the financial sector. Total public expenditure fell from 59 per cent of GDP in 1983 to 53½ per cent in 1986. However, these gains were gradually eroded in the subsequent protracted period of weak growth up to 1993, when the expenditure-to-GDP ratio once again ratcheted up towards 60 per cent.[2] This was mainly the result of unemployment rising to new highs and considerable inflow into other transfer schemes as well. These trends reversed again in the second half of the 1990s, when unemployment fell to the lowest level since the mid-1970s. By 2000, the public-expenditure-to-GDP ratio had returned to its 1980 level.

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The public sector's relative price and cost developments

One of the factors influencing the evolution of the overall expenditure-to-GDP ratio is price and cost developments in the public sector relative to those of the private sector. The public consumption deflator is likely to increase faster than the GDP deflator, at least in the long term. In part, this is due to the conventional problems of measuring public-sector productivity increases in the national accounts. Therefore, the deflator of public-sector production is measured on a cost basis and mainly follows wage increases. This implies a systematic upward bias in the deflator and an offsetting downward bias in the volume increases. Even so, the true – unmeasured – trend growth rate in public-sector productivity is generally accepted to be lower than the economy-wide trend productivity growth rate. This is because public-sector production is more labour intensive than production in the private sector, and the scope for technology-induced productivity increases lower. If at the same time public-sector wages increase in parallel with those in the private sector, this will cause even correctly measured unit labour costs to grow faster in the public sector than for the economy as a whole (Baumol's disease). All else being equal, this would imply a higher trend increase in the public consumption deflator than in the GDP deflator, thereby putting upward pressure on expenditure and taxes.

However, the actual evolution of the relative price of public consumption over the past 25 years does not, at first glance, conform to this hypothesis. While it did indeed increase rapidly up to the mid-1970s, it has mostly hovered around a constant level since then (Figure A1). This seems to have different explanations in the 15 years up to around 1990 and in the subsequent decade. Disregarding short-term fluctuations, unit labour costs in the public and the private sector increased at broadly the same rate between 1975 and 1990; the higher (measured) productivity growth in the private sector was offset in this period by lower nominal wage increases in the public sector. Against this background, it is not surprising that the deflators of public consumption and GDP also rose in parallel.

In contrast, during the 1990s the public sector's relative unit labour costs resumed the upward trajectory that would be expected to be the normal pattern. Two factors help to explain why the relative price of public consumption has nonetheless remained constant over the past decade. First, the rise in the relative unit labour costs did not manifest itself fully in the public sector's relative output price. This is attributable to the fact that the private sector's wage share declined, thereby causing output prices – and thus the GDP deflator – to increase faster than unit labour costs. Second, the deflator for non-wage public consumption declined significantly over the 1990s relative to the GDP deflator. These factors are probably of a temporary nature; the private sector's wage share is bound to stabilise sooner or later, and it is unrealistic to expect the relative price of non-wage public consumption to keep declining at the speed it has done in the past decade. If so – and if relative unit labour costs in the public sector continue to increase – the likelihood is that the public consumption deflator will increase more rapidly than the GDP deflator in the future, in contrast with recent developments.

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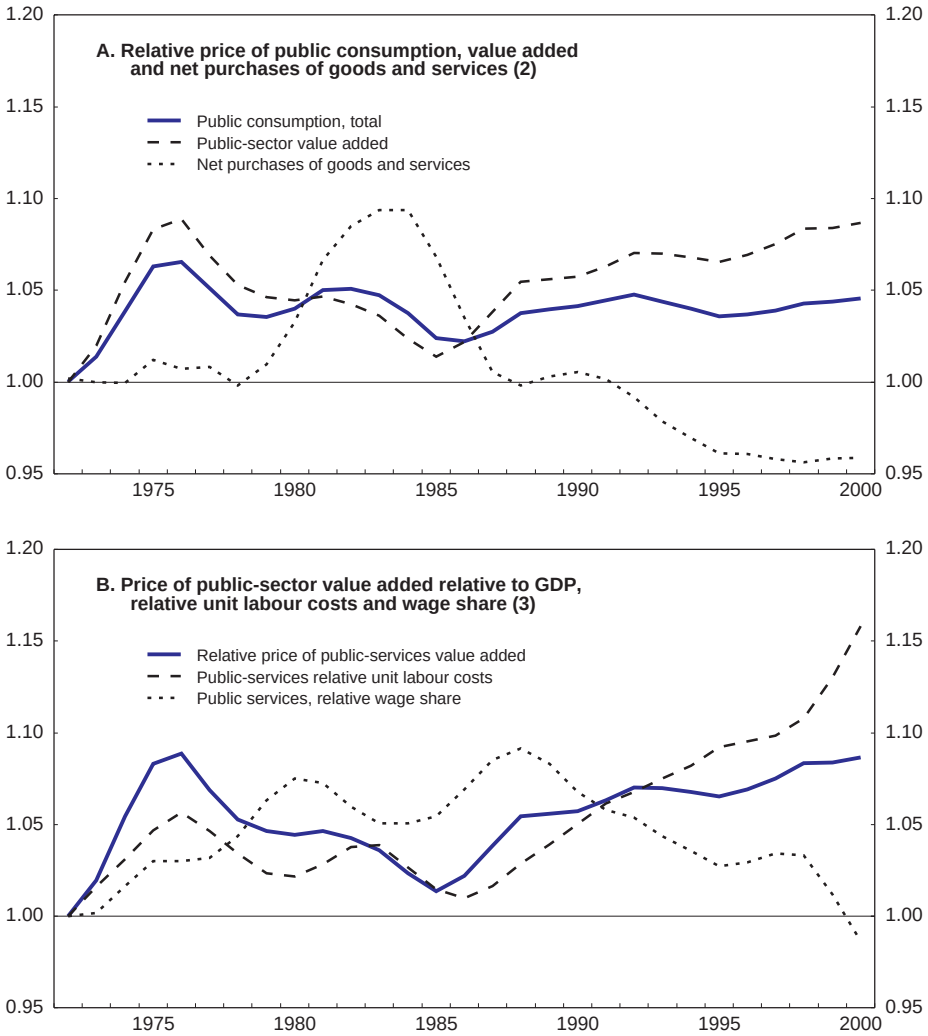
Figure A1. **Relative price and cost developments in the public sector**

1972 = 1[1]



1. All series are three-year centred moving averages.
2. National accounts deflators for the respective series relative to the GDP deflator.
3. The relative wage share is inverted, *i.e.* it is the economy-wide wage share over the public sector's wage share. The change in the relative price of public-sector value added is approximately the sum of the changes in relative unit labour costs and the inverse relative wage share.

Source: Statistics Denmark; OECD.

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An international comparison

Gross social outlays are significantly higher in Denmark than in the Anglo-Saxon and the Southern European countries (Table A1). In contrast, they are not particularly high when compared with the EU average, and slightly lower than in Germany, France and Italy. However, this simple comparison of gross social outlays is subject to a number of *caveats*:

- Transfers are subject to the same tax treatment as wage income in Denmark and several other countries, while they are predominantly untaxed in some countries. It is the net-of-tax value, which impinges on the amount of redistribution from workers to non-workers, even if it is the marginal tax rates that affects incentives.
- Indirect taxes are higher in Denmark than in most other countries, the VAT rate being 25 per cent and the base having few exemptions. Indirect taxes reduce the purchasing power value of social benefits. Therefore, as effective indirect tax rates vary across countries, comparing gross benefits without adjusting for the higher price level may be misleading.
- The collective economic support of individuals (or specific types of production) need not be accounted for on the expenditure side of the budget but may instead take the form of tax breaks for social purposes (TBSPs). Economic support of families with children, which exists in virtually all OECD countries, provides an example. While some countries (for example Denmark, Sweden and Norway) have child benefit schemes that appear on the expenditure side, other countries (for example, France, Germany, Italy and Ireland) give tax allowances to families with children.
- Governments can impose social expenditures on private-sector entities through legislation. Such mandatory private social benefits include employer payments to employees absent from work due to sickness and the financial value of employment protection legislation (EPL), for example severance pay and notice periods.

The OECD's work on net social expenditures (Adema, 1999 and 2001) seeks to address some of these problems in the area of social expenditures. Comparing gross and net social expenditures clearly demonstrates that accounting for the above mentioned factors can significantly alter the cross-country ranking of public expenditure levels. In general, net publicly mandated social expenditures vary much less across countries than do gross public expenditures. In 1997, Denmark had the highest level of gross public social expenditures amongst the 17 countries analysed. However, net publicly mandated expenditure in Denmark was below the levels in Belgium, Germany and Sweden and only moderately higher than in Finland, Italy and Norway (Table A2). Moreover, while net publicly mandated social expenditure to GDP was still 7 to 10 percentage points higher than in countries such as Australia, Canada, Ireland, New Zealand and the United States, the differences were considerably smaller than for gross social expenditure. A number of arguments would support that the much smaller gap between Denmark and other countries in net than in gross social expenditure is a result extendable to total public spending:[3]

- *First*, the distinction between net and gross social expenditure is of little importance to most other non-social budget components, and Denmark's gross spending on these is comparable to the levels elsewhere.
- *Second*, the adjustment for mandatory private social expenditures is probably incomplete in its treatment of the financial value of EPL. As the Danish EPL regime is closer to the liberal provisions of the Anglo-Saxon countries than to the stringent rules in most European countries, this might – all else being equal – tend to underestimate the "true" level of expenditure in the latter group of countries. However, there is no simple way to assess employees' financial value of, for example, long notice periods.

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Table A1. **The structure of public expenditures in selected OECD countries, 2000**

Per cent of GDP

	Current expenditure						Net capital expenditure		
	Consumption	Social security payments	Other current transfers	Subsidies	Property income	Total current expenditure	Net investment	Other net capital outlays	Total public expenditure
Sweden	26.6	18.9	1.9	2.0	5.4	54.7	0.5	–0.1	55.2
Denmark	**25.5**	**17.5**	**2.5**	**2.3**	**4.6**	**52.4**	**–0.4**	**–0.2**	**51.8**
France	23.7	18.4	1.7	1.3	3.3	48.4	2.9	0.8	52.1
Greece	15.0	15.8	8.5	0.2	8.8	48.3	4.1	–0.3	52.0
Austria	19.8	18.7	2.6	2.6	3.5	47.3	0.3	1.8	49.4
Belgium	21.4	14.7	2.2	1.5	7.2	46.9	0.2	0.8	47.9
Finland	21.5	13.3	6.8	1.6	3.1	46.3	0.5	0.0	46.8
Germany	19.0	18.9	1.6	1.7	3.5	44.8	0.2	0.9	45.9
Italy	18.1	17.2	1.3	1.2	6.8	44.6	1.2	0.9	46.7
Norway	21.2	15.5	2.3	2.8	2.1	43.9	2.0	0.3	46.2
Netherlands	22.8	12.5	1.4	1.6	4.4	42.7	0.5	–0.4	42.7
Japan	16.1	9.8	9.7	0.8	3.5	39.9	3.5	1.2	44.6
Portugal	20.0	11.8	2.5	1.0	3.2	38.6	2.2	0.1	40.9
United Kingdom	18.5	13.5	2.2	0.6	3.0	37.8	1.1	0.3	39.2
Spain	16.9	12.4	1.3	1.2	3.6	35.5	3.3	0.7	39.4
United States	14.3	10.7	0.0	0.3	3.8	29.1	1.2	–0.3	30.0
Ireland	14.1	9.8	0.0	1.0	2.5	27.4	2.5	2.9	32.8

Note: The countries are ranked in descending order according to their total current expenditure ratio.
Source: OECD.

Table A2. **Net social expenditure in selected countries, 1997**

Per cent of GDP at factor costs

	I Total gross public expenditures[1]	II Gross public social expenditures	III Net current public social expenditures[2]	IV Net publicly mandated social expenditures[3]	V Net total social expenditure[4]
Austria	49.7	28.5	23.4	23.9	24.6
Australia	33.5	18.7	17.9	18.8	21.9
Belgium	48.8	30.4	26.3	27.5	28.5
Canada	43.9	20.7	18.7	18.7	21.8
Czech Republic	42.4	21.7	19.3	19.3	19.3
Denmark[5]	**(2) 54.4**	**(1) 35.9**	**(3) 26.7**	**(4) 26.9**	**(4) 27.5**
Finland	51.3	33.3	24.8	24.8	25.6
Germany	46.4	29.2	27.2	27.9	28.8
Ireland	34.0	19.6	17.1	17.1	18.4
Italy	48.5	29.4	24.1	25.2	25.3
Japan	41.7	15.7	14.8	15.3	15.7
Netherlands	44.4	27.1	20.3	20.8	24.0
New Zealand	38.1	20.7	17.0	17.0	17.5
Norway	43.8	30.2	24.4	25.1	25.1
Sweden	58.2	35.7	28.5	28.7	30.6
UK	40.9	23.8	21.6	21.9	24.6
United States	31.4	15.8	16.4	16.8	23.4
Country average	44.2	25.6	21.7	22.1	23.7
EU average[6]	47.7	29.3	24.0	24.5	25.8

Note: Tax breaks for social purposes (TBSPs) are defined as "those reductions, exemptions, deductions or postponement of taxes which: *a*) perform the same policy function as transfer payments, which, if they existed, would be classified as social expenditures; or *b*) are aimed at stimulating private provision of benefits". TBSPs do not include tax advantages favouring married people, as these are considered an integral feature of the general tax system.

1. As a percentage of GDP at market prices.
2. II minus tax claw-back of transfers (including indirect taxes) + TBSPs.
3. III plus net mandatory private social expenditure.
4. IV plus net voluntary private social expenditure.
5. The numbers in the parentheses indicate Denmark's ranking among the 17 countries in the table.
6. Simple average of the 10 EU countries in the table.

Source: Adema (2001).

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– T*hird*, tax advantages to married people are not considered TBSPs but rather an inherent feature of some countries' tax systems. Such tax advantages are virtually non-existing in Denmark, but tangible in, for example, Ireland, the United Kingdom, Italy, France and Germany. If the purpose were to assess net total public expenditure, it would be natural to adjust for such tax advantages, and doing so would most likely tend to lower Denmark's relative expenditure level.

It should be noted, though, that most countries' transfer systems are less redistributive (even if the resulting outlays are recorded as *social* spending) than Denmark's. This is due to the fact that the Danish system is basically a flat rate system in contrast to the earnings related social security systems that dominate elsewhere.

Notes

1. Two examples may serve to illustrate this. F*irst*, a classic ministerial report from 1973 (P*erspektivplan* II) provided scenarios for the possible evolution of the public sector 15 years ahead but failed – even in its fastest growing scenario – to envisage the strength of the subsequent actual expansion. S*econd*, when the early retirement benefit scheme was introduced in 1979, the number of beneficiaries was not expected to exceed 1 per cent of total employment. However, it rose rapidly to above 5 per cent and stands at 7 per cent today.
2. The expenditure-to-GDP ratio continued to rise in 1994, when it exceeded 60 per cent. However, this was largely due to the fact that social pensions and social assistance were made taxable as from that year in order to strengthen the transparency of the transfer system. The move to higher but taxable pensions and social assistance benefits was neutral to the budget balance but raised both tax- and expenditure-to-GDP ratios by 1.9 percentage points.
3. It should be emphasised that this does not in itself imply anything about the severeness of overall distortions of incentives to work and educate in Denmark relative to other countries. This should be assessed on the basis of marginal tax rates and how public spending is channelled back.

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Annex III

Aspects of the central-government budget process

The budget process

Denmark does not have a Budget Act proper. The overall regulations concerning the budget and accounting processes are laid down in the constitution (Table A3) and are supplemented by the authority given to the parliament's finance committee to approve appropriations on behalf of the parliament. The more detailed provisions concerning, for example, budgeting and accounting principles, account structure and spending ministries' preparation of contributions to the budget proposal are specified in executive orders and circulars from the Ministry of Finance (MoF). The same is true for procedures for following up on expenditure developments during the fiscal year as well as for supplying information to the public and the parliament which have mostly emerged more informally. The single most important legal document is the Budget Guidelines (*Budgetvejledningen*), which is presented to and adopted by the parliament's finance committee and, therefore, has virtually the same status as a law. The most recent overhaul of the Budget Guidelines took place in the 1980s. Over the past decade, a number of less significant adjustments have been made, most recently in 2001.

The key elements in the 1980s reform of the Budget Guidelines were the introduction of ministerial expenditure ceilings (MECs) for total outlays within each spending ministry (*totalrammer*); greater fungibility and decentralisation of the budget process; and the so-called "expanded total balance principle" (*Det Udvidede Totalbalanceprincip*).

The system of ministerial expenditure ceilings aims to combine top-down control over expenditure developments with flexibility within each ministry and its agencies. The preliminary MECs appearing in the budget proposal are initially proposed by the MoF, discussed bilaterally with each spending ministry and eventually determined by the cabinet's economic committee. The MECs comprise separate envelopes for wages, other current expenditures, investments and transfers (including entitlement outlays). The permission to defray entitlement outlays is given in separate laws. Appropriations for such expenditures (typically social transfers) are, therefore, only considered to be estimates. Consequently, the expenditure levels specified in such appropriations can be exceeded. With this exception, once the budget has been passed and the fiscal year has commenced, each ministry must offset overruns in some accounts by savings in others, or apply for a supplementary appropriation. Such a request must – according to a circular from the Prime Minister's office dating back to 1924 – be endorsed by the MoF before its submission to the parliament's finance committee. While the MECs are both efficient and flexible in the internal government decision process, they have less restraining impact in the political negotiations leading up to the adoption of the budget. Thus, the MECs included in the budget proposal may be changed during the political process.

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Table A3. **Main features and characteristics of the budget process**

General legal framework	Government internal budget preparation	The parliamentary process: Amendments and adoption	Implementation of the budget
	The role and responsibilities of the actors in the budget process		
The Constitution Submission of the budget bill four months before the start of the fiscal year, *i.e.* by the end of August. No tax can be levied, and no outlays can be defrayed, unless provided for in a Finance Act passed by the parliament before the beginning of the fiscal year. If the reading of the Draft Finance Act is not completed before the start of the fiscal year, a Provisional Appropriation Bill shall be laid before the Parliament.	**Ministry/Minister of Finance (MoF)** Eventually update the budget code (executive orders and circulars). Set warranted fiscal stance and ministerial expenditure ceilings (MECs). Bilateral talks with spending ministries on required savings. Supervise and control spending ministries' budget contributions. Negotiate with local governments on next year's block grants. Report continuously to the Cabinet's Economic Committee. Submit the budget proposal by the end of August at the latest.	**Ministry/Minister of Finance** Serve, coordinate and head political negotiations. **Spending ministries/ministers** Serve the political negotiations on the issues under their legal responsibility. Prepare the legislation required to enact issues under discussion and, if necessary manage separate political talks at subordinate level.	**Ministry/Minister of Finance** Surveillance of actual expenditure developments and provision of information to the public and the parliament via quarterly Budget Surveys. Taking stands on spending ministries' requests for submissions of appropriation documents for the parliament's finance committee. Coordinate follow-up on the political agreements. Draft and submit the Supplementary Appropriation Act.
The Budget Guidelines (*Budgetvejledningen*) The detailed appropriation provisions are laid down in the Budget Guidelines, an executive order from the Ministry of Finance, which is approved by the parliament's finance committee in its capacity of acting on behalf of the parliament in most budget matters. The Budget Guidelines specify, for example, the account structure of the budget, the various types of appropriations and the amount of flexibility attached to them.	**Spending ministries/ministers** Accept MECs or ask for bilateral talks with the MoF. Bilateral talks with the MoF on required savings. Draft budget contributions subject to MECs and technical input from the MoF.	**Parliament's standing finance committee** Draft the committee's report after the first and the second parliamentary reading. Pass the "local-government appropriation document" (in June), thereby endorsing next year's block grants to local governments.	**Spending ministries/ministers** Monitor actual expenditure developments in their domain and report to the MoF. Initiate preparation of appropriation documents for submission to the finance committee whenever this is judged necessary. Prepare legislation, or other measures, required to implement political agreements in their domain.

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Table A3. **Main features and characteristics of the budget process** (*cont.*)

General legal framework	Government internal budget preparation	The parliamentary process: Amendments and adoption	Implementation of the budget
	The role and responsibilities of the actors in the budget process		
The State Accountancy Act Specifies the overall provisions on the state's accountancy and empowers the MoF to set the more detailed provisions.	**Cabinet's Economic Committee** Decides on strategic elements of the budget on the basis of recommendations mainly from the MoF.	**Parliament/opposition parties** Propose adjustments to the government's budget proposal. Engage in political discussions with the government. Vote on the budget in its entirety after the third reading.	**Parliament's standing finance committee** Decide on appropriation documents, including requests for supplementary funds, on behalf of the parliament.
The Auditor General's Act Specifies the competencies and the responsibilities of the Auditor General.			**Parliament** Adopt the Supplementary Appropriation Act, thereby formally enacting decisions already taken by the finance committee.
Characterisation There is no Budget Act proper, and the constitution merely sketches a framework for the budget process. Instead, the detailed provisions governing the budget process and the drafting of the budget, etc., are set in executive orders and circulars from the MoF. The latter are approved by the parliament's finance committee, and they broadly have the same legal effect as a law. While it might be argued that the reliance on ministerial regulations risks reducing transparency and accountability, this should be weighed against the high degree of flexibility it allows.	**Characterisation** The budget preparation is decentralised in the sense that decisions are taken on the basis of bilateral talks, while the cabinet does not take collective decisions on the budget as a whole. The MoF holds a dominant strategic position by setting MECs and the fiscal-stance target, thereby ensuring a top-down approach.	**Characterisation** Most aspects of the political process between the launching and the adoption of the budget is determined by a common understanding amongst the actors more than by codified law. In principle, any topic may be brought up (by all actors) during the negotiations; the scope of amendments may consequently be considerable. A party that moves an amendment would typically (according to common understanding) have to propose its financing as well, but no legal constraints prevent slippage.	**Characterisation** The execution of the budget is normally in close compliance with the MECs as the scope for supplementary appropriations is limited by the requirement of the MoF's formal acceptance. Spending ministries have some flexibility in the use of their budgets, for example in terms of keeping unforeseen revenue and carrying forward unused funds forward. Appropriations can also be, within rather narrow limits, redistributed for other purposes than those specified in the budget.

Source: OECD.

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The 1980s budget reform also aimed at facilitating greater fungibility and decentralisation. For example, greater flexibility for individual ministries and agencies to reallocate funds within their envelope was introduced, and the scope for carrying forward unused appropriations to later years was increased. Since the 1980s, further adjustments have been made to increase flexibility. Thus, carry-over of funds is now possible for all current expenditure, investments and some transfer programmes. In addition, the scope of net budgeting has been increased to include all current expenditure, thereby allowing higher commercial revenues to finance higher spending.

Finally, reflecting the newly established framework for budget co-operation between the central government and counties and municipalities (see Chapter III), the so-called "expanded total balance principle" was introduced. It implies that financial compensation to local governments for extra expenditures originating from new, or changed, legislation must be made from within individual ministries' expenditure ceilings as well.

In its technical design, the budget may be characterised by the following features:

- Cash budgeting and accounting have been standard practice in the Danish budget system since the 1950s. However, budgeting for the net present value of government commitments that go beyond the current fiscal year (for example, interest subsidies to social housing) were introduced in 2001. In addition, a MoF working group recently recommended that accrual accounting should be implemented in central-government balance sheets as from 2004 (Finansministeriet, 2000*e*).
- Some elements of accrual budgeting are already in place. Central government property has, partly, been transferred to central units, which in turn rent out office space to government institutions. Also, appropriations for higher education and vocational training are based on activity levels (unit costs).
- Multi-year budgeting is relatively weak. Appropriation calculations are included for three budget-forecast years subsequent to the fiscal year, but this framework is not fully integrated with the medium-term economic projections underlying the formulation of the overall economic strategy.
- Since 1990, all benefit levels in the transfer system have been linked to private-sector wage developments (with a lag of two years). Changes in the average private-sector wage level that arise from compositional effects – for example the decreasing share of unskilled workers –are not compensated. Otherwise, there is no mechanism in the indexing scheme itself that causes the transfer-to-GDP ratio to either increase or decrease over time. The percentage annual increase in nominal benefit levels is announced routinely by the MoF in an executive order and involves no political decision-taking.

Public investment policies

Government investment policies have changed in several respects in the recent past. In particular, the responsibility for big infrastructure projects is increasingly being assigned to typically fully state-owned companies. Examples include the Danish natural gas grid, bridge and tunnel projects and the current metro project in Copenhagen. The responsibilities of the established companies typically comprise financing, building and operating the projects. Funds are raised in the capital markets against a state guarantee, while user charges are intended to facilitate the subsequent amortisation of the loans. In addition, several capital-intensive government activities, including postal services and railways, have been separated out from the budget itself and converted into independent companies. As the services from such investment are eventually mainly user-financed, the investment expenditure itself

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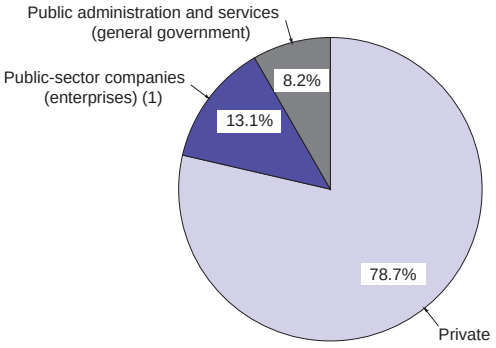
appears as business rather than general-government investment in the national accounts. Information on public-sector companies' investments is not available before 1993, but in the period since then, it has risen some 20 per cent in constant prices. The public-enterprise sector accounted for 13 per cent of total gross fixed investment in 1998, while the general government's share was 8 per cent (Figure A2). Companies owned by the central government account for the dominant part of the investments by public-sector enterprises. In contrast, local-government investment activity remains largely tax-financed, due, in part, to legal restrictions on their borrowing powers.

Inspired by the ongoing expansion of public-private partnerships (PPP) in the United Kingdom, a recent government report (Regeringen, 2001*b*) recommended analysing further the advantages and disadvantages of more direct private-sector involvement in investment activity, including the subsequent ownership and responsibility for maintenance activity. Indeed, private ownership might potentially have a number of advantages including a clearer governance structure, shifting the risk assessment to the market and higher productivity in the provision of services. In addition, past and recent experience with projects that have been undertaken by publicly owned companies are mixed, at best, as delays and cost overruns have been frequent. The arguments for allowing and encouraging (partial) private ownership – where appropriate – are quite similar to those underlying the policy in the recent years to privatise state-owned enterprises.

As in other countries, general-government (tax-financed) investment has declined steadily from some 4½ per cent of GDP in the early 1970s to 1¾ per cent in recent years. Investment expenditure funded by the central government is treated separately in the budget process. Following a new procedure introduced in 1999, spending ministries are not allocated funds for investment except to continue projects already started. New projects are

Figure A2. **Private and public gross fixed investments**

Per cent



1. Public-sector companies' investments figure in the national accounts as investments in the enterprise sector, rather than in general government.

Source: Regeringen (2001*b*).

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prioritised centrally within a budget envelope by the government's economic committee amongst proposals submitted by individual ministries. Submissions must provide certain background information, ranging from proper cost-benefit analysis for major infrastructure projects to considerations for existing capacity and future operational costs for cultural institutions. Cost-benefit analyses must comply with MoF guidelines (Finansministeriet, 1999*b*), which, amongst other things, recommend using a real rate of return of 6 to 7 per cent. In contrast, local governments' investment decisions (covering three-quarters of general-government investment) are subject to only very few regulatory constraints. For example, there exist no requirements to undertake cost-benefit analysis.

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Annex IV

Financing local governments

Extensive devolution of responsibilities to local governments

The most important functions carried out at the county level are hospital services, health insurance, upper secondary education, care for disabled people and regional public transport. The municipalities' tasks comprise the provision of services such as primary and lower-secondary education, childcare, elderly care and labour-market programmes for the unemployed without benefit entitlement (Table A4). In addition, the extent of local-government involvement in the social-security schemes is unique amongst the OECD countries; most importantly, the municipalities are responsible for assessing the eligibility of applicants for

Table A4. **The responsibilities of counties and municipalities**

Municipalities

Municipalities	Countries
Services	Services
Primary and lower secondary education	Hospitals
Childcare	Health insurance
Elderly care	Upper secondary education
Libraries, local sports facilities	Care of mentally and physically disabled people
Public utilities	Certain cultural activities
Environmental supervision	Regional public transport
Emergency services	Regional physical planning
Labour-market programmes for cash-benefit recipients	
Integration of refugees	
Social transfers	
Assessment of eligibility for:	
Early retirement (disability) pension	
Cash benefit	
Rehabilitation benefit	
Wage subsidies to people with reduced working capacity	
Follow-up measures *vis-à-vis* sickness-benefit recipients	

Source: OECD; Ministry of the Interior (1999).

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early retirement pension, which is the second-largest social-security scheme after old-age pension.

The 1970 municipal reform resulted in a local-government structure with 14 counties and 275 municipalities, a structure that has since remained almost unchanged.[1] The decentralisation process was made possible by this reform, but it has gone much further than what was envisaged at that time (Indenrigsministeriet, 1998). Decentralisation has also continued far beyond what could reasonably be labelled as the provision of "local public goods", and thereby beyond the arguments of the classical theory of fiscal federalism (Oates, 1990 and Lotz, 1998). Among the larger spending areas only tertiary education and labour-market policies are still managed at the national level along with central administration, defence and the judicial system.[2]

Local governments' revenue structure

Taxes account for 58 per cent of local governments' total revenue. In addition to income taxes local-government taxes also comprise land and property taxes:

- The income tax base is set by the central government. The average tax rate (the total of municipal and county rates) has risen from less than 20 per cent in the early 1970s to 32 per cent in 2001.
- Both counties and municipalities levy land tax at rates that are allowed to vary between 0.6 and 2.4 per cent of the land value. Around one-quarter of the municipalities have set their rate at – or near – the minimum level, and very few apply the highest rate possible. The average land-tax rate was 1.6 per cent in 2001 and has risen much less over the recent past than have income-tax rates.
- In addition to the land tax, the former imputed rents tax was converted into a municipal property tax as part of the 1998 tax reform. However, the property-tax rate is uniform across municipalities and is fixed by the central government.

The second largest revenue component is sales of goods and services at 23 per cent (Figure A3). This latter includes primarily households' and companies' payments for public utilities' services but also user charges for childcare and certain auxiliary services. Block grants from the central government, which are not tied to any specific purpose, amount to some 11 per cent of total revenues. These grants are distributed to local governments according to their tax bases and do not, therefore, imply any redistribution. Finally, reimbursements amount to 8 per cent of revenues (excluding old-age pension, which is fully reimbursed by the central government). Reimbursements apply primarily to social transfer schemes such as early retirement (disability) pension, cash benefit and sickness benefit. Municipalities' outlays under these schemes are partly compensated directly by the central government at rates that vary between 35 and 50 per cent. The differences between total outlays and aggregate reimbursements are paid by the central government according to the same distribution as the block grants.

Inter-municipal equalisation

Fiscal conditions vary considerably across counties and municipalities, as both tax bases and expenditure needs are related to age profiles and social structure. For example, taxable income per inhabitant in the wealthiest municipality is almost 2½ times the level in the poorest. If each individual local authority were to fund its outlays itself, tax rates and/or service

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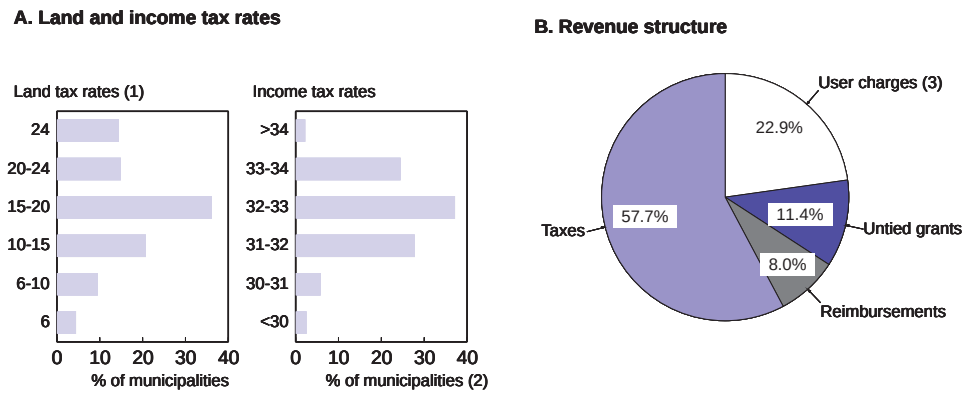
Figure A3. **Local governments' revenue structure, 2000**

Per cent



1. Per thousand of land values.
2. If measured as a percentage of all inhabitants, the distribution of income tax rates is more concentrated in the 32 to 33 per cent range.
3. This includes primarily payments for public utilities services but also user charges for child care and certain auxiliary services.

Source: Statistics Denmark; Ministry of the Interior (1999).

standards would consequently be very uneven across the country. This is generally not considered acceptable, given the typically non-local-goods character of the services for which local governments are responsible. The inter-municipal equalisation schemes, which were introduced in the early 1980s, were designed with a view to ensuring that relatively even standards can be obtained across local governments without major differences in tax rates. Basically, the equalisation schemes are zero-sum processes in which funds are transferred from rich to poor municipalities. Most municipalities have, afterall, tax bases and expenditure needs that are quite close to the national average. Therefore, the funds being moved from one local authority to another are relatively modest, amounting to some 1¼ per cent of GDP. However, for individual municipalities the net payments to – or net receipts from – the system can be very significant. The biggest (relative) net contributing municipalities could reduce its tax rate by some 8 to 9 percentage points without the equalisation schemes, while the municipalities receiving the biggest relative benefits would need to raise their tax rates similarly (Figure A4, Panel A).

Redistribution *via* equalisation is essential, given the combination of extensive taxing powers, non-redistributing central-government grants and the devolution of non-local-goods services to local governments (Lotz, 2001). As only per-inhabitant differences are equalised, the schemes do not reduce individual municipalities' incentives to increase tax bases by attracting new residents. The equalisation schemes aim to strike a balance between their main purpose of making roughly equal standards possible at equal tax rates and providing reasonable incentives for individual local authorities to strengthen tax bases and operate efficiently. While they imply considerable narrowing of tax-rate dispersion, differences are far from being eliminated altogether (Figure A4, Panel B). Moreover, the Danish schemes imply

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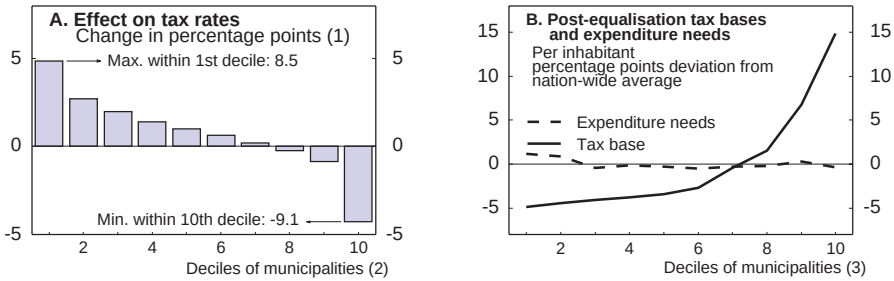
Figure A4. **Effects of inter-municipal equalisation**
2000



1. Increase in the tax rate that would be necessary to maintain current spending if equalisation were absent and central government grants were distributed according to the number of inhabitants instead of taxable per-inhabitant income. Median municipality within each decile.
2. Ranked in descending order according to the tax-rate effect.
3. Increasing according to per-inhabitant taxable income.

Source: Ministry of the Interior; Lotz (2001).

less redistribution than those in some other countries, for example, Norway, Finland, Sweden and Australia; in the last two, equalisation is approximately 100 per cent on both the revenue and the expenditure side. The schemes appear to work without too much tension, although some dissatisfaction is frequently voiced from both the largest contributors and the biggest beneficiaries. Nonetheless, the design of the current equalisation system has some peculiarities. In particular, the range of different equalisation percentages and the criteria upon which they hinge (see box below) are somewhat arbitrary, and, in particular, it is not obvious why equalisation in the Copenhagen metropolitan area should be greater than elsewhere (Table A5). Furthermore, the more generous grants scheme to poor municipalities (with per capita tax bases below 90 per cent of the national average) implies that these have very little incentive to strengthen their tax base.

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Inter-municipal equalisation

Equalisation takes place on both the revenue and the expenditure sides of local governments' budgets, and it applies to both municipalities and counties. The equalisation base on the revenue side is the sum of the income tax base and a share of taxable land values, all measured on a per-inhabitant basis. The equalisation for expenditure needs is more complicated; it is based on a composite index, which takes three elements into consideration. The first is an "estimate" of the minimum amount required to operate a reasonable professional administration in a small municipality (DKr 7½ million). The second is the age profile in each local authority, thereby taking account of demographic expenditure pressures. Finally, a number of indicators of the social structure (for example the number of children of single parents and the number of rental dwellings) enter the calculation of the composite index. In the selection of these indicators of expenditure needs, the intent has been to ensure that they are "exogenous" in the sense that individual local authorities cannot manipulate the outcome, at least not in the short term.

The county schemes equalise 80 per cent of the difference from the national averages for tax bases as well as expenditure needs (Table A5). The national municipal scheme equalises only 45 per cent of the deviations from average tax bases and expenditure needs. However, an additional 40 per cent of the differences is equalised within the greater Copenhagen area, and there is a separate scheme that compensates "poor" municipalities (defined as those with a tax base per inhabitant of less than 90 per cent of the national average) for an additional 40 per cent of the gap. In sum, the equalisation is generally greater at the county than at the municipal level and greater in the Copenhagen metropolitan area than in the rest of the country.

Table A5. **Inter-municipal equalisation schemes**[1]

Municipalities

	Tax base equalisation[2]		Equalisation of expenditures needs[3]	
	Counties	Municipalities	Counties	Municipalities
National scheme	80 per cent	45 per cent	80 per cent	45 per cent
Copenhagen area scheme	–	40 per cent on top of the national scheme	–	40 per cent on top of the national scheme
Grants to poor municipalities[4]	–	40 per cent on top of the national scheme	–	

Note: The percentages refer to the proportion of the difference between an individual local government's tax base or expenditure needs and the average, which is equalised.

1. The Ministry of the Interior operates a number of other minor equalisation schemes in addition to those reported here. For example, separate schemes are in place for sharing costs of refugees or supporting municipalities on small islands.
2. The tax base comprises the income tax base plus 6.5 per cent of taxable land values. Municipalities also receive a share of corporate tax revenue, but the latter is not equalised.
3. Expenditure needs are proxied by a composite index comprising three basic elements: *i*) a basic amount of DKr 7.5 billion; *ii*) a number of demographic characteristics; and *iii*) a number of social criteria.
4. Municipalities with a tax base per inhabitant that is at least 10 per cent below the national average get an additional 40 per cent of the difference equalised, *i.e.* on the top of the national scheme. This grant, for which close to 100 municipalities qualified in 2000, is financed collectively *via* the block grant scheme.

Source: Ministry of the Interior (1999).

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Notes

1. The only change is the imminent amalgamation of the five municipalities of the island of Bornholm into one administrative unit, which was decided upon in a referendum in May 2001. Before 1970, Denmark was divided into 86 boroughs, and around 1 300 parishes within 25 counties. In many areas, however, the urban expansion had extended across the boundaries of neighbouring parishes, and many parishes were too small to near the costs of professional administration, let alone managing basic services like primary education or elderly care.
2. In the case of active labour-market policies, the social partners have been allowed greater influence on policies than users in other areas. For example, the unemployment insurance funds are managed by the social partners, who also hold considerable discretion over decisions regarding the content of adult-education programmes. Even the appeal system is different from that in other sectors. Thus, for social services and in the health sector, the highest appeal body is a purely state authority whose decisions may be contested in the ordinary judicial system. For labour-market policies, the highest appeal instance is indeed formally independent, but the social partners appoint the majority of the members of its board, and its decisions cannot be appealed.

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Annex V

Policies on user charges in selected areas

As argued in Chapter III user charges are not, at present, applied consistently. F*irst*, while distributional concerns are allegedly important in shaping policies, it is not always obvious that these are more compelling for services provided free of charge than for others on which user charges are levied. Labour-market training for employed, for example, is generally free, while dental care is subject to considerable user co-payment. S*econd*, where charges exist, they do not appear well aligned with the likely price sensitivity of demand. The optimal level of user charges might tentatively be defined as that which "best" balances the trade-off between unwarranted distributional effects and beneficial impacts in curbing (excessive) demand relative to free provision.[1] Defining the optimal charge is obviously much easier than determining it in practice, if only because information on the price elasticities is limited. A general guideline is, nevertheless, that charges should be linked directly to individuals' actual consumption,[2] in contrast with fixed fees (earmarked taxes). However, distributional concerns could often provide an argument for a cap on user charges, in particular if the price sensitivity declines with increasing demand, as might often be the case with health services, for example.

This annex provides some information on current policies in the areas of childcare and pharmaceuticals, where user co-payment schemes exist. In addition, it outlines some arguments for and against broadening the application of user charges, even if only of modest size, to a range of other services currently provided free.

Childcare

For childcare, municipalities are entitled to levy user charges subject to limits set by the central government. The statutory co-payment percentages vary between 30 and 35 per cent of the full costs, while effective co-payments are around 20 per cent of full costs. The introduction of the "brothers-and-sisters" discount early in the 1990s implied a significant reduction of effective user charges. The discount, which is compulsory for the municipalities, implies that parents pay normal tariff for one child, while for all subsequent children, they enjoy a 50 per cent discount. For example, parents with one child under three years of age and one between three and six years of age will typically pay half charge for the latter child, which in an average municipality would be some DKr 750 per month (the equivalent of € 100) for a full-time place.

For children aged three to five years (in kindergarten) the coverage ratio has increased from below 80 per cent in 1991 to almost 95 per cent in 2001. Over the same period, half-time day-care places[3] have largely been abolished and converted into full-time places. In part, this standardisation is likely to result from a demand response to the "brothers-and-sisters" discount.[4] Thus, if only the marginal price is low enough parents may demand full-time day

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care even if their "objective need" is well below this and despite children's actual attendance (*i.e.* the number of hours per week in day-care) being well below full time on average. Considering the extra flexibility full-time places provide, the marginal price is indeed quite low, in particular if the "brothers-and-sisters" discount applies; in the last case parents' monthly payment would typically only be DKr 300 higher for a full-time place relative to a half-time place. The general point is that families' true demand for childcare is likely to vary considerably according to working time and working patterns, and the availability of help from relatives, etc. This does not conform to the observed standardised supply structure.

Experience from Sweden over the past decade provides an interesting contrast to the standardisation observed in Denmark. The structure of the Swedish childcare sector around 1990 showed considerable resemblance with its Danish counterpart both in having a limited degree of diversity in the types of places provided (per hour, half-time, full-time, etc.) and in the level of parents' payment. However, over the course of the 1990s the sector evolved very differently in the two countries. Unlike in Denmark, user charges were not reduced in Sweden and municipalities have not been bound by legal constraints in setting the level of charges. Against this backdrop, it is interesting to observe that the supply structure became much more diversified during the course of the 1990s and user charges became more closely linked to the actual attendance. In 1993, 60 per cent of the municipalities in Sweden had no or only a weak link between the size charges and the number of hours of attendance. In that year, one-fifth of the municipalities had four different rates or more. In 1999, the equivalent figure was 67 per cent, and 18 per cent of the municipalities had moved to charging on a per-hour basis (Skolverket, 1999).

The design of charges in Denmark appears inappropriate from an economic point of view, as they relate to average rather than marginal consumption. There is virtually no relation between the individual child's actual time spent in day care and the parents payment, partly because full-time places are the only type of service on offer. Thus, the present system neither encourages nor enables parents with reduced need[5] for care hours to align demand with their actual need. Redesigning user charges by relating them to marginal consumption, *i.e.* moving to charges that vary with actual attendance could facilitate a reduction in (excessive) demand without compromising labour supply. Some caution is necessary, though, as excessive time dependence might add to effective marginal tax rates, thereby discouraging high or increasing working hours.

Pharmaceuticals

Demand for pharmaceuticals is normally considered relatively price elastic,[6] at least for small-scale users, indicating that private co-payment could be effectual in curbing public outlays. However, discouraging excessive consumption should be reconciled with safeguarding those in need of expensive drugs and long-term treatment. The level of private co-payment in Denmark is a little above the OECD average, and per capita consumption is somewhat lower (Jacobzone, 2000). The latter reflects cautious prescription habits and the success of the public sector in using its monopsony power to impose the use of generics and keep a tight rein on prices through reference pricing. Nonetheless, public expenditure has increased rapidly over the past decade. The subsidy scheme in place during under this period may have given insufficient economic incentive for households to restrain consumption, as the co-payment was fully proportional to the final price.

The "stop-loss" type of subsidy scheme that came into force in March 2000 aimed at improving the system on this count. It effectively increased payment for consumption under a certain threshold, while providing better coverage for high users. Unlike its predecessor, the current scheme does not embody any differentiation of various types of drugs. Instead,

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consumption below DKr 500 per year is fully paid by the consumer, while 85 per cent of consumption above DKr 2 800 is reimbursed; intermediate consumption is partially reimbursed at lower percentages.[7] It is still too early to assess the impact of the new system and the threshold in particular. However, when a similar system was in place in 1990 and 1991, the trend increase in public reimbursements reversed to a significant decline. At that time, the "stop-loss" scheme was abolished less than two years after its introduction, as its administration proved too cumbersome. This drawback has been avoided in the new system under which all purchases are instantaneously stored electronically in a central register.

Broadening user charges to other services

Elderly care

Public spending for elderly care is higher in Denmark than elsewhere, while informal care providers play a less important role. In large part this is attributable to within-family resources being limited by factors such as high labour-force participation rates and the fact that a higher proportion of the elderly live alone. While broad, publicly financed provision of elderly care therefore seems natural, the absence of any user charges is nevertheless remarkable. Following the abolition of user charges in 1989, Denmark stands out in an international comparison by providing elderly care, *i.e.* personal care and practical home help free (Rostgaard and Fridberg, 1998; Jacobzone, 1999). As part of the agreement with the National Association of Local Governments in 1998, the previous government committed itself to introducing income-dependent user charges for practical home help. However, it subsequently retreated from this commitment due to both political and administrative difficulties. As the demand for home care is likely to quite price sensitive excessive demand and spending pressures could probably be better contained if even modest user charges were introduced.

Health care

Primary health care is free and fully tax financed. Resistance against user charges is profound, not only amongst policy makers; in a recent analysis of the Danish health-care sector the Economic Council concluded, mainly on the ground of the implied distributional effects, that user charges are inappropriate in this area (Det økonomiske Råd, 1999). Consumption of primary health services is indeed higher for people in the lowest deciles of the income distribution than for those with high incomes, in particular because pensioners typically have above-average consumption and below-average incomes. Thus, introducing user charges and cutting taxes would increase inequality, but could be alleviated to some extent by exempting pensioners or making charges subject to a ceiling. Nevertheless, many countries do apply user charges for primary health care.[8] This includes the other Scandinavian countries, where equity objectives are probably as strong as in Denmark.[9] The fact that private co-payment for these services has a long history in Sweden and Norway but are considered unthinkable in Denmark illustrates the ambiguity of the distribution argument.

Moreover, unlimited user charges (*i.e.* without a ceiling) exist for dental care, and the effective private co-payment rate has increased over the past 15 years from 25 to 40 per cent. Although the new subsidy scheme for pharmaceuticals described above has now reduced co-payments for high users, unlimited charges have also been in place in this area until recently, which is somewhat surprising, given the emphasis on equity objectives in other areas.

A number of studies (for example, Pedersen *et al.* (1995) suggest that user payment for visits to general practitioners, subject to an annual ceiling of some DKr 1 000 (€ 135) might

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reduce public spending by some 25 per cent – one-quarter of which is attributable to the demand response and the remaining part due to the revenue from the user charge itself. These savings are relatively modest in absolute terms and should be balanced against other considerations such as the risk of deterring preventive cares. In addition, it should be kept in mind that the Danish health care sector is relatively cheap exactly because of the gate-keeper function served by the general practitioners.

Tertiary education

While the private return on human capital investment is reduced by the compressed wage structure and the high progressivity of the tax system, the financial terms of acquiring tertiary education are generous. There are no tuition fees and the effective, private co-payment rate is very low by international standards (see Table 18). In addition, student grants are more generous than elsewhere. Introducing tuition fees could probably improve outcomes in several respects. As discussed in the previous S*urvey*, the amount of excessive time spent in the education system is considerable, and, on average, Danish students graduate at a higher age than in almost any other OECD country. This causes significant fiscal costs because of the implied extra education spending and the tax revenue foregone because of the postponed entry into the workforce. Including some element of payment would strengthen incentives to complete studies on time. The fact that the return on human capital investment is primarily private (despite the aforementioned effects) is an argument for such tuition fees being substantial. However, if this is deemed unacceptable because of the implied distributional effects, even linking modest charges to examinations (thereby putting a price on not attending planned exams or re-doing exams to improve previously obtained results) might in itself encourage a faster pass-through.

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Notes

1. If the overriding objectives were to ensure economic efficiency, marginal cost pricing would theoretically provide the "best" user charges. If fees are set below marginal costs, and demand is price sensitive, individuals will consume more than they otherwise would.
2. Insurance companies use co-payments in precisely the same way to limit "moral hazard".
3. Opening hours in half- and full-time places may vary somewhat, but typical opening hours would be daily between 7 h 00 and 13 h 30 in half-time places and between 6 h 30 and 17 h 30 in full-time places.
4. Empirical findings suggest that the demand for out-of-home childcare is quite responsive to the price when demand is not rationed. For example, Hofferth and Wissoker, (1991) estimate a price elasticity of –3 to –4 on American data, and Gustafsson and Stafford (1991) find an elasticity of –2.7 on Swedish data between the joint decision to work and use public childcare one the one hand and parents' payment for childcare on the other.
5. Such as part-time and shift workers, parents on leave, temporarily non-employed or parents with non-working relatives living nearby.
6. Jacobzone (2000) refers to results from the Rand Health Insurance Experiment, considered the most comprehensive study available on price effects on demand for pharmaceuticals. The study found that co-payments of 25 per cent reduce demand by a quarter, thereby suggesting a price elasticity of –1.
7. These are 2000 levels which are adjusted annually according to private-sector wage developments. For the chronically ill, the annual payment is capped at DKr 3 600.
8. Manning *et al.* (1987) find that the demand response from introducing user charges is bigger for primary than for secondary care and that it weakens the greater is the share of private co-financing.
9. Another group of counties, including Canada, Germany, the Netherlands (public), Spain and the United Kingdom follow the Danish example of providing primary health care free.

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Annex VI

Calendar of main economic events

2000

August

The Minister for Finance presents the proposed Budget for the fiscal year 2001 to Parliament.

The National Bank raised its lending and certificates of deposits' rates by 40 basis points during the course of the month.

September

The National Bank raised the discount rate by 25 basis points to 4.5 per cent, in line with the rise in ECB rates.

Referendum held on Danish participation in the third phase of the European Monetary Union, with 53 per cent voting against adopting the euro.

Immediately following the referendum, the National Bank raised its lending and certificates of deposits rates by a further 50 basis points to reach a margin of 110 basis points above the discount rate. (This margin was gradually unwound in multiple small steps over the subsequent months and by May 2001 it was only 50 basis points).

October

The National Bank raises the discount rate to 4.75 per cent in line with ECB moves.

Local governments adopt their budgets for 2001 with an average increase in income tax rates of 0.4 per cent.

November

The government concludes its negotiations on the 2001 Budget and is able to present an agreement entered into with the Socialist People's Party and the Unity List – with the Centre Democratic Party taking part in a number of sub-agreements. Among other things, the overall agreement includes initiatives in the area of education, research, environment, expansion of the green jobs pool, increased benefits and services for some categories of children and young people, and senior citizens, food security and veterinary quality, transport, and health. Some environmental taxes were raised.

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December

European Union adopts the Nice Treaty.

The Prime Minister announces a reshuffle of ministerial portfolios.

2001

March

The United States ban imports of meat from the European Union as foot-and-mouth disease is found in France.

May

United States lift import ban on Danish meat products

The National Bank lowers the discount rate by 25 basis points to 4.5 per cent, matching the reduction of the European Central Bank.

June

Denmark ratifies the Nice Treaty.

August

Agreements are reached with the Associations of Local Authorities on the budgets for 2002.

The Minister of Finance introduces Budget Bill for 2002 to Parliament.

September

The National Bank lowers the discount rate by 50 basis points to 3.75 per cent, in line with reductions by most central banks including the European Central Bank to facilitate the reopening of the New York Stock Exchange after the terrorist attacks.

October

Local government budgets are finalised for 2002, with an average increase in income tax rates of 0.1 per cent.

Parliament is dissolved and a snap-election is called for 20 November.

November

The Social Democratic-led coalition government headed by Poul Nyrup Rassmussen is defeated by an alliance led by Liberal leader Anders Fogh Rasmussen.

The National Bank lowers the discount rate by 50 basis points to 3.25 per cent, in line with the European Central Bank reduction.

December

Prime Minister Anders Fogh Rasmussen outlines his government's programme in his opening address to Parliament.

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